

we have

$$\begin{aligned}
 \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_{C_1} \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy \\
 &= \int_0^{2\pi} \frac{-a \sin t}{a^2} (-a \sin t) dt + \frac{a \cos t}{a^2} (a \cos t) dt \\
 &= \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \int_0^{2\pi} dt = 2\pi. \quad \blacksquare
 \end{aligned}$$

The conclusion of Green's Theorem (18.17) may be expressed in vector form. To see how this is accomplished, first note that if $\mathbf{F}$ is a vector field in two dimensions, then we may write

$$\mathbf{F}(x, y) = M\mathbf{i} + N\mathbf{j} + 0\mathbf{k}$$

where $M = M(x, y)$ and $N = N(x, y)$. The curl of $\mathbf{F}$ is

$$\begin{aligned}
 \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & 0 \end{vmatrix} \\
 &= 0\mathbf{i} + 0\mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k}.
 \end{aligned}$$

Observe that the coefficient of $\mathbf{k}$ has the same form as the double integral in (18.17). If we consider the unit tangent vector

$$\mathbf{T} = \frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} + \frac{dz}{ds} \mathbf{k}$$

to C , where s represents arc length, then Green's Theorem may be written

$$(18.19) \quad \oint_C \mathbf{F} \cdot \mathbf{T} ds = \iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} dA.$$

Since $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ is the component of $\text{curl } \mathbf{F}$ in the direction of the z -axis we shall refer to it as the **normal component** (to R) of $\text{curl } \mathbf{F}$. In words, (18.19) may be phrased as follows: *the line integral of the tangential component of $\mathbf{F}$ taken along C once in the positive direction is equal to the double integral over R of the normal component of $\text{curl } \mathbf{F}$.* The three-dimensional analogue of this result is *Stokes' Theorem*, which will be discussed in Section 18.7. Physical interpretations will be given at that time.

Green's Theorem is important in the area of mathematics known as *complex variables*, a subject that is fundamental for many applications in the physical sciences and engineering.

18.4 Exercises

Use Green's Theorem to evaluate the line integrals in Exercises 1–14.

- 1 $\oint_C (x^2 + y) dx + xy^2 dy$, where C is the closed curve determined by $y^2 = x$ and $y = -x$ from $(0, 0)$ to $(1, -1)$.
- 2 $\oint_C (x + y^2) dx + (1 + x^2) dy$, where C is the closed curve determined by $y = x^3$ and $y = x^2$ from $(0, 0)$ to $(1, 1)$.
- 3 $\oint_C x^2 y^2 dx + (x^2 - y^2) dy$ where C is the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$, and $(0, 1)$.
- 4 $\oint_C \sqrt{y} dx + \sqrt{x} dy$ where C is the triangle with vertices $(1, 1)$, $(3, 1)$, and $(2, 2)$.
- 5 $\oint_C xy dx + (y + x) dy$ where C is the unit circle with center at $(0, 0)$.
- 6 $\oint_C e^x \sin y dx + e^x \cos y dy$ where C is the ellipse $3x^2 + 8y^2 = 24$.
- 7 $\oint_C xy dx + \sin y dy$ where C is the triangle with vertices $(1, 1)$, $(2, 2)$, and $(3, 0)$.
- 8 $\oint_C \tan^{-1} x dx + 3x dy$ where C is the rectangle with vertices $(1, 0)$, $(0, 1)$, $(2, 3)$, and $(3, 2)$.
- 9 $\oint_C y^2(1 + x^2)^{-1} dx + 2y \tan^{-1} x dy$ where C is the graph of $x^{2/3} + y^{2/3} = 1$.
- 10 $\oint_C (x^2 + y^2) dx + 2xy dy$ where C is the boundary of the region bounded by the graphs of $y = \sqrt{x}$, $y = 0$, and $x = 4$.
- 11 $\oint_C (x^4 + 4) dx + xy dy$, where C is the cardioid $r = 1 + \cos \theta$.
- 12 $\oint_C xy dx + (x^2 + y^2) dy$, where C is the first quadrant loop of $r = \sin 2\theta$.
- 13 $\oint_C (x + y) dx + (y + x^2) dy$, where C is the boundary of the region between the graphs of $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.
- 14 $\oint_C (1 - x^2 y) dx + \sin y dy$, where C is the boundary of the region that lies inside the square with vertices $(\pm 2, \pm 2)$ and outside of the square with vertices $(\pm 1, \pm 1)$.

In Exercises 15 and 16 use Theorem (18.18) to find the area of the region bounded by the curve C with the given parametric equations.

- 15 $x = a \cos^3 t$, $y = a \sin^3 t$; $0 \leq t \leq 2\pi$
- 16 $x = a \cos t$, $y = a \sin t$; $0 \leq t \leq 2\pi$

In Exercises 17 and 18 use (18.18) to find the area of the region bounded by the graphs of the given equations.

- 17 $y = 4x^2$, $y = 16x$
- 18 $y = x^3$, $y^2 = x$
- 19 Prove formula (b) in the partial proof of Green's Theorem.
- 20 If $\mathbf{F}(x, y)$ is a two-dimensional vector field and $\int_A^B \mathbf{F} \cdot d\mathbf{r}$ is independent of path in a region D , use Green's Theorem to prove that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every piecewise smooth simple closed curve in D .
- 21 Let R be the region bounded by a piecewise smooth simple closed curve C in the xy -plane. If the area of R is A , use Green's Theorem to prove that the centroid $(\bar{x}, \bar{y})$ is

$$\bar{x} = \frac{1}{2A} \oint_C x^2 dy, \quad \bar{y} = -\frac{1}{2A} \oint_C y^2 dx.$$

- 22 Suppose a homogeneous lamina of density k has the shape of a region in the xy -plane that is bounded by a piecewise smooth simple closed curve C . Prove that the moments of inertia with respect to the x - and y -axes are

$$I_x = -\frac{k}{3} \oint_C y^3 dx, \quad I_y = \frac{k}{3} \oint_C x^3 dy.$$

- 23 Use Exercise 21 to find the centroid of a semicircular region of radius a .
- 24 Use Exercise 22 to find the moment of inertia of a homogeneous circular disc of radius a with respect to a diameter.
- 25 Suppose f and g are differential functions. Prove that $\oint_C f(x) dx + g(y) dy = 0$ for every piecewise smooth simple closed curve C .
- 26 Let $M = y/(x^2 + y^2)$ and $N = -x/(x^2 + y^2)$. If R is the region bounded by the unit circle C with center at the origin, show that

$$\oint_C M dx + N dy \neq \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

Explain why Green's Theorem is not true in this instance.

- 27 Suppose $\mathbf{F}(x, y)$ has continuous first partial derivatives in a simply connected region R . If $\text{curl } \mathbf{F} = \mathbf{0}$ in R , prove that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every piecewise smooth simple closed curve C in R .

18.5 Surface Integrals

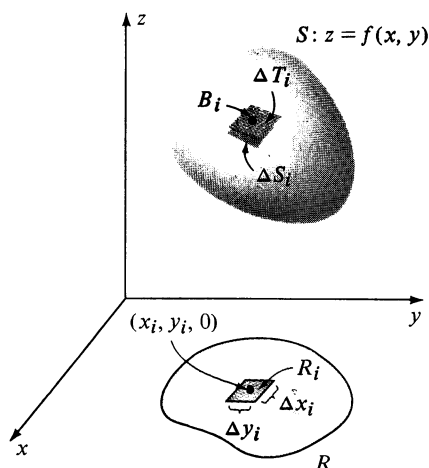


FIGURE 18.15

Line integrals are evaluated along curves. Double and triple integrals are defined on regions in two and three dimensions, respectively. It is also possible to consider an integral of a function over a surface. For simplicity we shall restrict our discussion to rather well-behaved surfaces, and our demonstrations will be at an intuitive level. Rigorous treatments may be found in texts on advanced calculus.

If the projection of a surface S on a coordinate plane is a region of the type we considered for double integrals, then S is said to have a **regular projection** on the coordinate plane. Suppose that S is the graph of $z = f(x, y)$ and that S has a regular projection R on the xy -plane. Suppose further that f has continuous first partial derivatives on R . In Chapter 17 we defined the area A of S . A similar technique may be used to define an integral of $g(x, y, z)$ over the surface S , where the function g is continuous throughout a region containing S . We shall employ the notation used in Section 17.9. In particular, as illustrated in Figure 18.15, ΔS_i and ΔT_i will denote areas of parts of S and the tangent plane to S at $B_i(x_i, y_i, z_i)$, respectively, that project onto the rectangle R_i of an inner partition P of R . As in the definitions of all previous integrals, we evaluate g at B_i for each i and form the sum $\sum_i g(x_i, y_i, z_i) \Delta T_i$. By definition, the **surface integral of g over S** is

$$(18.20) \quad \iint_S g(x, y, z) \, dS = \lim_{\|P\| \rightarrow 0} \sum_i g(x_i, y_i, z_i) \Delta T_i$$

where the limit of the sum is defined in the usual way. In a manner similar to the development of formula (17.31) for surface area, this integral may be evaluated by means of the formula

$$(18.21) \quad \iint_S g(x, y, z) \, dS = \iint_R g(x, y, f(x, y)) \sqrt{(f_x(x, y))^2 + (f_y(x, y))^2 + 1} \, dA.$$

If S is the union of several surfaces of the type we are considering, then the surface integral is defined as the sum of the individual surface integrals. Observe that if $g(x, y, z) = 1$ for all (x, y, z) , then (18.21) reduces to (17.31) and hence the surface integral equals the surface area of S .

An elementary physical application may be obtained by considering a thin metal sheet, such as tinfoil, that has the shape of S . If the area density at (x, y, z) is $g(x, y, z)$, then (18.21) is the mass of the sheet. The center of gravity and moments of inertia may be obtained by employing the methods used for solids in the preceding chapter.

Example 1 Evaluate $\iint_S x^2 z \, dS$ where S is the portion of the circular cone $z^2 = x^2 + y^2$ that lies between the planes $z = 1$ and $z = 4$.

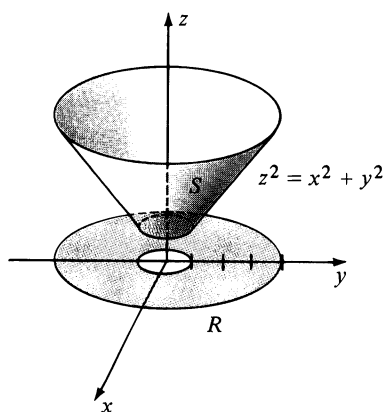


FIGURE 18.16

Solution As shown in Figure 18.16, the projection R of S onto the xy -plane is the annular region bounded by circles of radii 1 and 4 with centers at the origin.

If we write the equation for S in the form

$$z = (x^2 + y^2)^{1/2} = f(x, y)$$

then

$$f_x(x, y) = \frac{x}{(x^2 + y^2)^{1/2}} \quad \text{and} \quad f_y(x, y) = \frac{y}{(x^2 + y^2)^{1/2}}.$$

Applying (18.21) and noting that the radical reduces to $\sqrt{2}$, we obtain

$$\iint_S x^2 z \, dS = \iint_R x^2 (x^2 + y^2)^{1/2} \sqrt{2} \, dx \, dy.$$

Using polar coordinates to evaluate the double integral,

$$\begin{aligned} \iint_S x^2 z \, dS &= \int_0^{2\pi} \int_1^4 (r^2 \cos^2 \theta) r \sqrt{2} r \, dr \, d\theta = \sqrt{2} \int_0^{2\pi} \cos^2 \theta \left[\frac{r^5}{5} \right]_1^4 d\theta \\ &= \frac{1023\sqrt{2}}{5} \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta = \frac{1023\sqrt{2}}{10} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{2\pi} \\ &= \frac{1023\sqrt{2}\pi}{5} \approx 909.0. \end{aligned}$$

As at the beginning of this section, suppose S is the graph of $z = f(x, y)$, where S has a regular projection R on the xy -plane, and f has continuous first partial derivatives on R . Consider a unit normal vector $\mathbf{n}$ to S , that is, a unit vector that is normal to the tangent plane to S at B_i , as illustrated in (i) of Figure 18.17. Let α , β , and γ be the angles between $\mathbf{n}$ and the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, respectively. As in the figure, we shall choose the upper normal to S , in the sense that $0 \leq \gamma \leq \pi/2$.

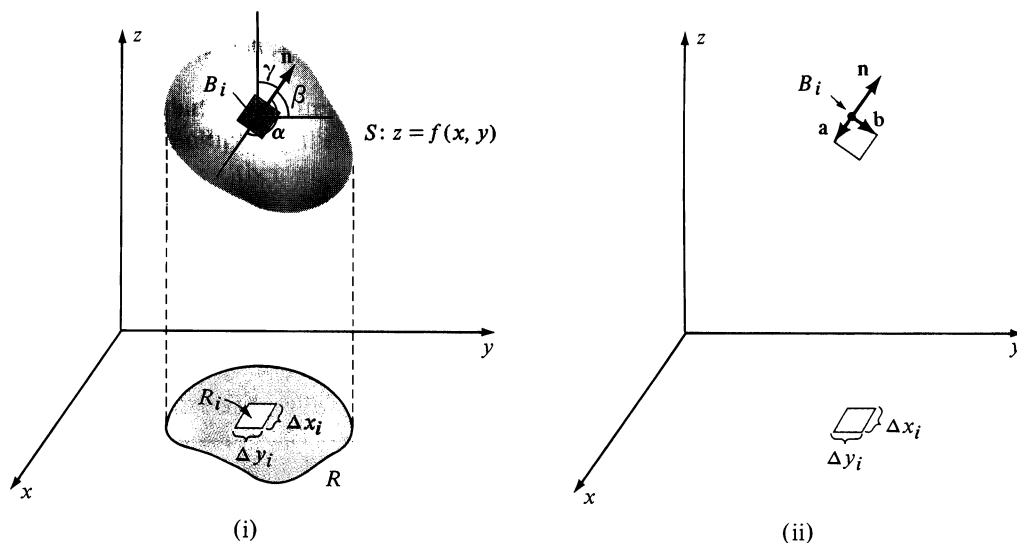


FIGURE 18.17

Since the components of $\mathbf{n}$ on $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are

$$\mathbf{n} \cdot \mathbf{i} = \cos \alpha, \quad \mathbf{n} \cdot \mathbf{j} = \cos \beta, \quad \mathbf{n} \cdot \mathbf{k} = \cos \gamma$$

we have

$$(18.22) \quad \mathbf{n} = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}.$$

If B_i , $\mathbf{a}$ and $\mathbf{b}$ are as shown in (ii) of Figure 18.17, then

$$\mathbf{n} = \frac{1}{|\mathbf{a} \times \mathbf{b}|} \mathbf{a} \times \mathbf{b}.$$

Using the formulas for $\mathbf{a} \times \mathbf{b}$ and $|\mathbf{a} \times \mathbf{b}|$ obtained on page 825, and abbreviating $f_x(x_i, y_i)$ and $f_y(x_i, y_i)$ by f_x and f_y gives us

$$(18.23) \quad \mathbf{n} = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}} (-f_x \mathbf{i} - f_y \mathbf{j} + \mathbf{k}).$$

Comparing this formula for $\mathbf{n}$ with (18.22) we see that

$$\cos \gamma = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}}$$

and hence, if γ is acute, (18.21) may be written

$$(18.24) \quad \iint_S g(x, y, z) dS = \iint_R g(x, y, f(x, y)) \sec \gamma dx dy.$$

In like manner, if an equation for S is $y = h(x, z)$ where h has continuous first partial derivatives, and S has a regular projection R_1 on the xz -plane (see Figure 18.18), then

$$(18.25) \quad \begin{aligned} \iint_S g(x, y, z) dS &= \iint_{R_1} g(x, h(x, z), z) \sec \beta dx dz \\ &= \iint_{R_1} g(x, h(x, z), z) \sqrt{(h_x(x, z))^2 + (h_z(x, z))^2 + 1} dx dz. \end{aligned}$$

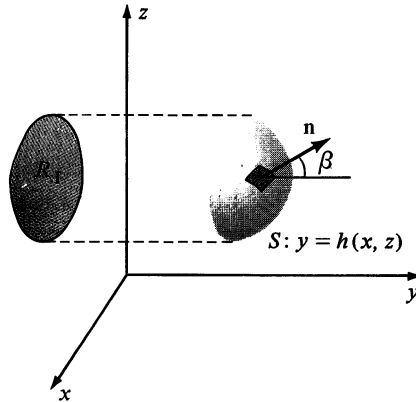


FIGURE 18.18

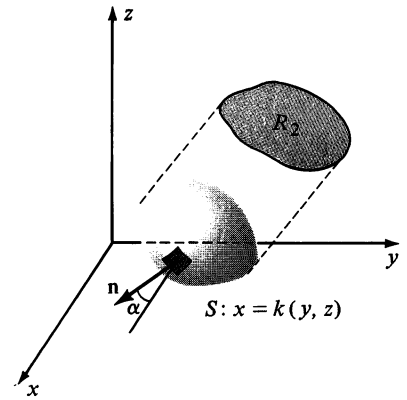


FIGURE 18.19

Similarly, if S is given by $x = k(y, z)$ where k has continuous first partial derivatives and if S has a regular projection R_2 on the yz -plane (see Figure 18.19), then

$$(18.26) \quad \iint_S g(x, y, z) \, dS = \iint_{R_2} g(k(y, z), y, z) \sec \alpha \, dy \, dz \\ = \iint_{R_2} g(k(y, z), y, z) \sqrt{(k_y(y, z))^2 + (k_z(y, z))^2 + 1} \, dy \, dz.$$

Example 2 Evaluate $\iint_S (xz/y) \, dS$, where S is the part of the cylinder $x = y^2$ that lies in the first octant between the planes $z = 0$, $z = 5$, $y = 1$, and $y = 4$.

Solution The surface S is sketched in Figure 18.20, where for clarity we have used a different unit of length on the x -axis. Since the projection of S on the yz -plane is the rectangle with vertices $(0, 1, 0)$, $(0, 4, 0)$, $(0, 4, 5)$, and $(0, 1, 5)$, we may use formula (18.26) with $k(y, z) = y^2$, as follows:

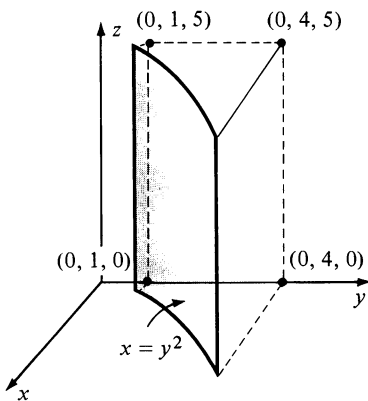


FIGURE 18.20

$$\iint_S \frac{xz}{y} \, dS = \int_1^4 \int_0^5 \frac{y^2 z}{y} \sqrt{(2y)^2 + 0^2 + 1} \, dz \, dy \\ = \int_1^4 \int_0^5 yz \sqrt{4y^2 + 1} \, dz \, dy = \int_1^4 y \sqrt{4y^2 + 1} \left[\frac{z^2}{2} \right]_0^5 \, dy \\ = \frac{25}{2} \int_1^4 y \sqrt{4y^2 + 1} \, dy = \frac{25}{24} (4y^2 + 1)^{3/2} \Big|_1^4 \\ = \frac{25}{24} [65^{3/2} - 5^{3/2}] \approx 534.2. \quad \blacksquare$$

It follows from our work in Section 17.9 (see page 825) and the previous discussion, that

$$\Delta T_i = \sec \gamma \, \Delta x_i \, \Delta y_i$$

that is, the area on the tangent plane in (i) of Figure 18.17 can be found by multiplying the area $\Delta x_i \, \Delta y_i$ of R_i by $\sec \gamma$. If we represent the area of R_i by $dx \, dy$, then we are motivated to define the **differential dS of surface area** by

$$dS = \sec \gamma \, dx \, dy.$$

In applications we may regard dS as the element of surface area on S that projects onto a rectangular region in the xy -plane of area $dx \, dy$. Similarly, for the surface integrals corresponding to Figures 18.18 and 18.19 we write

$$dS = \sec \beta \, dx \, dz \quad \text{and} \quad dS = \sec \alpha \, dy \, dz$$

respectively.

In certain cases we may consider surface integrals that involve a vector function $\mathbf{F}$, where

$$\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

and M , N , and P are continuous functions. If $\mathbf{n}$ is the unit normal (18.22), then $\mathbf{F}(x, y, z) \cdot \mathbf{n}$ is called the **normal component** of $\mathbf{F}$ at the point (x, y, z) . By definition, the **surface integral of the normal component of $\mathbf{F}$ over S** is

$$(18.27) \quad \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S (M \cos \alpha + N \cos \beta + P \cos \gamma) \, dS$$

where we have used the symbols M , N , and P for the functions to denote their values at (x, y, z) .

If S has regular projections R , R_1 , and R_2 on the xy -, xz -, and yz -planes, respectively, and if α , β , and γ are acute, then using the three forms for dS obtained previously we may write (18.27) as

$$(18.28) \quad \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{R_2} M \, dy \, dz + \iint_{R_1} N \, dx \, dz + \iint_R P \, dx \, dy$$

To attach a physical significance to (18.27), suppose that S is submerged in a fluid having a velocity field $\mathbf{F}(x, y, z)$. Let dS represent a small element of area on S . If $\mathbf{F}$ is continuous, then it is almost constant on dS and, as illustrated in Figure 18.21, the amount of fluid crossing dS in a unit of time may be approximated by the volume of a cylinder of base area dS and altitude $\mathbf{F} \cdot \mathbf{n}$.

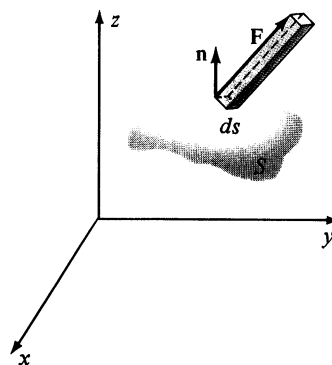


FIGURE 18.21

If dV denotes the volume of the cylinder in the figure, then $dV = \mathbf{F} \cdot \mathbf{n} \, dS$. Since dV represents the amount of fluid crossing dS per unit time, the surface integral (18.27) is the volume of fluid crossing S per unit time. In this context it is called the *flux* of $\mathbf{F}$ *through* (or *over*) S . Thus we have the following, where it is assumed that $\mathbf{F}$ and S satisfy the conditions we have imposed.

Definition (18.29)

The **flux** of a vector field $\mathbf{F}$ through (or over) a surface S is $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$.

If the fluid in the previous discussion has density $\rho = \rho(x, y, z)$, then the value of the flux integral $\iint_S \rho \mathbf{F} \cdot \mathbf{n} \, dS$ is the *mass* of the fluid crossing S .

Surface integrals may be defined over closed surfaces, such as spheres, ellipsoids, etc. In this case if we choose $\mathbf{n}$ as the *outer normal*, then the flux measures the *net outward flow* per unit time. If the integral in Definition (18.29) is positive, then the flow out of S exceeds the flow into S and we say there is a **source** of $\mathbf{F}$ within S . If the integral is negative, then the flow into S exceeds the flow out of S and we say there is a **sink** within S . If the integral is 0, then the flow into and the flow out of S are equal, that is, the sources and sinks balance one another.

Example 3 Let S be the part of the graph of $z = 9 - x^2 - y^2$ such that $z \geq 0$. If $\mathbf{F}(x, y, z) = 3x\mathbf{i} + 3y\mathbf{j} + z\mathbf{k}$, find the flux of $\mathbf{F}$ through S .

Solution The graph is sketched in Figure 18.22, together with a typical unit upper normal $\mathbf{n}$ at $P(x, y, z)$ and the vector $\mathbf{F}(x, y, z)$. According to (18.23), a unit upper normal to S at the point (x, y, z) is

$$\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}.$$

Hence by (18.29) the flux of $\mathbf{F}$ through S is

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S \frac{6x^2 + 6y^2 + z}{\sqrt{4x^2 + 4y^2 + 1}} \, dS.$$

Applying (18.21) gives us

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_R \frac{6x^2 + 6y^2 + 9 - x^2 - y^2}{\sqrt{4x^2 + 4y^2 + 1}} \sqrt{4x^2 + 4y^2 + 1} \, dx \, dy \\ &= \iint_R (5x^2 + 5y^2 + 9) \, dx \, dy \end{aligned}$$

where R is the circular region in the xy -plane bounded by the graph of $x^2 + y^2 = 9$. Changing to polar coordinates,

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^{2\pi} \int_0^3 (5r^2 + 9)r \, dr \, d\theta = \frac{567\pi}{2} \approx 890.6.$$

If, for example, $\mathbf{F}$ is the velocity field of an expanding gas, and $|\mathbf{F}|$ is measured in cm/sec, then the unit of flux would be cm³/sec. Hence the flow of gas across the surface S in Figure 18.22 is approximately 890.6 cm³/sec. ■

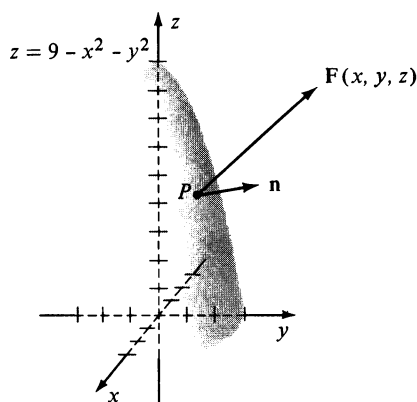


FIGURE 18.22

18.5 Exercises

In Exercises 1–4 evaluate $\iint_S f(x, y, z) \, dS$.

- 1 $f(x, y, z) = x^2$; S the upper half of the sphere $x^2 + y^2 + z^2 = a^2$.
- 2 $f(x, y, z) = x^2 + y^2 + z^2$; S the part of the plane $z = y + 4$ that is inside the cylinder $x^2 + y^2 = 4$.
- 3 $f(x, y, z) = x + y$; S the first octant portion of the plane $2x + 3y + z = 6$.
- 4 $f(x, y, z) = (x^2 + y^2 + 1)^{1/2}$; S the part of the paraboloid $2z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 2y$.

In Exercises 5–8 set up, but do not evaluate, the given surface integral by using a projection of S on (a) the yz -plane; (b) the xz -plane.

- 5 $\iint_S xy^2z^3 \, dS$; S the part of the plane $2x + 3y + 4z = 12$ that is in the first octant.
- 6 $\iint_S (xz + 2y) \, dS$; S the part of the graph of $y = x^3$ between the planes $y = 0$, $y = 8$, $z = 2$, and $z = 0$.
- 7 $\iint_S (x^2 - 2y + z) \, dS$; S the part of the graph of $4x + y = 8$ bounded by the coordinate planes and the plane $z = 6$.
- 8 $\iint_S (x^2 + y^2 + z^2) \, dS$; S the first octant part of the graph of $x^2 + y^2 = 4$ bounded by the coordinate planes and the plane $x + z = 2$.
- 9 Interpret $\iint_S f(x, y, z) \, dS$ geometrically if f is the constant function $f(x, y, z) = c$, where $c > 0$, and S has a regular projection on the xy -plane.
- 10 Show that a double integral $\iint_R f(x, y) \, dA$ of the type considered in Theorem 17.9 is a special case of a surface integral.

In Exercises 11–14 find $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, where $\mathbf{n}$ is a unit upper normal to S .

- 11 $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; S the upper half of the sphere $x^2 + y^2 + z^2 = a^2$.
- 12 $\mathbf{F} = x\mathbf{i} - y\mathbf{j}$; S the part of the sphere $x^2 + y^2 + z^2 = a^2$ lies in the first octant.
- 13 $\mathbf{F} = 2\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$; S the part of the cone $z = (x^2 + y^2)^{1/2}$ that is inside the cylinder $x^2 + y^2 = 1$.
- 14 $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; S the part of the plane $3x + 2y + z = 12$ cut out by the planes $x = 0$, $y = 0$, $x = 1$, and $y = 2$.

In Exercises 15 and 16 use (18.28) to find the flux of F through S .

- 15 $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; S the part of the plane $2x + 3y + z = 6$ that lies in the first octant.

- 16 $\mathbf{F}(x, y, z) = (x^2 + z)\mathbf{i} + y^2z\mathbf{j} + (x^2 + y^2 + z)\mathbf{k}$; S the part of the paraboloid $z = x^2 + y^2$ in the first octant that is cut off by the plane $z = 4$.

In Exercises 17 and 18 find the flux of $\mathbf{F}$ over the closed surface S . (Be sure to use the *outer* normal to S .)

- 17 $\mathbf{F}(x, y, z) = (x + y)\mathbf{i} + z\mathbf{j} + xz\mathbf{k}$; S the surface of the cube having vertices $(\pm 1, \pm 1, \pm 1)$.
- 18 $\mathbf{F}(x, y, z) = x\mathbf{i} - y\mathbf{j} + z\mathbf{k}$; S the surface of the solid bounded by the graphs of $z = x^2 + y^2$ and $z = 4$.
- 19 Prove that if S is given by $z = f(x, y)$, then (18.29) can be written

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R (-Mf_x - Nf_y + P) \, dx \, dy.$$

- 20 By Coulomb's Law (see page 832), if a point charge of q coulombs is located at the origin, then the force exerted on a unit charge at (x, y, z) is given by $\mathbf{F}(x, y, z) = (cq/|\mathbf{r}|^3)\mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. If S is any sphere with center at the origin, show that the flux of $\mathbf{F}$ through S is $4\pi cq$.
- 21 If a thin metal sheet T has the shape of a surface S and has area density $f(x, y, z)$, then the *moments* of T with respect to the coordinate planes are defined by

$$M_{xy} = \iint_S zf(x, y, z) \, dS$$

$$M_{xz} = \iint_S yf(x, y, z) \, dS$$

$$M_{yz} = \iint_S xf(x, y, z) \, dS.$$

Use limits of sums to show that these are “natural” definitions. The *center of mass* $(\bar{x}, \bar{y}, \bar{z})$ is given by

$$\bar{x}M = M_{yz}, \quad \bar{y}M = M_{xz}, \quad \bar{z}M = M_{xy}$$

where M is the mass of T . Suppose a metal funnel has the shape of the surface S described in Example 1 (see Figure 18.16). If the unit of length is centimeters, and the area density at the point (x, y, z) is z^2 gm/cm², find the mass and the center of mass of the funnel. Find the moment of inertia of the funnel with respect to the z -axis.

- 22 Refer to Exercise 21. A thin metal sheet of constant density k has the shape of the hemisphere $z = \sqrt{a^2 - x^2 - y^2}$. Find its center of mass.

18.6 The Divergence Theorem

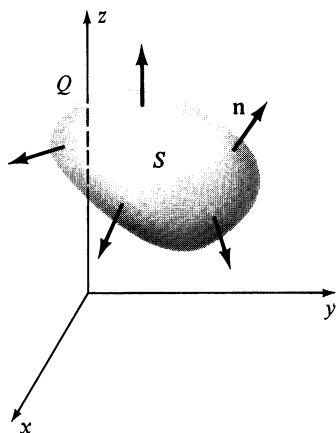


FIGURE 18.23

One of the most important theorems of vector calculus is the *Divergence Theorem*. It is also called *Gauss' Theorem*, in honor of Karl Friedrich Gauss (1777–1855), whom many consider the greatest mathematician of all time. The theorem has to do with a surface S that completely encloses a region Q in three dimensions. A surface of this type will be called a **closed surface**. For example, S could be the surface of a sphere, an ellipsoid, a cube, or a tetrahedron. The Divergence Theorem can be proved for very complicated regions; however, the proof would take us into the field of advanced calculus. Instead, *throughout this section*, we shall assume that Q is a region over which triple integrals can be evaluated using the techniques developed in Chapter 17. It will also be assumed that the resulting surface integrals can be evaluated over S . Most of the regions that are encountered in applications, such as quadric surfaces, polyhedra, closed cylinders, etc., are of this type. In this section $\mathbf{n}$ will denote a unit *outer* normal to S . Illustrations of a region Q and a surface S of the types to be considered, together with some typical positions of $\mathbf{n}$ are illustrated in Figure 18.23.

Under the restrictions we have placed on Q and S , the Divergence Theorem may be stated as follows.

The Divergence Theorem (18.30)

Let Q be a region in three dimensions bounded by a closed surface S , and let $\mathbf{n}$ denote the unit outer normal to S at (x, y, z) . If $\mathbf{F}$ is a vector function that has continuous partial derivatives on Q , then

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_Q \nabla \cdot \mathbf{F} \, dV.$$

In physical applications the formula in Theorem (18.30) states that *the flux of $\mathbf{F}$ over S equals the triple integral of the divergence of $\mathbf{F}$ over Q* . If $\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$, then using (18.27) and the definition of $\nabla \cdot \mathbf{F}$, the formula may be written

$$\iint_S (M \cos \alpha + N \cos \beta + P \cos \gamma) \, dS = \iiint_Q \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} \right) dV.$$

To prove this equality it is sufficient to show that

$$\iint_S M \cos \alpha \, dS = \iiint_Q \frac{\partial M}{\partial x} \, dV$$

$$\iint_S N \cos \beta \, dS = \iiint_Q \frac{\partial N}{\partial y} \, dV$$

$$\iint_S P \cos \gamma \, dS = \iiint_Q \frac{\partial P}{\partial z} \, dV.$$

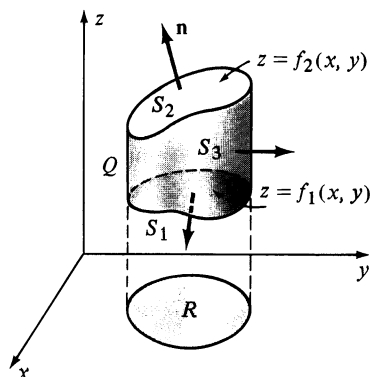


FIGURE 18.24

Since the proofs for all three formulas are basically similar, we shall prove only the third. It is quite difficult to prove the theorem for the most general case, and hence we shall specialize Q and S . Specifically, let R be a region in the xy -plane that satisfies the conditions of Green's Theorem, and suppose Q lies over R and between two surfaces having the equations $z = f_1(x, y)$ and $z = f_2(x, y)$, where the functions f_1 and f_2 have continuous first partial derivatives and $f_1(x, y) \leq f_2(x, y)$ for all (x, y) in R . As illustrated in Figure 18.24, the surface S that bounds Q consists of a bottom surface S_1 , a top surface S_2 , and a lateral surface S_3 .

On S_3 we have $\gamma = 90^\circ$ and, therefore, $\cos \gamma = 0$. Consequently,

$$\iint_S P \cos \gamma \, dS = \iint_{S_1} P \cos \gamma \, dS + \iint_{S_2} P \cos \gamma \, dS.$$

Applying (18.24),

$$\iint_{S_2} P \cos \gamma \, dS = \iint_R P(x, y, f_2(x, y)) \, dx \, dy.$$

Since the formulas for surface integrals in Section 18.5 were derived under the assumption that $\mathbf{n}$ is always an *upper* normal, and since the outer normal $\mathbf{n}$ on S_1 is a *lower* normal, it is necessary to change the sign on the right side of (18.24) when working with S_1 . Indeed, a formula for $\mathbf{n}$ on S_1 may be found by taking the negative of the vector in (18.23). Thus

$$\iint_{S_1} P \cos \gamma \, dS = - \iint_R P(x, y, f_1(x, y)) \, dx \, dy$$

and hence

$$\begin{aligned} \iint_S P \cos \gamma \, dS &= \iint_R [P(x, y, f_2(x, y)) - P(x, y, f_1(x, y))] \, dx \, dy \\ &= \iint_R \left[\int_{f_1(x, y)}^{f_2(x, y)} \frac{\partial P}{\partial z} \, dz \right] \, dx \, dy \\ &= \iiint_Q \frac{\partial P}{\partial z} \, dV \end{aligned}$$

which is what we wished to prove. It is not difficult to extend our proof to finite unions of regions of the type we have considered. The proofs of the formulas for $\iint_S M \cos \alpha \, dS$ and $\iint_S N \cos \beta \, dS$ are similar, provided Q and S are suitably restricted.

Example 1 Let Q be the region bounded by the graphs of $x^2 + y^2 = 4$, $z = 0$, and $z = 3$. Let S denote the surface of Q . If $\mathbf{F}(x, y, z) = x^3 \mathbf{i} + y^3 \mathbf{j} + z^3 \mathbf{k}$, use the Divergence Theorem to evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$.

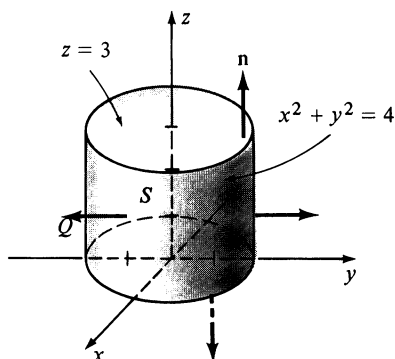


FIGURE 18.25

Solution The surface S and some typical positions of $\mathbf{n}$ are sketched in Figure 18.25. Since

$$\nabla \cdot \mathbf{F} = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$$

we have, by the Divergence Theorem,

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = 3 \iiint_Q (x^2 + y^2 + z^2) \, dV.$$

It is convenient to use cylindrical coordinates to evaluate the triple integral. Thus

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= 3 \int_0^{2\pi} \int_0^2 \int_0^3 (r^2 + z^2) r \, dz \, dr \, d\theta \\ &= 3 \int_0^{2\pi} \int_0^2 \left[r^2 z + \frac{1}{3} z^3 \right]_0^3 r \, dr \, d\theta = 3 \int_0^{2\pi} \int_0^2 (3r^2 + 9)r \, dr \, d\theta \\ &= 3 \int_0^{2\pi} \left[\frac{3}{4} r^4 + \frac{9}{2} r^2 \right]_0^2 d\theta = 3 \int_0^{2\pi} 30 \, d\theta = 90\theta \Big|_0^{2\pi} = 180\pi. \quad \blacksquare \end{aligned}$$

Example 2 Let Q be the region bounded by the cylinder $z = 4 - x^2$, the plane $y + z = 5$, and the xy - and xz -planes. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ if $\mathbf{F}(x, y, z) = (x^3 + \sin z)\mathbf{i} + (x^2y + \cos z)\mathbf{j} + e^{x^2+y^2}\mathbf{k}$.

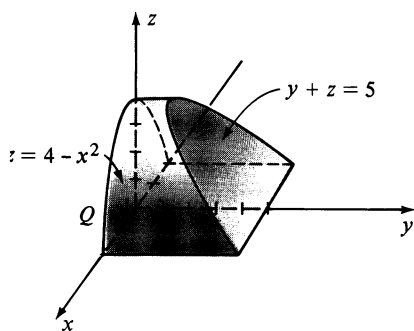


FIGURE 18.26

Solution The region Q is sketched in Figure 18.26. It would be an incredibly difficult job to evaluate the integral directly; however, using the Divergence Theorem we can obtain the value from the triple integral

$$\iiint_Q (3x^2 + x^2) \, dV.$$

Referring to Figure 18.26 and using (17.21), we see that this integral equals

$$\begin{aligned} \int_{-2}^2 \int_0^{4-x^2} \int_0^{5-z} 4x^2 \, dy \, dz \, dx &= \int_{-2}^2 \int_0^{4-x^2} 4x^2(5-z) \, dz \, dx \\ &= \int_{-2}^2 \left[20x^2z - 2x^2z^2 \right]_0^{4-x^2} dx \\ &= \int_{-2}^2 (48x^2 - 4x^4 - 2x^6) \, dx \\ &= \frac{4608}{35} \approx 131.7. \quad \blacksquare \end{aligned}$$

We may use the Divergence Theorem to obtain a physical interpretation for the divergence of a vector field. Let us begin by recalling from the Mean Value Theorem for Definite Integrals (5.20) that if a function f of one variable

is continuous on a closed interval $[a, b]$, then there is a number c in (a, b) such that

$$\int_a^b f(x) dx = f(c)L$$

where $L = b - a$ is the length of $[a, b]$. An analogous result may be proved for triple integrals. Specifically, if a function f of three variables is continuous throughout a spherical region Q , then there is a point $A(u, v, w)$ in the interior of Q such that

$$\iiint_Q f(x, y, z) dV = \left[f(x, y, z) \right]_A V$$

where V is the volume of Q and $[f(x, y, z)]_A$ denotes $f(u, v, w)$. This result is sometimes called the **Mean Value Theorem for Triple Integrals**. It follows that if $\mathbf{F}$ is a continuous vector function, then

$$\iiint_Q \nabla \cdot \mathbf{F} dV = \left[\nabla \cdot \mathbf{F} \right]_A V$$

or

$$\left[\nabla \cdot \mathbf{F} \right]_A = \frac{1}{V} \iiint_Q \nabla \cdot \mathbf{F} dV.$$

Applying the Divergence Theorem,

$$\left[\nabla \cdot \mathbf{F} \right]_A = \frac{1}{V} \iint_S \mathbf{F} \cdot \mathbf{n} dS$$

where S is the surface of Q . The ratio on the right may be interpreted as *the flux of $\mathbf{F}$ per unit volume over the sphere*.

Next let P be an arbitrary point and suppose $\mathbf{F}$ is continuous throughout a region containing P in its interior. Let S_k be the surface of a sphere of radius k with center at P . From the previous discussion, for each k there is a point P_k within S_k such that

$$\left[\nabla \cdot \mathbf{F} \right]_{P_k} = \frac{1}{V_k} \iint_{S_k} \mathbf{F} \cdot \mathbf{n} dS$$

where V_k is the volume of the sphere (see Figure 18.27).

If we let $k \rightarrow 0$, then $P_k \rightarrow P$ and we obtain

$$(18.31) \quad \left[\operatorname{div} \mathbf{F} \right]_P = \lim_{k \rightarrow 0} \frac{1}{V_k} \iint_{S_k} \mathbf{F} \cdot \mathbf{n} dS$$

that is, *the divergence of $\mathbf{F}$ at P is the limiting value of the flux per unit volume over a sphere with center P , as the radius of the sphere approaches 0*. In particular, if $\mathbf{F}$ represents the velocity of a fluid, then $\operatorname{div} \mathbf{F}|_P$ can be interpreted as the rate of loss or gain of fluid per unit volume at P . If $\rho = \rho(x, y, z)$ is the density

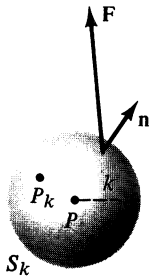


FIGURE 18.27

at (x, y, z) , then $[\operatorname{div} \rho \mathbf{F}]_P$ is the change in *mass per unit volume*. It follows from the remarks at the end of Section 18.5 that there is a source or sink at P if $\operatorname{div} \mathbf{F}]_P > 0$ or $\operatorname{div} \mathbf{F}]_P < 0$, respectively. If the fluid is incompressible and no sources or sinks are present, then there can be no gain or loss within the volume element V_k and hence at every point P ,

$$\operatorname{div} \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} = 0.$$

This formula is called the **equation of continuity** for incompressible fluids.

The limit formulation of divergence given in (18.31) may also be applied to physical concepts such as magnetic or electric flux, since these entities have many characteristics that are similar to those of fluids. In like manner, if $\mathbf{F}$ represents the flow of heat in a region, and if $\nabla \cdot \mathbf{F}]_P > 0$, then there is a source of heat at P (or heat is *leaving* P and, therefore, the temperature at P is decreasing). If $\nabla \cdot \mathbf{F}]_P < 0$, then heat is being *absorbed* at P (or the temperature at P is increasing).

Example 3 Suppose a closed surface S forms the boundary of a region Q and the origin O is an interior point of Q . If an inverse square field is $\mathbf{F} = (q/r^3)\mathbf{r}$ where q is a constant, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, and $r = |\mathbf{r}|$, prove that the flux of $\mathbf{F}$ over S is $4\pi q$ regardless of the shape of Q .

Solution Since $\mathbf{F}$ is not continuous at O the Divergence Theorem cannot be applied directly; however, we may proceed as follows. Let S_1 be a sphere of radius a and center O that lies completely within S (see Figure 18.28), and let Q_1 denote the region that lies outside of S_1 and inside of S . Since $\mathbf{F}$ is continuous throughout Q_1 we may apply the Divergence Theorem, obtaining

$$\iiint_{Q_1} \nabla \cdot \mathbf{F} \, dV = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS + \iint_{S_1} \mathbf{F} \cdot \mathbf{n} \, dS.$$

It can be shown that if $\mathbf{F}$ is an inverse square field, then $\nabla \cdot \mathbf{F} = 0$ (see Exercise 33 in Section 18.1) and, therefore,

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = - \iint_{S_1} \mathbf{F} \cdot \mathbf{n} \, dS.$$

Since the unit normal $\mathbf{n}$ on S_1 is an *outer* normal, it points *away* from the region Q_1 that has S_1 as part of its boundary. Hence, on S_1 the unit normal $\mathbf{n}$ points *toward* the origin. Thus $\mathbf{n} = (-1/r)\mathbf{r}$, where $r = a$, and

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= - \iint_{S_1} \left(\frac{q}{r^3} \right) \mathbf{r} \cdot \left(-\frac{1}{r} \mathbf{r} \right) dS \\ &= \iint_{S_1} \frac{q}{r^4} (\mathbf{r} \cdot \mathbf{r}) \, dS = \iint_{S_1} \frac{q}{r^2} \, dS \\ &= \frac{q}{a^2} \iint_{S_1} dS = \frac{q}{a^2} (4\pi a^2) = 4\pi q. \end{aligned}$$

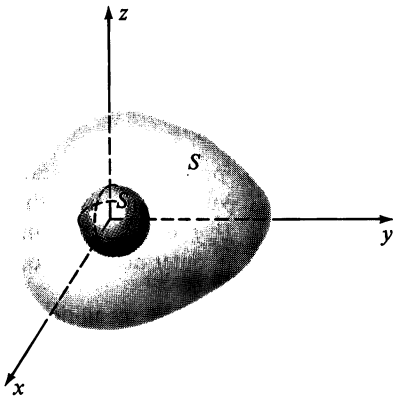


FIGURE 18.28

Example 3 has an important application in the theory of electricity. By Coulomb's Law (see page 832), if a point charge of q coulombs is located at the origin, then the force exerted on a unit charge at (x, y, z) is $\mathbf{F}(x, y, z) = (cq/|\mathbf{r}|^3)\mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and c is a constant. It follows that the electric flux of $\mathbf{F}$ over any closed surface containing the origin is $4\pi cq$. Thus, the electric flux is independent of the size or shape of S and depends only on q . It is not difficult to extend this to the case where there is more than one point charge within S . This result is called *Gauss' Law*, and has far-reaching consequences in the study of electric fields.

18.6 Exercises

In Exercises 1–4 use the Divergence Theorem to find $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$.

- $\mathbf{F} = y \sin x \mathbf{i} + y^2 z \mathbf{j} + (x + 3z)\mathbf{k}$; S the surface of the region bounded by the planes $x = \pm 1$, $y = \pm 1$, $z = \pm 1$.
- $\mathbf{F} = y^3 e^z \mathbf{i} - xy \mathbf{j} + x \arctan y \mathbf{k}$; S the surface of the region bounded by the coordinate planes and the plane $x + y + z = 1$.
- $\mathbf{F} = (x^2 + \sin yz)\mathbf{i} + (y - xe^{-z})\mathbf{j} + z^2 \mathbf{k}$; S the surface of the region bounded by the graphs of $x^2 + y^2 = 4$, $x + z = 2$, $z = 0$.
- $\mathbf{F} = 2xy \mathbf{i} + z \cosh x \mathbf{j} + (z^2 + y \sin^{-1} x)\mathbf{k}$; S the surface of the region bounded by the graphs of $z = x^2 + y^2$ and $z = 9$.

In Exercises 5–10 use the Divergence Theorem to find the flux of $\mathbf{F}$ through S .

- $\mathbf{F}(x, y, z) = yz \mathbf{i} + xz \mathbf{j} + xy \mathbf{k}$;
 S the graph of $x^{2/3} + y^{2/3} + z^{2/3} = 1$.
- $\mathbf{F}(x, y, z) = (x^2 + z^2)\mathbf{i} + (y^2 - 2xy)\mathbf{j} + (4z - 2yz)\mathbf{k}$; S the part of the cone $x = \sqrt{y^2 + z^2}$ between the planes $x = 0$ and $x = 9$.
- $\mathbf{F}(x, y, z) = 3x \mathbf{i} + xz \mathbf{j} + z^2 \mathbf{k}$; S the surface of the region bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy -plane.
- $\mathbf{F}(x, y, z) = xy^2 \mathbf{i} + yz^2 \mathbf{j} + xz^2 \mathbf{k}$; S the surface of the region that lies between the cylinders $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$, and the planes $z = -1$ and $z = 2$.
- $\mathbf{F}(x, y, z) = 2xz \mathbf{i} + xyz \mathbf{j} + yz \mathbf{k}$; S the surface of the region bounded by the coordinate planes and the graphs of $x + 2z = 4$ and $y = 2$.
- $\mathbf{F}(x, y, z) = x^3 \mathbf{i} + y^3 \mathbf{j} + z^3 \mathbf{k}$; S the surface of the region that is inside both the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 25$.

In Exercise 11–14 verify the Divergence Theorem (18.30) by evaluating both the surface integral and the triple integral.

- $\mathbf{F} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$; S the graph of $x^2 + y^2 + z^2 = a^2$.
- $\mathbf{F} = x^2 \mathbf{i} + y^2 \mathbf{j} + z^2 \mathbf{k}$; S the surface of the cube bounded by the coordinate planes and the planes $x = a$, $y = a$, $z = a$, where $a > 0$.
- $\mathbf{F} = (x + z)\mathbf{i} + (y + z)\mathbf{j} + (x + y)\mathbf{k}$; S the surface of the region
 $Q = \{(x, y, z) : 0 \leq y^2 + z^2 \leq 1, 0 \leq x \leq 2\}$.
- $\mathbf{F} = |\mathbf{r}|^2 \mathbf{r}$, where $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$; S the graph of $x^2 + y^2 + z^2 = a^2$.

- If $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = 0$ for every closed surface of the type considered in the Divergence Theorem, prove that $\operatorname{div} \mathbf{F} = 0$.
- Use the Divergence Theorem to prove that if a scalar function f has continuous second partial derivatives, then $\iiint_Q \nabla^2 f \, dV = \iint_S D_{\mathbf{n}} f \, dS$ where $D_{\mathbf{n}} f$ is the directional derivative of f in the direction of an outer normal $\mathbf{n}$ to S .

In Exercises 17–22 assume that S and Q satisfy the conditions of the Divergence Theorem.

- If f and g are scalar functions which have continuous second partial derivatives, prove that

$$\iiint_Q (f \nabla^2 g + \nabla f \cdot \nabla g) \, dV = \iint_S (f \nabla g) \cdot \mathbf{n} \, dS.$$

(Hint: Let $\mathbf{F} = f \nabla g$ in (18.30).)

- If f and g are scalar functions which have continuous second partial derivatives, prove that

$$\iiint_Q (f \nabla^2 g - g \nabla^2 f) \, dV = \iint_S (f \nabla g - g \nabla f) \cdot \mathbf{n} \, dS.$$

(Hint: Use the identity in Exercise 17 together with that obtained by interchanging f and g .)

- 19 If $\mathbf{F}$ is a conservative vector field with potential function f , and if $\operatorname{div} \mathbf{F} = 0$, prove that $\iiint_Q \mathbf{F} \cdot \mathbf{F} \, dV = \iint_S f \mathbf{F} \cdot \mathbf{n} \, dS$.
- 20 If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and V is the volume of Q , prove that $V = \frac{1}{3} \iint_S \mathbf{r} \cdot \mathbf{n} \, dS$.
- 21 If $\mathbf{F}$ has continuous second partial derivatives prove that $\iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n} \, dS = 0$.
- 22 If $\mathbf{a}$ is a constant vector prove that $\iint_S \mathbf{a} \cdot \mathbf{n} \, dS = 0$.

A surface (or triple) integral of a vector function is defined as the sum of the surface (or triple) integrals of each component of the function. With this understanding, establish the results in

Exercises 23 and 24, where S and Q satisfy the conditions of the Divergence Theorem.

- 23 $\iint_S \mathbf{F} \times \mathbf{n} \, dS = - \iiint_Q \nabla \times \mathbf{F} \, dV$ (Hint: Apply the Divergence Theorem and Exercise 26 of Section 18.1 to $\mathbf{c} \times \mathbf{F}$, where $\mathbf{c}$ is an arbitrary constant vector.)
- 24 $\iint_S f \mathbf{n} \, dS = \iiint_Q \nabla f \, dV$ (Hint: Apply the Divergence Theorem to $f\mathbf{c}$ where $\mathbf{c}$ is an arbitrary constant vector.)
- 25 If $\mathbf{F}$ is orthogonal to S at each point (x, y, z) , prove that $\iiint_Q \operatorname{curl} \mathbf{F} \, dV = 0$.
- 26 If $\mathbf{r}$ is the position vector of (x, y, z) and $r = |\mathbf{r}|$, prove that $\iiint_Q \mathbf{r} \, dV = \frac{1}{2} \iint_S r^2 \mathbf{n} \, dS$.

18.7 Stokes' Theorem

In (18.19) we stated the following vector form for Green's Theorem:

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_R \operatorname{curl} \mathbf{F} \cdot \mathbf{k} \, dA$$

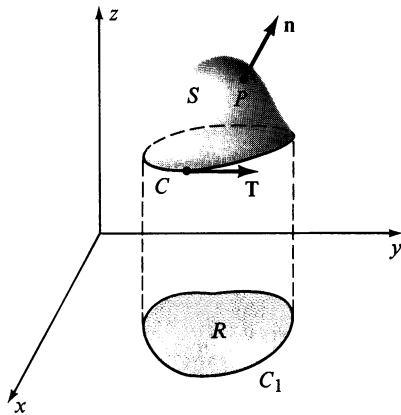


FIGURE 18.29

where the plane curve C is the boundary of R . This result may be extended to a piecewise smooth simple closed curve C in three dimensions that forms the boundary of a surface S . The sketch in Figure 18.29 illustrates a special case in which S is the graph of $z = f(x, y)$, where f has continuous first partial derivatives and the projection C_1 of C on the xy -plane is a curve that bounds a region R of the type considered in Green's Theorem. In the figure, $\mathbf{n}$ represents a unit *upper* normal to S . We take the positive direction along C as that which corresponds to the positive direction along C_1 . The vector $\mathbf{T}$ is a unit tangent vector to C . If $\mathbf{F}$ is a vector function that has continuous partial derivatives in a region containing S , then we have the following theorem, named after the English mathematical physicist George G. Stokes (1819–1903).

Stokes' Theorem (18.32)

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n} \, dS$$

The formula in Theorem (18.32) may be stated as follows: *the line integral of the tangential component of $\mathbf{F}$ taken along C once in the positive direction equals the surface integral of the normal component of $\operatorname{curl} \mathbf{F}$ over S .* Of course, the line integral may also be written $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{r}$ is the position vector of the point (x, y, z) on C . To consider more general situations than the one pictured in Figure 18.29 it is necessary to consider what is called an *oriented surface* S ; that is, a surface on which a unit normal $\mathbf{n}$ and positive direction

along C may be defined in a suitable manner. The interested reader should consult more advanced texts for a proof of Stokes' Theorem.

Example 1 Let S be the part of the paraboloid $z = 9 - x^2 - y^2$ where $z \geq 0$, and let C be the trace of S on the xy -plane. Verify Stokes' Theorem for the vector field $\mathbf{F} = 3z\mathbf{i} + 4x\mathbf{j} + 2y\mathbf{k}$.

Solution We wish to show that the two integrals in Theorem (18.32) have the same value.

The surface is the same as that considered in Example 3 of Section 18.5 (see Figure 18.22) where we found that

$$\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}.$$

By (18.6)

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3z & 4x & 2y \end{vmatrix} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}.$$

Consequently,

$$\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S \frac{4x + 6y + 4}{\sqrt{4x^2 + 4y^2 + 1}} \, dS.$$

Using (18.21) to evaluate this surface integral gives us

$$\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R (4x + 6y + 4) \, dx \, dy$$

where R is the region in the xy -plane bounded by the circle of radius 3 with center at the origin. Changing to polar coordinates, we obtain

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS &= \int_0^{2\pi} \int_0^3 (4r \cos \theta + 6r \sin \theta + 4)r \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^3 [r^2(4 \cos \theta + 6 \sin \theta) + 4r] \, dr \, d\theta \\ &= \int_0^{2\pi} (36 \cos \theta + 54 \sin \theta + 18) \, d\theta \\ &= \left[36 \sin \theta - 54 \cos \theta + 18\theta \right]_0^{2\pi} = 36\pi. \end{aligned}$$

The line integral in Theorem (18.32) may be written

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C 3z \, dx + 4x \, dy + 2y \, dz$$

where C is the circle in the xy -plane having equation $x^2 + y^2 = 9$. Since $z = 0$ on C , this reduces to

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C 4x \, dy.$$

Using the parametric equations $x = 3 \cos t$, $y = 3 \sin t$ for C , we obtain

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} 4(3 \cos t) 3 \cos t \, dt = \int_0^{2\pi} 36 \cos^2 t \, dt \\ &= 18 \int_0^{2\pi} (1 + \cos 2t) \, dt = 36\pi \end{aligned}$$

which is the same as the value of the surface integral. ■

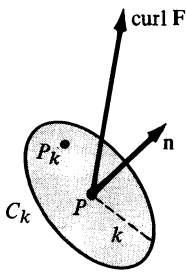


FIGURE 18.30

We may use Stokes' Theorem to obtain a physical interpretation for $\text{curl } \mathbf{F}$ in a manner similar to that in which the Divergence Theorem was used for $\text{div } \mathbf{F}$ in (18.31). If P is any point, let S_k be a circular disc of radius k with center at P and let C_k denote the circumference of S_k (see Figure 18.30). Applying Stokes' Theorem and a mean value theorem for double integrals leads to

$$\oint_{C_k} \mathbf{F} \cdot \mathbf{T} \, ds = \iint_{S_k} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = [\text{curl } \mathbf{F} \cdot \mathbf{n}]_{P_k} (\pi k^2)$$

where P_k is some point in S_k and πk^2 is the area of S_k . Thus

$$[\text{curl } \mathbf{F} \cdot \mathbf{n}]_{P_k} = \frac{1}{\pi k^2} \oint_{C_k} \mathbf{F} \cdot \mathbf{T} \, ds.$$

If we let $k \rightarrow 0$, then $P_k \rightarrow P$ and hence

$$(18.33) \quad [\text{curl } \mathbf{F} \cdot \mathbf{n}]_P = \lim_{k \rightarrow 0} \frac{1}{\pi k^2} \oint_{C_k} \mathbf{F} \cdot \mathbf{T} \, ds.$$

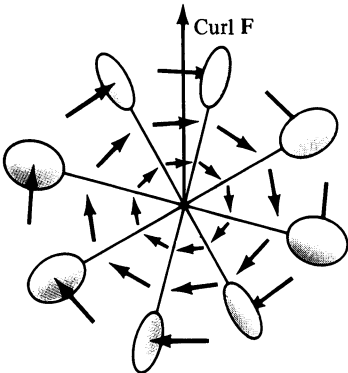


FIGURE 18.31

If $\mathbf{F}$ is the velocity field of a fluid, then the line integral $\oint_C \mathbf{F} \cdot \mathbf{T} \, ds$ is called the **circulation around C** . It measures the average tendency of the fluid to move, or *circulate*, around the curve. We see from (18.33) that $[\text{curl } \mathbf{F} \cdot \mathbf{n}]_P$ provides information about the motion of the fluid around the circumference of a circular disc that is perpendicular to $\mathbf{n}$, as the disc shrinks to a point. Since $[\text{curl } \mathbf{F} \cdot \mathbf{n}]_P$ will have its maximum value when $\mathbf{n}$ is parallel to $\text{curl } \mathbf{F}$, it follows that the direction of $\text{curl } \mathbf{F}$ at P is that for which the circulation around the boundary of a disc perpendicular to $\text{curl } \mathbf{F}$ will have its maximum value as the disc shrinks to a point. If we replace the disc S_k in Figure 18.30 by a miniature paddlewheel of the type illustrated in Figure 18.31, then the wheel will rotate most rapidly when $\mathbf{n}$ is parallel to $\text{curl } \mathbf{F}$. Occasionally $\text{curl } \mathbf{F} \cdot \mathbf{n}$ is called the **rotation of $\mathbf{F}$ about $\mathbf{n}$** and is denoted by $\text{rot } \mathbf{F}$, since it measures the tendency of the vector field to *rotate* about P .

Example 2 Let $\mathbf{F}(x, y, z) = 3(y^2 + 1)^{-1}\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$, where $|y| \leq 4$. Describe the vector field $\mathbf{F}$ and discuss the circulation of $\mathbf{F}$ around several circles C_k that lie in the xy -plane. Where does $\text{curl } \mathbf{F} \cdot \mathbf{n}$ achieve its maximum value?

Solution If we restrict our attention to the xy -plane, the vectors in $\mathbf{F}$ form the pattern shown in Figure 18.32. (Verify this fact.) The same pattern is repeated in any plane parallel to the xy -plane.

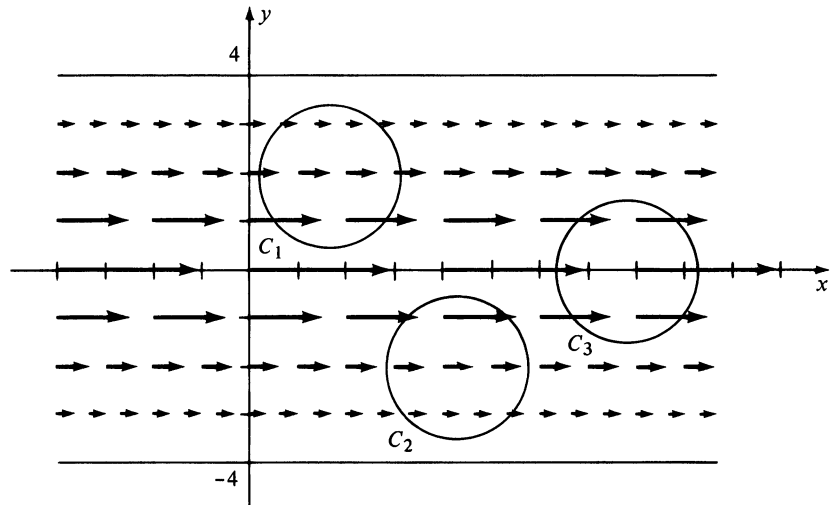


FIGURE 18.32

Let us regard $\mathbf{F}$ as the velocity field of a fluid that is flowing in a culvert. Note that the speed of the fluid is greatest along the x -axis and decreases as the point (x, y) approaches either of the lines $y = \pm 4$. If we consider the circle C_1 , then evidently $\oint_{C_1} \mathbf{F} \cdot \mathbf{T} \, ds > 0$, since as C_1 is traversed in the positive direction, the tangential component $\mathbf{F} \cdot \mathbf{T}$ is large and positive on the lower half of C_1 and negative (but close to 0) on the upper half. If C_1 represents a small paddle wheel, it would rotate in the counterclockwise direction. By again considering values of $\mathbf{F} \cdot \mathbf{T}$ it is evident that $\oint_{C_2} \mathbf{F} \cdot \mathbf{T} \, ds < 0$ and $\oint_{C_3} \mathbf{F} \cdot \mathbf{T} \, ds = 0$. The paddlewheel corresponding to C_2 would rotate in a clockwise direction, and that for C_3 would not rotate at all. It is easy to show that

$$\text{curl } \mathbf{F} = \frac{6y}{(y^2 + 1)^2} \mathbf{k}.$$

Since the unit normal $\mathbf{n}$ is $\mathbf{k}$ we have

$$\text{curl } \mathbf{F} \cdot \mathbf{n} = \text{curl } \mathbf{F} \cdot \mathbf{k} = \frac{6y}{(y^2 + 1)^2}.$$

Observe that $\text{curl } \mathbf{F} \cdot \mathbf{n}$ is positive for points above the x -axis, negative for points below the x -axis, and 0 for points on the x -axis. Differentiating,

$$D_y(\text{curl } \mathbf{F} \cdot \mathbf{n}) = \frac{6(1 - 3y^2)}{(y^2 + 1)^3}$$

and hence critical numbers for $\text{curl } \mathbf{F} \cdot \mathbf{n}$ are $y = \pm 1/\sqrt{3} \approx \pm 0.577$. It follows from the first derivative test that these lead to maximum values. Thus the maximum values for $\text{curl } \mathbf{F} \cdot \mathbf{n}$ occur at any point on either of the horizontal lines $y = \pm 1/\sqrt{3}$. ■

To bring out another connection between $\text{curl } \mathbf{F}$ and rotational aspects of motion, consider the special case where a fluid is rotating uniformly about an axis, as if it were a rigid body. If we concentrate on a single fluid particle at the point $P(x, y, z)$, where x , y , and z are functions of time t , then we have a situation similar to that illustrated in Figure 15.12, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector of P and $\boldsymbol{\omega} = \omega_1\mathbf{i} + \omega_2\mathbf{j} + \omega_3\mathbf{k}$ is the (constant) angular velocity. If we denote the velocity $d\mathbf{r}/dt$ of P by $\mathbf{F}$, then as in Section 15.3,

$$\mathbf{F} = \boldsymbol{\omega} \times \mathbf{r} = (\omega_2 z - \omega_3 y)\mathbf{i} + (\omega_3 x - \omega_1 z)\mathbf{j} + (\omega_1 y - \omega_2 x)\mathbf{k}.$$

By (18.6),

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \omega_2 z - \omega_3 y & \omega_3 x - \omega_1 z & \omega_1 y - \omega_2 x \end{vmatrix}.$$

Using the fact that $\boldsymbol{\omega}$ is a constant vector it may be verified that

$$\text{curl } \mathbf{F} = 2\omega_1\mathbf{i} + 2\omega_2\mathbf{j} + 2\omega_3\mathbf{k} = 2\boldsymbol{\omega}.$$

This shows that the magnitude of $\text{curl } \mathbf{F}$ is twice the angular velocity, and the direction of $\text{curl } \mathbf{F}$ is along the axis of rotation.

In our discussion of line integrals that are independent of path we introduced the notion of a simply connected region in the plane (see page 849). This concept can be extended to three dimensions; however, the usual definition requires properties of curves and surfaces studied in more advanced courses. For our purposes a region D in three dimensions will be called **simply connected** if every simple closed curve C in D is the boundary of a surface S that satisfies the conditions of Stokes' Theorem. In regions of this type any simple closed curve can be continuously shrunk to a point in D without crossing the boundary of D . For example, the region inside of a sphere or a rectangular parallelepiped is simply connected. The region inside of a torus (a doughnut-shaped surface) is not simply connected. Using this rather restrictive definition we can establish the following result.

Theorem (18.34)

If $\mathbf{F}(x, y, z)$ has continuous partial derivatives throughout a simply connected region D , then $\text{curl } \mathbf{F} = \mathbf{0}$ in D if and only if $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every simple closed curve C in D .

Proof If $\text{curl } \mathbf{F} = \mathbf{0}$, then by Stokes' Theorem,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = 0.$$

Conversely, suppose $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every simple closed curve C . If at some point P , $\text{curl } \mathbf{F} \neq \mathbf{0}$, then by continuity there is a subregion of D containing P throughout which $\text{curl } \mathbf{F} \neq \mathbf{0}$. If, in this subregion, we choose a circular disc of the type illustrated in Figure 18.30, where $\mathbf{n}$ is parallel to $\text{curl } \mathbf{F}$, then

$$\oint_{C_k} \mathbf{F} \cdot d\mathbf{r} = \iint_{S_k} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS > 0,$$

a contradiction. Consequently, $\text{curl } \mathbf{F} = \mathbf{0}$ throughout D . $\square$

If $\text{curl } \mathbf{F} = \mathbf{0}$ in D , then the circulation around every closed curve C is 0. (Why?) This also means that the *rotation* of $\mathbf{F}$ about any unit vector $\mathbf{n}$ is 0. For this reason a vector field $\mathbf{F}$ such that $\text{curl } \mathbf{F} = \mathbf{0}$ is often called **irrotational**.

It can be shown that if $\mathbf{F}$ is continuous on a suitable region, then the vanishing of the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ around every simple closed curve is equivalent to independence of path. Moreover, as in Theorem (18.14), independence of path is equivalent to $\mathbf{F} = \nabla f$ for some function f (that is, $\mathbf{F}$ is conservative). Combining these facts with Theorem (18.34), we obtain the following.

Theorem (18.35)

If $\mathbf{F}(x, y, z)$ has continuous first partial derivatives throughout a simply connected region, then the following conditions are equivalent.

- (i) $\mathbf{F}$ is conservative.
- (ii) $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path.
- (iii) $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every simple closed curve C .
- (iv) $\text{curl } \mathbf{F} = \mathbf{0}$.

Example 3 Prove that if $\mathbf{F}(x, y, z) = (3x^2 + y^2)\mathbf{i} + 2xy\mathbf{j} - 3z^2\mathbf{k}$, then $\mathbf{F}$ is conservative.

Solution The function $\mathbf{F}$ has continuous partial derivatives for all (x, y, z) . Since

$$\begin{aligned} \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 + y^2 & 2xy & -3z^2 \end{vmatrix} \\ &= (0 - 0)\mathbf{i} + (0 - 0)\mathbf{j} + (2y - 2y)\mathbf{k} = \mathbf{0}, \end{aligned}$$

$\mathbf{F}$ is conservative by (iv) of Theorem (18.35). $\blacksquare$

In advanced mathematics and applications it is useful to study pairs of vector fields $\mathbf{F}$ and $\mathbf{G}$ that have the form

$$\mathbf{F}(x, y) = v(x, y)\mathbf{i} + u(x, y)\mathbf{j}$$

$$\mathbf{G}(x, y) = u(x, y)\mathbf{i} - v(x, y)\mathbf{j}$$

where the functions u and v have continuous partial derivatives. Note that $\mathbf{F}(x, y)$ and $\mathbf{G}(x, y)$ are orthogonal at every point since

$$\mathbf{F} \cdot \mathbf{G} = vu - uv = 0.$$

Moreover, under suitable restrictions (see Theorem 18.16) it can be shown that $\mathbf{F}$ and $\mathbf{G}$ are both conservative if and only if

$$(18.36) \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

These conditions on u and v are called the **Cauchy–Riemann equations** and are exploited extensively in the study of complex variables. We see from (18.36) that

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x} = -\frac{\partial^2 u}{\partial y^2}$$

and hence
$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Similarly, we can show that

$$\nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

Consequently, u and v satisfy Laplace's equation and hence are harmonic. (See Exercise 36 of Section 18.1).

18.7 Exercises

In Exercises 1–4 verify Stokes' Theorem for the given $\mathbf{F}$ and S .

- 1 $\mathbf{F} = y^2\mathbf{i} + z^2\mathbf{j} + x^2\mathbf{k}$; S the first octant portion of the plane $x + y + z = 1$.
- 2 $\mathbf{F} = 2y\mathbf{i} - z\mathbf{j} + 3\mathbf{k}$; S the part of the paraboloid $z = 4 - x^2 - y^2$ that lies inside the cylinder $x^2 + y^2 = 1$.
- 3 $\mathbf{F} = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$; S the hemisphere $z = (a^2 - x^2 - y^2)^{1/2}$.
- 4 $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$; S the part of the cone $z = (x^2 + y^2)^{1/2}$ cut off by the plane $z = 1$.

- 5 If $\mathbf{F} = (3z - \sin x)\mathbf{i} + (x^2 + e^y)\mathbf{j} + (y^3 - \cos z)\mathbf{k}$, use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is the curve $x = \cos t$, $y = \sin t$, $z = 1$; $0 \leq t \leq 2\pi$.
- 6 If $\mathbf{F} = yz\mathbf{i} + xy\mathbf{j} + xz\mathbf{k}$, use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where C is the square with vertices $(0, 0, 2)$, $(1, 0, 2)$, $(1, 1, 2)$, $(0, 1, 2)$.
- 7 If $\mathbf{F} = 2y\mathbf{i} + e^z\mathbf{j} - \arctan x\mathbf{k}$, use Stokes' Theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS$, where S is the part of the paraboloid $z = 4 - x^2 - y^2$ cut off by the xy -plane.

- 8 If $\mathbf{F} = xy^2\mathbf{i} + yz^2\mathbf{j} + zx^2\mathbf{k}$, use Stokes' Theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS$, where S is the triangle with vertices $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$.

Establish the identities in Exercises 9–11 under the assumption that C and S satisfy the conditions of Stokes' Theorem.

- 9 $\oint_C f \nabla g \cdot d\mathbf{r} = \iint_S (\nabla f \times \nabla g) \cdot \mathbf{n} \, dS$, where f and g are scalar functions.
- 10 $\oint_C \mathbf{a} \times \mathbf{r} \cdot d\mathbf{r} = 2\mathbf{a} \cdot \iint_S \mathbf{n} \, dS$, where $\mathbf{a}$ is a constant vector.
- 11 $\oint_C f \, d\mathbf{r} = \iint_S \mathbf{n} \times \nabla f \, dS$, where f is a scalar function. (Hint: Compare with Exercise 18 of Section 15.7.)
- 12 If $\mathbf{F}(x, y, z) = y\mathbf{i} + (x + e^z)\mathbf{j} + (1 + ye^z)\mathbf{k}$, use (ii) of Theorem (18.35) to prove that $\mathbf{F}$ is conservative.
- 13 If u and v have continuous partial derivatives and satisfy the Cauchy–Riemann equations (18.36), prove that v satisfies Laplace's equation $\nabla^2 v = 0$.
- 14 Let $\mathbf{n}$ denote the unit outer normal at any point P on the surface of a sphere S . If $\mathbf{F}$ has continuous first partial derivatives within and on S , prove that $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = 0$ by using (a) the Divergence Theorem; (b) Stokes' Theorem.
- 15 If M , N , and P are functions of x , y , and z that have continuous first partial derivatives on a simply connected region, prove that

$$\int_C M(x, y, z) \, dx + N(x, y, z) \, dy + P(x, y, z) \, dz$$

is independent of path if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}.$$

- 16 If S and C satisfy the conditions of Stoke's Theorem, and if $\mathbf{F}$ is a constant vector function, use (18.32) to prove that $\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = 0$.

18.8 Transformations of Coordinates

In previous sections of this text we studied many different types of scalar and vector functions. Let us now consider a function T whose domain and range are subsets of $\mathbb{R} \times \mathbb{R}$. Thus, to each ordered pair (x, y) in the domain there corresponds a unique ordered pair (u, v) in the range such that $T(x, y) = (u, v)$. We may represent T geometrically as illustrated in Figure 18.33, where the point (u, v) in the uv -plane corresponds to the point (x, y) in the xy -plane. As usual, (u, v) is referred to as the **image** of (x, y) under T .

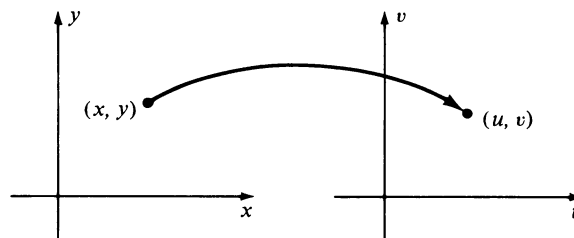
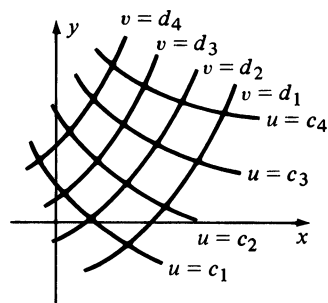


FIGURE 18.33 $T(x, y) = (u, v)$

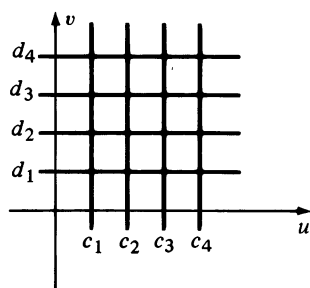
Since each pair (u, v) is determined by (x, y) , it follows that u and v are functions of x and y ; that is,

$$(18.37) \quad u = f(x, y), \quad v = g(x, y)$$

where f and g have the same domain as T . We shall refer to equations (18.37), or to the function T , as a **transformation of coordinates** from the xy -plane to the uv -plane.



(i)



(ii)

FIGURE 18.34

Given the transformation of coordinates (18.37), let us partition a region in the uv -plane by means of vertical lines $u = c_1, u = c_2, u = c_3, \dots$, and horizontal lines $v = d_1, v = d_2, v = d_3, \dots$. The corresponding level curves for f and g , that is

$$u = f(x, y) = c_i, \quad v = g(x, y) = d_j$$

where $i = 1, 2, 3, \dots$ and $j = 1, 2, 3, \dots$ determine a corresponding “curvilinear” partition of a region in the xy -plane. Figure 18.34 illustrates four curves of each type associated with a certain transformation T .

We shall refer to the curves $u = f(x, y) = c_i$ and $v = g(x, y) = d_j$ in the xy -plane as **u -curves** and **v -curves**, respectively. Of course, the types of curves obtained depend on the nature of the functions f and g . The next example illustrates a case where each u -curve and v -curve is a straight line.

Example 1 Let T be the transformation of coordinates defined by

$$u = x + 2y, \quad v = x - 2y.$$

- (a) Find the images of $(0, 1)$, $(1, 2)$, and $(2, -3)$.
 (b) Sketch, in the uv -plane, the vertical lines $u = 2, u = 4, u = 6, u = 8$, and the horizontal lines $v = -1, v = 1, v = 3, v = 5$. Sketch the corresponding u -curves and v -curves in the xy -plane.

Solution

- (a) To find the images of the given ordered pairs we may use the formula

$$T(x, y) = (u, v) = (x + 2y, x - 2y).$$

This gives us

$$T(0, 1) = (2, -2), \quad T(1, 2) = (5, -3), \quad T(2, -3) = (-4, 8).$$

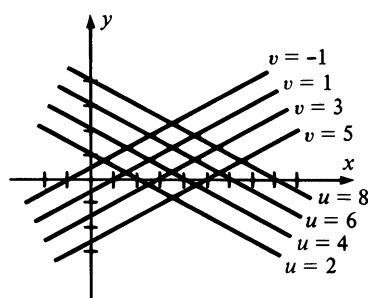
- (b) The desired vertical and horizontal lines in the uv -plane are sketched in (ii) of Figure 18.35. The u -curves in the xy -plane are the lines

$$x + 2y = 2, \quad x + 2y = 4, \quad x + 2y = 6, \quad x + 2y = 8$$

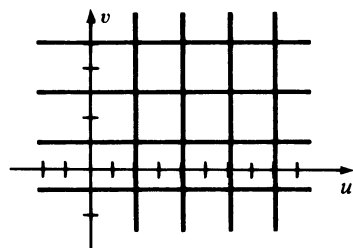
and the v -curves are the lines

$$x - 2y = -1, \quad x - 2y = 1, \quad x - 2y = 3, \quad x - 2y = 5$$

as illustrated in (i) of the figure. ■



(i)



(ii)

FIGURE 18.35

The transformation T in Example 1 is one-to-one, in the sense that different ordered pairs in the xy -plane have different images in the uv -plane (see Exercise 13). In general, if T is a one-to-one transformation of coordinates, then by reversing the correspondence we obtain a transformation T^{-1} from the

uv -plane to the xy -plane called the **inverse** of T . We may specify T^{-1} by means of equations of the form

$$x = \varphi(u, v), \quad y = \psi(u, v).$$

Note that T , followed by T^{-1} , or vice versa, is an identity function.

Example 2

- (a) Find the inverse of the transformation T defined in Example 1.
 (b) Find the curve in the uv -plane that maps into the ellipse $x^2 + 4y^2 = 1$ under T^{-1} .

Solution

- (a) The transformation T is given by

$$u = x + 2y, \quad v = x - 2y.$$

If we add corresponding sides of these equations we obtain $u + v = 2x$. Subtracting corresponding sides leads to $u - v = 4y$. Thus T^{-1} is given by

$$x = \frac{1}{2}(u + v), \quad y = \frac{1}{4}(u - v).$$

- (b) Since x and y are related by means of the last two equations in part (a), the points (u, v) that map into $x^2 + 4y^2 = 1$ must satisfy the equation

$$\left[\frac{1}{2}(u + v)\right]^2 + 4\left[\frac{1}{4}(u - v)\right]^2 = 1.$$

This simplifies to

$$u^2 + v^2 = 2.$$

It follows that the circle of radius $\sqrt{2}$ with center at the origin in the uv -plane maps into the ellipse $x^2 + 4y^2 = 1$ under T^{-1} . ■

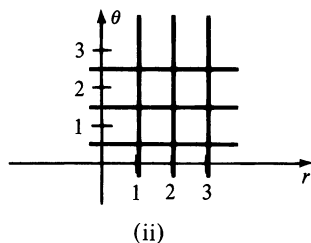
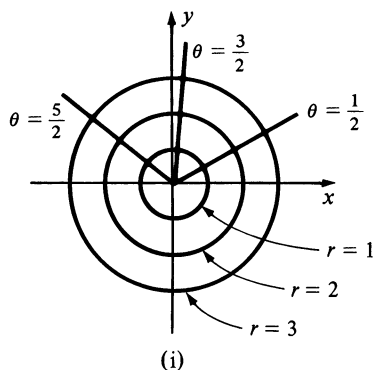


FIGURE 18.36

Example 3 If $P(x, y)$ is any point other than $(0, 0)$ in the xy -plane, let r and θ be defined as follows:

$$r = \sqrt{x^2 + y^2}$$

θ = the smallest nonnegative angle from the positive x -axis to the position vector $\overline{OP}$.

Thus, r and θ are polar coordinates for P , and the preceding equalities define a transformation of coordinates T from the xy -plane to the $r\theta$ -plane.

- (a) Sketch, in the $r\theta$ -plane, the graphs of $r = 1$, $r = 2$, $r = 3$ and $\theta = \frac{1}{2}$, $\theta = \frac{3}{2}$, $\theta = \frac{5}{2}$. Sketch the corresponding r -curves and θ -curves in the xy -plane.
 (b) Find T^{-1} .

Solution

- (a) Note that we must exclude points (r, θ) in the $r\theta$ -plane such that $r \leq 0$, $\theta < 0$, or $\theta \geq 2\pi$. (Why?) The desired graphs are sketched in Figure 18.36.

- (b) It is not difficult to show that T is one-to-one (see Exercise 14). The familiar polar coordinate formulas

$$x = r \cos \theta, \quad y = r \sin \theta$$

define the inverse transformation T^{-1} . ■

The discussion in this section can be extended to three dimensions. Thus, a transformation of coordinates T from an xyz -coordinate system to a uvw -coordinate system is determined by three equations of the form

$$u = f(x, y, z), \quad v = g(x, y, z), \quad w = h(x, y, z)$$

where the functions f , g , and h are defined on some region R . In this case the planes $u = c_i$, $v = d_j$, $z = e_k$, where c_i , d_j , e_k are real numbers, determine **u -surfaces**, **v -surfaces**, and **w -surfaces** in the xyz -coordinate system (see Exercises 15 and 16).

18.8 Exercises

In Exercises 1–10 let T be the transformation from the xy -plane to the uv -plane defined by the given equations.

(a) Describe the u -curves and the v -curves.

(b) If T is one-to-one, find equations $x = \varphi(u, v)$, $y = \psi(u, v)$ that specify T^{-1} .

1 $u = 3x, v = 5y$

2 $u = e^x, v = e^y$

3 $u = x - y, v = 3y + 2x$

4 $u = 5x - 4y, v = 2x + 3y$

5 $u = 2x + y, v = xy$

6 $u = x^3, v = x + y$

7 $u = x^2 + 4y^2, v = 4x^2 - y^2$

8 $u = x + y, v = x^2 - y$

9 $u = x^2 - y^2, v = 2xy$

10 $u = y/(x^2 + y^2), v = x/(x^2 + y^2), x^2 + y^2 \neq 0$

11 If T is the transformation defined in Exercise 1, find the image of the rectangle with vertices $(0, 0)$, $(0, 1)$, $(2, 1)$, and $(2, 0)$. What is the image of the unit circle $x^2 + y^2 = 1$?

12 If T is the transformation defined in Exercise 3, find the image of the triangle with vertices $(0, 0)$, $(0, 1)$, $(2, 0)$. What is the image of the line $x + 2y = 1$?

13 Prove that the transformation T of Example 1 is one-to-one.

14 Prove that the transformation T of Example 3 is one-to-one.

In Exercises 15 and 16 let T be the indicated transformation from the xyz -coordinate system to the uvw -coordinate system. Find the u -surfaces, the v -surfaces, and the w -surfaces. If T is one-to-one, determine T^{-1} .

15 $u = 2x + y, v = y + z, w = x + y + z$

16 $u = x^2 + y^2 + z^2, v = x^2 + y^2 - z^2, w = x^2 - y^2 - z^2$

18.9 Change of Variables in Multiple Integrals

In Section 5.6 we developed a technique for making a change of variables in a definite integral $\int_a^b f(x) dx$. Under suitable conditions, if we let $x = g(u)$, then $dx = g'(u) du$ and

$$\int_a^b f(x) dx = \int_c^d f(g(u))g'(u) du$$

where $a = g(c)$ and $b = g(d)$. In this section we shall obtain an analogous formula for a change of variables in a double integral $\iint_R F(x, y) dA$. In particular, suppose we let

$$x = f(u, v), \quad y = g(u, v)$$

where f and g have continuous second partial derivatives. These equations define a transformation of coordinates W from the uv -plane to the xy -plane. If

we use these equations to substitute for x and y in the double integral, then the integrand becomes a function of u and v . Our objective is to find a region S in the uv -plane that maps onto R under W , as illustrated in Figure 18.37, and such that

$$\iint_R F(x, y) dA = \iint_S F(f(u, v), g(u, v)) dA.$$

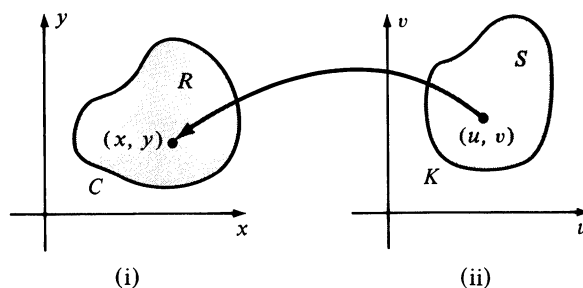


FIGURE 18.37

To change variables in a double integral $\iint_R F(x, y) dA$ it is necessary to impose suitable restrictions on the region R and the function F . Specifically, we shall assume that R consists of all points that are either inside or on a piecewise smooth simple closed curve C , and that F has continuous first partial derivatives throughout an open region containing R . We shall also require that the transformation W maps a region S of the uv -plane in a one-to-one manner onto R , and that S is bounded by a piecewise smooth simple closed curve K that W maps onto C . Finally, we impose the condition that as (u, v) traces K once in the positive direction, the corresponding point (x, y) traces C once, either in the positive or negative direction. It is possible to weaken these conditions; however, that is beyond the scope of our work.

The function of u and v introduced in the next definition will be used in the change of variable process. It is named after the mathematician C. G. Jacobi (1804–51).

Definition (18.38)

If $x = f(u, v)$ and $y = g(u, v)$, then the **Jacobian** of x and y with respect to u and v , denoted by $\partial(x, y)/\partial(u, v)$, is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}.$$

In the next theorem all symbols have the same meanings as before, and it is assumed that the regions and functions satisfy the conditions we have discussed. In the statement of the theorem the notations $dx dy$ and $du dv$ are used in place of dA so that there is no misunderstanding about the region of integration.

Theorem (18.39)

If $x = f(u, v)$, $y = g(u, v)$ is a transformation of coordinates, then

$$\iint_R F(x, y) dx dy = \pm \iint_S F(f(u, v), g(u, v)) \frac{\partial(x, y)}{\partial(u, v)} du dv.$$

The $+$ sign or the $-$ sign is chosen if, as (u, v) traces the boundary K of S once in the positive direction, the corresponding point (x, y) traces the boundary C of R once in the positive or negative direction, respectively.

Proof Let us begin by choosing $G(x, y)$ such that $\partial G / \partial x = F$. Applying Green's Theorem (18.17) with $G = N$ gives us

$$(a) \quad \iint_R F(x, y) dx dy = \iint_R \frac{\partial}{\partial x} [G(x, y)] dx dy = \oint_C G(x, y) dy.$$

Suppose the curve K in the uv -plane is given parametrically by

$$u = \varphi(t), v = \psi(t), \quad \text{where } a \leq t \leq b.$$

From our assumptions on the transformation, parametric equations for the curve C in the xy -plane are

$$(b) \quad \begin{aligned} x &= f(u, v) = f(\varphi(t), \psi(t)) \\ y &= g(u, v) = g(\varphi(t), \psi(t)) \end{aligned}$$

where $a \leq t \leq b$. We may, therefore, evaluate the line integral $\oint_C G(x, y) dy$ in (a) by formal substitutions for x and y . To simplify the notation, let

$$H(t) = G[f(\varphi(t), \psi(t)), g(\varphi(t), \psi(t))].$$

Applying the Chain Rule to y in (b) gives us

$$\frac{dy}{dt} = \frac{\partial y}{\partial u} \frac{du}{dt} + \frac{\partial y}{\partial v} \frac{dv}{dt} = \frac{\partial y}{\partial u} \varphi'(t) + \frac{\partial y}{\partial v} \psi'(t).$$

Consequently,

$$\begin{aligned} \oint_C G(x, y) dy &= \oint_C H(t) \frac{dy}{dt} dt \\ &= \int_a^b H(t) \left[\frac{\partial y}{\partial u} \varphi'(t) + \frac{\partial y}{\partial v} \psi'(t) \right] dt. \end{aligned}$$

Since $du = \varphi'(t) dt$ and $dv = \psi'(t) dt$, we may regard the last line integral as a line integral around the curve K in the uv -plane. Thus

$$(c) \quad \oint_C G(x, y) dy = \pm \oint_K G \frac{\partial y}{\partial u} du + G \frac{\partial y}{\partial v} dv$$

where for simplicity we have used G as an abbreviation for $G(f(u, v), g(u, v))$. The choice of the $+$ sign or the $-$ sign is determined by letting t vary from a to b and noting whether (x, y) traces C in the same or opposite direction, respectively, as (u, v) traces K .

The line integral on the right in (c) has the form

$$\oint_K M du + N dv$$

where $M = G \frac{\partial y}{\partial u}$ and $N = G \frac{\partial y}{\partial v}$.

Applying Green's Theorem, we obtain

$$\begin{aligned} \oint_K M du + N dv &= \iint_S \left(\frac{\partial N}{\partial u} - \frac{\partial M}{\partial v} \right) du dv \\ &= \iint_S \left(G \frac{\partial^2 y}{\partial u \partial v} + \frac{\partial G}{\partial u} \frac{\partial y}{\partial v} - G \frac{\partial^2 y}{\partial v \partial u} - \frac{\partial G}{\partial v} \frac{\partial y}{\partial u} \right) du dv \\ &= \iint_S \left[\left(\frac{\partial G}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial G}{\partial y} \frac{\partial y}{\partial u} \right) \frac{\partial y}{\partial v} - \left(\frac{\partial G}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial G}{\partial y} \frac{\partial y}{\partial v} \right) \frac{\partial y}{\partial u} \right] du dv \\ &= \iint_S \frac{\partial G}{\partial x} \left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right) du dv. \end{aligned}$$

Using the fact that $\partial G / \partial x = F(x, y)$, together with the definition of Jacobian (18.38), gives us

$$\oint_K M du + N dv = \iint_S F(f(u, v), g(u, v)) \frac{\partial(x, y)}{\partial(u, v)} du dv.$$

Combining this formula with (a) and (c) leads to the desired result. $\square$

Example 1 Evaluate $\iint_R e^{(y-x)/(y+x)} dx dy$ where R is the region in the xy -plane bounded by the trapezoid with vertices $(0, 1)$, $(0, 2)$, $(2, 0)$, and $(1, 0)$.

Solution Since $y - x$ and $y + x$ appear in the integrand, it is convenient to let

$$u = y - x, \quad v = y + x.$$

These equations define a transformation T from the xy -plane to the uv -plane. To apply Theorem (18.39), it is necessary to use the inverse transformation T^{-1} . To find T^{-1} we solve the preceding equations for x and y in terms of u and v , obtaining

$$x = \frac{1}{2}(v - u), \quad y = \frac{1}{2}(v + u).$$

The Jacobian of this transformation from the uv -plane to the xy -plane is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}.$$

The region R is sketched in (i) of Figure 18.38.

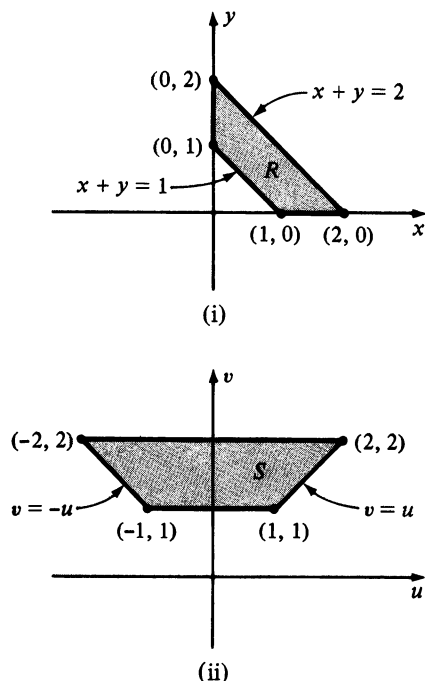


FIGURE 18.38

To determine the region S in the uv -plane that corresponds to R , we note that the sides of R lie on the lines

$$x = 0, \quad y = 0, \quad x + y = 1, \quad x + y = 2.$$

Using the equations for the transformations T and T^{-1} we see that the corresponding curves in the uv -plane are

$$v = u, \quad v = -u, \quad v = 1, \quad v = 2,$$

respectively. These lines form the boundary of the trapezoidal region S shown in (ii) of Figure 18.38. It is left to the reader to verify that interior points of S correspond to interior points of R , and that as (u, v) traces the boundary of S once in the *positive* (counterclockwise) direction, the corresponding point (x, y) traces the boundary of R once in the *negative* (clockwise) direction. Applying Theorem 18.39,

$$\begin{aligned} \iint_R e^{(y-x)/(y+x)} dx dy &= - \int_1^2 \int_{-v}^v e^{u/v} \left(-\frac{1}{2} \right) du dv \\ &= \frac{1}{2} \int_1^2 \left[v e^{u/v} \right]_{-v}^v dv = \frac{1}{2} \int_1^2 v(e - e^{-1}) dv \\ &= \frac{1}{2} (e - e^{-1}) \left[\frac{v^2}{2} \right]_1^2 = \frac{3}{4} (e - e^{-1}). \quad \blacksquare \end{aligned}$$

Example 2 Evaluate $\iint_R e^{-(x^2+y^2)} dx dy$ where R is the region in the xy -plane bounded by the circle $x^2 + y^2 = a^2$.

Solution The polar coordinate substitutions

$$x = r \cos \theta, \quad y = r \sin \theta$$

determine a transformation from the $r\theta$ -plane to the xy -plane, where

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r.$$

A transformation similar to that defined in Example 3 of Section 18.8 maps the rectangle S in the $r\theta$ -plane bounded by $r = 0$, $r = a$, $\theta = 0$, and $\theta = 2\pi$ onto R . Moreover, as (r, θ) traces the boundary of S once in the positive direction, the corresponding point (x, y) traces R once in the positive direction. Hence, from Theorem (18.39),

$$\begin{aligned} \iint_R e^{-(x^2+y^2)} dx dy &= \int_0^{2\pi} \int_0^a e^{-r^2} r dr d\theta = \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_0^a d\theta \\ &= -\frac{1}{2} (e^{-a^2} - 1) 2\pi = \pi(1 - e^{-a^2}). \quad \blacksquare \end{aligned}$$

The theory we have developed can be extended to triple integrals. Given a transformation

$$x = f(u, v, w), \quad y = g(u, v, w), \quad z = h(u, v, w)$$

from a uvw -coordinate system to an xyz -coordinate system, we define the **Jacobian** $\partial(x, y, z)/\partial(u, v, w)$ of the transformation by

$$(18.40) \quad \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

If R and S are corresponding regions in the xyz - and uvw -coordinate systems, then under suitable restrictions,

$$(18.41) \quad \iiint_R F(x, y, z) \, dx \, dy \, dz = \pm \iiint_S G(u, v, w) \frac{\partial(x, y, z)}{\partial(u, v, w)} \, du \, dv \, dw$$

where $G(u, v, w)$ is the expression obtained by substituting for x, y , and z in $F(x, y, z)$.

To illustrate, if we use the spherical coordinate formulas (14.50), then

$$\begin{aligned} x &= \rho \sin \phi \cos \theta \\ y &= \rho \sin \phi \sin \theta \\ z &= \rho \cos \phi. \end{aligned}$$

It can be shown (see Exercise 17), that

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2 \sin \phi.$$

Hence (18.41) takes on the form

$$\iiint_R F(x, y, z) \, dx \, dy \, dz = \pm \iiint_S G(\rho, \theta, \phi) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

which is in agreement with formula (17.29), obtained in an intuitive manner.

18.9 Exercises

In Exercises 1–4 find the Jacobian $\partial(x, y)/\partial(u, v)$.

- 1 $x = u^2 - v^2, y = 2uv$
- 2 $x = e^u \sin v, y = e^u \cos v$
- 3 $x = ve^{-2u}, y = u^2e^{-v}$
- 4 $x = u/(u^2 + v^2), y = v/(u^2 + v^2)$

In Exercises 5 and 6 find $\partial(x, y, z)/\partial(u, v, w)$.

- 5 $x = 2u + 3v - w, y = v - 5w, z = u + 4w$
- 6 $x = u^2 + vw, y = 2v + u^2w, z = uvw$

In Exercises 7–10 use the indicated change of variables to express the integral as a double integral over a region S in the uv -plane. (Do not evaluate.)

- 7 $\iint_R (y - x) \, dx \, dy$; R the region bounded by $y = 2x, y = 0, x = 2; x = u + v, y = 2v$
- 8 $\iint_R (3x - 4y) \, dx \, dy$; R the region bounded by $y = 3x, 2y = x, x = 4; x = u - 2v, y = 3u - v$
- 9 $\iint_R (\frac{1}{4}x^2 + \frac{1}{9}y^2) \, dx \, dy$; $R = \{(x, y): \frac{1}{4}x^2 + \frac{1}{9}y^2 \leq 1\}; x = 2u, y = 3v$
- 10 $\iint_R xy \, dx \, dy$; R the region bounded by $y = 2\sqrt{1 - x}, x = 0, y = 0; x = u^2 - v^2, y = 2uv$

In Exercises 11–16 evaluate the given integral by making the indicated change of variable.

- 11 $\iint_R (x - y)^2 \cos^2(x + y) \, dx \, dy$; R the region bounded by the square with vertices $(0, 1)$, $(1, 2)$, $(2, 1)$, $(1, 0)$; $u = x - y$, $v = x + y$.
- 12 $\iint_R \sin[(y - x)/(y + x)] \, dx \, dy$; R the trapezoid with vertices $(1, 1)$, $(2, 2)$, $(4, 0)$, $(2, 0)$; $u = y - x$, $v = y + x$.
- 13 $\iint_R (x^2 + 2y^2) \, dx \, dy$; R the region in the first quadrant bounded by the graphs of $xy = 1$, $xy = 2$, $y = x$, $y = 2x$; $x = u/v$, $y = v$.
- 14 $\iint_R (4x - 4y + 1)^{-2} \, dx \, dy$; R the region bounded by $x = \sqrt{-y}$, $x = y$, $x = 1$; $x = u + v$, $y = v - u^2$.
- 15 $\iint_R (2y + x)/(y - 2x) \, dx \, dy$; R the trapezoid with vertices $(-1, 0)$, $(-2, 0)$, $(0, 4)$, $(0, 2)$; $u = y - 2x$, $v = 2y + x$.
- 16 $\iint_R (\sqrt{x - 2y} + \frac{1}{4}y^2) \, dx \, dy$; R the triangle with vertices $(0, 0)$, $(4, 0)$, $(4, 2)$; $u = \frac{1}{2}y$, $v = x - 2y$.
- 17 Verify that for the spherical coordinate transformation,

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2 \sin \phi.$$

- 18 Use Theorem (18.39) to derive a formula for

$$\iiint_R F(x, y, z) \, dx \, dy \, dz$$

for a transformation from rectangular to cylindrical coordinates.

- 19 If the transformation of coordinates $x = f(u, v)$, $y = g(u, v)$ is one-to-one, show that

$$\frac{\partial(x, y)}{\partial(u, v)} \frac{\partial(u, v)}{\partial(x, y)} = 1.$$

(Hint: Use the following property of determinants:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} p & q \\ r & s \end{vmatrix} = \begin{vmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{vmatrix}.$$

- 20 Given the transformation $x = f(u, v)$, $y = g(u, v)$ and the transformation $u = h(r, s)$, $v = k(r, s)$ show that

$$\frac{\partial(x, y)}{\partial(u, v)} \frac{\partial(u, v)}{\partial(r, s)} = \frac{\partial(x, y)}{\partial(r, s)}.$$

(Hint: Use the hint given in Exercise 19.)

18.10 Review: Concepts

Define or discuss each of the following.

- 1 Vector field
- 2 Vector function
- 3 Scalar function
- 4 Inverse square field
- 5 Conservative vector field
- 6 Divergence
- 7 Curl
- 8 Line integral
- 9 Applications of line integrals
- 10 Independence of path
- 11 Green's Theorem
- 12 Surface integrals
- 13 The Divergence Theorem (Gauss' Theorem)
- 14 Stokes' Theorem
- 15 Transformations of coordinates
- 16 Jacobians
- 17 Change of variables in multiple integrals

Review: Exercises

In Exercises 1–4 give a geometric description of the vector field $\mathbf{F}$.

- 1 $\mathbf{F}(x, y) = 2x\mathbf{i} + y\mathbf{j}$
- 2 $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
- 3 $\mathbf{F}(x, y, z) = -z\mathbf{k}$

$$4 \quad \mathbf{F}(x, y, z) = \nabla(x^2 + y^2 + z^2)^{-1/2}$$

- 5 Find a two-dimensional conservative vector field that has the potential function $f(x, y) = y^2 \tan x$.
- 6 Find a three-dimensional conservative vector field that has the potential function $f(x, y, z) = \ln(x + y + z)$.

Given the points $A(1, 0)$ and $B(-1, 4)$, evaluate the line integral $\int_C y^2 dx + xy dy$ if C is the curve described in each of Exercises 7–10.

- 7 C is given parametrically by $x = 1 - t$, $y = t^2$; $0 \leq t \leq 2$.
- 8 C consists of the line segments from A to the point $D(-1, 0)$, and from D to B .
- 9 C is the line segment from A to B .
- 10 C is the graph of $y = 2 - 2x^3$ from A to B .
- 11 Evaluate $\int_C xy ds$, where C is the part of the graph of $y = x^4$ between $(-1, 1)$ and $(2, 16)$.

Given $A(0, 0, 0)$ and $B(2, 4, 8)$ evaluate

$$\int_C x dx + (x + y) dy + (x + y + z) dz$$

if C is the curve described in each of Exercises 12–14.

- 12 C consists of three line segments, the first parallel to the z -axis, the second parallel to the x -axis, and the third parallel to the y -axis.
- 13 C consists of the line segment from A to B .
- 14 C is given parametrically by $x = t$, $y = t^2$, $z = t^3$.
- 15 If $\mathbf{F}(x, y) = (x + y)\mathbf{i} + (x - y)\mathbf{j}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C has parametric equations $x = \cos t$, $y = \sin t$, $-\pi \leq t \leq 0$.
- 16 The force $\mathbf{F}$ at a point (x, y, z) in three dimensions is given by $\mathbf{F}(x, y, z) = xy\mathbf{i} + y^2z\mathbf{j} + xz^2\mathbf{k}$. If C is the square with vertices $A_1(1, 1, 1)$, $A_2(-1, 1, 1)$, $A_3(-1, -1, 1)$, and $A_4(1, -1, 1)$, find the work done by $\mathbf{F}$ as its point of application moves around C once in the direction determined by increasing subscripts on the A_i .

Prove that the line integrals in Exercises 17 and 18 are independent of path and find their values.

- 17 $\int_{(1, -1)}^{(2, 3)} (x + y) dx + (x + y) dy$
- 18 $\int_{(0, 0, 0)}^{(2, 1, 3)} (8x^3 + z^2) dx - 3z dy + (2xz - 3y) dz$
- 19 If $\mathbf{F}(x, y, z) = 2xe^{2y}\mathbf{i} + 2(x^2e^{2y} + y \cot z)\mathbf{j} - y^2 \csc^2 z\mathbf{k}$, prove that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path and find a potential function f for $\mathbf{F}$.

Use Green's Theorem to evaluate the line integral

$$\oint_C xy dx + (x^2 + y^2) dy$$

if C is the curve described in each of Exercises 20–22.

- 20 C is the closed curve determined by $y = x^2$ and $y - x = 2$ between $(-1, 1)$ and $(2, 4)$.
- 21 C is the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$.
- 22 C is the circle with equation $x^2 + y^2 - 2x = 0$.
- 23 Find $\text{div } \mathbf{F}$ and $\text{curl } \mathbf{F}$ if $\mathbf{F}(x, y, z) = x^3z^4\mathbf{i} + xyz^2\mathbf{j} + x^2y^2\mathbf{k}$.
- 24 If f and g are scalar functions of three variables prove that $\nabla \cdot (f \nabla g) = f \nabla^2 g + \nabla f \cdot \nabla g$.
- 25 Evaluate $\iint_S xyz dS$, where S is the part of the plane $z = x + y$ that lies over the triangular region in the xy -plane having vertices $(0, 0, 0)$, $(1, 0, 0)$, and $(0, 2, 0)$.
- 26 Evaluate $\iint_S x^2z^2 dS$ where S is the top half of the cylinder $y^2 + z^2 = 4$ between $x = 0$ and $x = 1$.
- 27 Let Q be the region bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = 1$. If $\mathbf{F} = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$, use the Divergence Theorem to find $\iint_S \mathbf{F} \cdot \mathbf{n} dS$, where S is the surface of Q and $\mathbf{n}$ is the unit outer normal to S .
- 28 Verify the Divergence Theorem if $\mathbf{F} = 2x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$ and S is the surface of the parallelepiped bounded by the planes $x = \pm 1$, $y = \pm 2$, $z = \pm 3$.

Verify Stokes' Theorem for the given $\mathbf{F}$ and S in Exercises 29 and 30.

- 29 $\mathbf{F} = y^2\mathbf{i} + 2x\mathbf{j} + 5y\mathbf{k}$ and S is the hemisphere $z = (4 - x^2 - y^2)^{1/2}$.
- 30 $\mathbf{F} = (x + y)\mathbf{i} + (y + z)\mathbf{j} + (x + z)\mathbf{k}$ and S is the region bounded by the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.
- 31 If a transformation T of the xy -plane to the uv -plane is given by $u = 2x + 5y$, $v = 3x - 4y$, find:
 - (a) the u -curves.
 - (b) the v -curves.
 - (c) T^{-1} .
 - (d) $\partial(x, y)/\partial(u, v)$.
 - (e) the image of the line $ax + by + c = 0$.
 - (f) the image of the circle $x^2 + y^2 = a^2$.
- 32 Evaluate the integral $\int_0^1 \int_y^{2-y} e^{(x-y)/(x+y)} dx dy$ by means of the change of variables $u = x - y$, $v = x + y$.

19 Differential Equations

A *differential equation* is an equation that involves derivatives or differentials. If only derivatives of a function of one variable occur, then a differential equation is called *ordinary*. A *partial differential equation* contains partial derivatives. The primary

objective of this chapter is to develop techniques for solving certain basic types of ordinary differential equations. The discussion is not intended to be a treatise on the subject, but rather to serve as an introduction to this vast and important area of mathematics.

19.1 Introduction

An equation that involves x , y , y' , y'' , $\dots$, and $y^{(n)}$, where y is a function of x , and $y^{(n)}$ is the n th derivative of y with respect to x , is called an **ordinary differential equation of order n** . The following are examples of ordinary differential equations of orders 1, 2, 3, and 4, respectively.

$$y' = 2x$$

$$\frac{d^2y}{dx^2} + x^2\left(\frac{dy}{dx}\right)^3 - 15y = 0$$

$$(y''')^4 - x^2(y'')^5 + 4xy = xe^x$$

$$\left(\frac{d^4y}{dx^4}\right)^2 - 1 = x^3 \frac{dy}{dx}$$

If a function f has the property that substitution of $f(x)$ for y in a differential equation results in an identity for all x in some interval, then $f(x)$ (or f) is called a **solution** of the differential equation. For example, if C is any real number, then a solution of the differential equation $y' = 2x$ is

$$f(x) = x^2 + C$$

because substitution of $f(x)$ for y leads to the identity $2x = 2x$. We call $x^2 + C$ the **general solution** of $y' = 2x$, since every solution has this form.

Example 1 If C_1 and C_2 are any real numbers, prove that

$$f(x) = C_1 e^{5x} + C_2 e^{-5x}$$

is a solution of $y'' - 25y = 0$.

Solution Since

$$f'(x) = 5C_1 e^{5x} - 5C_2 e^{-5x}$$

and

$$f''(x) = 25C_1 e^{5x} + 25C_2 e^{-5x}$$

we have

$$y'' - 25y = (25C_1 e^{5x} + 25C_2 e^{-5x}) - 25(C_1 e^{5x} + C_2 e^{-5x}) = 0$$

for every x . This shows that $f(x)$ is a solution of $y'' - 25y = 0$. ■

It can be shown that the solution given in Example 1 is the general solution of $y'' - 25y = 0$. Observe that the differential equation is of order 2 and the general solution contains two arbitrary constants (called **parameters**) C_1 and C_2 . The precise definition of general solution involves the concept of **independent parameters** and is left for more advanced courses. It can be shown that general solutions of n th-order differential equations contain n independent parameters $C_1, C_2, \dots, C_n$. A **particular solution** is obtained by assigning specific values to the parameters. Some differential equations have solutions that are not special cases of the general solution. These so-called **singular solutions** will not be discussed in this text.

Example 2 Find the particular solution of $y' = 2x$ that satisfies the condition that $y = 5$ if $x = 2$.

Solution The general solution of $y' = 2x$ may be written $y = x^2 + C$. If $y = 5$ when $x = 2$, then $5 = 4 + C$, or $C = 1$. Hence the desired particular solution is $y = x^2 + 1$. ■

Conditions of the type stated in Example 2 are often called **boundary conditions** for the differential equation. If the general solution contains one parameter, then one boundary condition is sufficient to find a particular solution. If, as in Example 1, the general solution contains two parameters, then two boundary conditions are needed for particular solutions. Similar statements hold if more than two parameters are involved. Values of $y, y', \dots$, that correspond to “first” values of the independent variable x are sometimes referred to as **initial conditions**. For example, if x represents time, we often take $x = 0$ and the values of $y, y', \dots$ at $x = 0$ as initial conditions.

The solutions we have considered express y explicitly in terms of x . Solutions of certain differential equations are stated implicitly. In this case,

implicit differentiation is used to check the solution, as illustrated in the following example.

Example 3 Show that $x^3 + x^2y - 2y^3 = C$ is an implicit solution of

$$(x^2 - 6y^2)y' + 3x^2 + 2xy = 0.$$

Solution If the first equation is satisfied by $y = f(x)$, then differentiating implicitly,

$$3x^2 + 2xy + x^2y' - 6y^2y' = 0$$

$$\text{or} \quad (x^2 - 6y^2)y' + 3x^2 + 2xy = 0.$$

Thus $f(x)$ is a solution of the differential equation. ■

One of the simplest types of differential equations is

$$M(x) + N(y)y' = 0, \quad \text{or} \quad M(x) + N(y) \frac{dy}{dx} = 0$$

where M and N are continuous functions. If $y = f(x)$ is a solution, then

$$M(x) + N(f(x))f'(x) = 0.$$

If $f'(x)$ is continuous, then indefinite integration leads to

$$\int M(x) dx + \int N(f(x))f'(x) dx = C$$

$$\text{or} \quad \int M(x) dx + \int N(y) dy = C.$$

The last equation is an (implicit) solution of the differential equation. A device that is useful for remembering this method of solution is to change the equation $M(x) + N(y)(dy/dx) = 0$ to the following *differential form*:

$$M(x) dx + N(y) dy = 0.$$

The solution is then found by formally integrating each term. The differential equation $M(x) + N(y)y' = 0$ is said to be **separable**, since the variables x and y may be separated as we have indicated.

Example 4 Solve the differential equation $y^4 e^{2x} + \frac{dy}{dx} = 0$.

Solution Writing the equation in differential form and then separating the variables gives us

$$y^4 e^{2x} dx + dy = 0$$

$$\text{and} \quad e^{2x} dx + \frac{1}{y^4} dy = 0.$$

Integrating each term, we obtain the (implicit) solution

$$\frac{1}{2}e^{2x} - \frac{1}{3y^3} = C.$$

If we multiply both sides by 6, then another form of the solution is

$$3e^{2x} - \frac{2}{y^3} = D$$

where $D = 6C$. If an explicit form is desired we may solve for y , obtaining

$$y = \left(\frac{2}{3e^{2x} - D} \right)^{1/3}. \quad \blacksquare$$

Example 5 Solve the differential equation $2xy + 6x + (x^2 - 4)y' = 0$.

Solution The equation may be written

$$2x(y + 3) + (x^2 - 4)\frac{dy}{dx} = 0.$$

Assuming that $(y + 3)(x^2 - 4) \neq 0$, we may divide both sides by this product and express the equation in differential form as follows:

$$\frac{2x}{x^2 - 4} dx + \frac{1}{y + 3} dy = 0.$$

Thus the given equation is separable, and integration of each term gives us

$$\ln |x^2 - 4| + \ln |y + 3| = C_1$$

or

$$\ln |(x^2 - 4)(y + 3)| = C_1.$$

This may also be written

$$|(x^2 - 4)(y + 3)| = e^{C_1} = C_2.$$

It can be shown, furthermore, that the solution is

$$(x^2 - 4)(y + 3) = C$$

for every nonzero C . In explicit form,

$$y = \frac{C}{x^2 - 4} - 3.$$

Since it was assumed that $(x^2 - 4)(y + 3) \neq 0$, the cases $x = \pm 2$ and $y = -3$ require special attention. Direct substitution shows that $y = -3$ is a solution of the given differential equation; however, y' is undefined if $x = \pm 2$. Consequently, $y = C/(x^2 - 4) - 3$ is the solution if C is any real number. ■

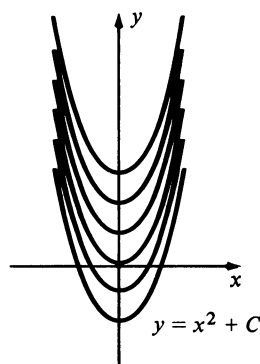


FIGURE 19.1

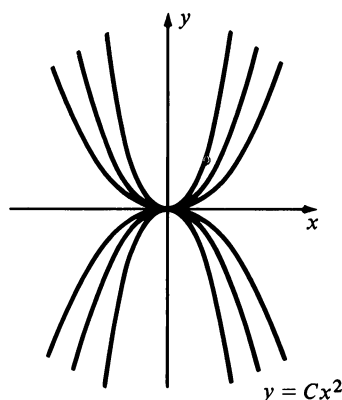


FIGURE 19.2

Since general solutions of differential equations contain arbitrary constants, the graphs of these solutions are families of curves. For example, the solution $y = x^2 + C$ of $y' = 2x$ leads to a family of parabolas of the type illustrated in Figure 19.1. Each member of the family is obtained by assigning a specific value to C . Conversely, if we *begin* with the equation of a family of curves, then implicit differentiation may be used to obtain a differential equation called a **differential equation of the family**.

Example 6 Find a differential equation of the family of parabolas defined by $y = Cx^2$ where C is a nonzero real number.

Solution The family consists of all parabolas with vertex at the origin and focus on the y -axis, as illustrated in Figure 19.2. Differentiating the given equation we obtain $y' = 2Cx$ and therefore

$$y = Cx^2 = (Cx)x = (\tfrac{1}{2}y')x.$$

Hence a differential equation for the family is

$$2y - y'x = 0.$$

Letting $y' = dy/dx$ we obtain the differential form

$$2y \, dx - x \, dy = 0. \quad \blacksquare$$

An equation in x and y that contains two arbitrary constants is said to define a **two-parameter family of curves**. Similar terminology is used if there are more than two constants. To find a differential equation when several constants are involved, it is necessary to differentiate the given equation several times.

Example 7 Find a differential equation for the two-parameter family of curves

$$y = C_1 \cos x + C_2 \sin x.$$

Solution Differentiating twice, we obtain

$$y' = -C_1 \sin x + C_2 \cos x$$

$$y'' = -C_1 \cos x - C_2 \sin x.$$

Comparing with the given equation we see that

$$y'' = -y, \quad \text{or} \quad y'' + y = 0.$$

This is a differential equation for the family. \(\blacksquare\)

19.1 Exercises

In Exercises 1–8 prove that y is a solution of the indicated differential equation.

- 1 $y = C_1 e^x + C_2 e^{2x}; y'' - 3y' + 2y = 0$
- 2 $y = C e^{-3x}; y' + 3y = 0$
- 3 $y = e^x(C_1 \cos x + C_2 \sin x); y'' - 2y' + 2y = 0$
- 4 $y = (C_1 + C_2 x)e^{-2x}; y'' + 4y' + 4y = 0$
- 5 $y = Cx^{-2/3}; 2xy^3 dx + 3x^2 y^2 dy = 0$
- 6 $y = Cx^3; x^3 y''' + x^2 y'' - 3xy' - 3y = 0$
- 7 $y^2 - x^2 - xy = C; (x - 2y)y' + 2x + y = 0$
- 8 $x^2 - y^2 = C; yy' = x$

In Exercises 9–14 find a differential equation that has the given general solution.

- 9 $y^2 = x^3 + C$
- 10 $x^2 + y^2 = C^2$
- 11 $y = e^{Cx}$
- 12 $y = C_1 x^3 + C_2 x$
- 13 $y = C_1 e^x + C_2 x e^x$
- 14 $y = C_1 e^x + C_2 x e^x + C_3 e^{-x}$

In Exercises 15–18 find a differential equation of the indicated family of curves.

- 15 All nonvertical lines through the origin
- 16 All circles with centers at the origin
- 17 All circles with centers on the x -axis
- 18 All parabolas having focus and vertex on the x -axis

Solve the differential equations in Exercises 19–32.

- 19 $x dy - y dx = 0$
- 20 $(1 + y^2) dx + (1 + x^2) dy = 0$
- 21 $2y dx + (xy + 5x) dy = 0$
- 22 $(xy - 4x) dx + (x^2 y + y) dy = 0$
- 23 $y' = x - 1 + xy - y$
- 24 $(y + yx^2) dy + (x + xy^2) dx = 0$
- 25 $e^{x+2y} dx - e^{2x-y} dy = 0$
- 26 $\cos x dy - y dx = 0$
- 27 $y(1 + x^3)y' + x^2(1 + y^2) = 0$
- 28 $x^2 y' - yx^2 = y$
- 29 $x \tan y - y' \sec x = 0$
- 30 $xy + y' e^{x^2} \ln y = 0$
- 31 $e^y \sin x dx - \cos^2 x dy = 0$
- 32 $\sin y \cos x dx + (1 + \sin^2 x) dy = 0$

In Exercises 33–40 find the particular solution of the differential equation that satisfies the given boundary conditions.

- 33 $2y^2 y' = 3y - y'; x = 3, y = 1$
- 34 $\sqrt{x} y' - \sqrt{y} = x\sqrt{y}; x = 9, y = 4$
- 35 $x dy - (2x + 1)e^{-y} dx = 0; x = 1, y = 2$
- 36 $\sec 2y dx - \cos^2 x dy = 0; x = \pi/4, y = \pi/6$
- 37 $(xy + x) dx + \sqrt{4 - x^2} dy = 0; x = 0, y = 1$
- 38 $x dy - \sqrt{1 - y^2} dx = 0; x = 1, y = \frac{1}{2}$
- 39 $\cot x dy - (1 + y^2) dx = 0; x = 0, y = 1$
- 40 $\csc y dx - e^x dy = 0; x = 0, y = 0$

19.2 Exact Differential Equations

The separable differential equations considered in the previous section are special cases of the first-order equation

$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0$$

where P and Q are functions of x and y . This equation may be expressed in terms of differentials as

$$P(x, y) dx + Q(x, y) dy = 0.$$

In this section we shall restrict our attention to differential equations that are *exact*, in the sense of the following definition.

Definition (19.1)

Let P and Q have continuous first partial derivatives. The differential equation

$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0, \quad \text{or} \quad P(x, y) dx + Q(x, y) dy = 0$$

is an **exact differential equation** provided

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Example 1 Show that the following differential equation is exact:

$$(3x^2y - 2y^3 + 3) dx + (x^3 - 6xy^2 + 2y) dy = 0.$$

Solution If $P(x, y) = 3x^2y - 2y^3 + 3$ and $Q(x, y) = x^3 - 6xy^2 + 2y$, we see that

$$P_y(x, y) = 3x^2 - 6y^2 = Q_x(x, y).$$

Hence the equation is exact by Definition (19.1). ■

Theorem (19.2)

If $P(x, y) + Q(x, y)y' = 0$ is an exact differential equation, then there exists a function F of x and y such that

$$\frac{\partial F}{\partial x} = P \quad \text{and} \quad \frac{\partial F}{\partial y} = Q.$$

Proof
 F by

If (x_1, y_1) is in the domain of P and Q , let us define the function

$$F(x, y) = \int_{x_1}^x P(t, y) dt + \int_{y_1}^y Q(x_1, t) dt.$$

It is left as an exercise to prove that $F_x(x, y) = P(x, y)$ and $F_y(x, y) = Q(x, y)$. □

The next result is the fundamental existence theorem for solutions of exact differential equations.

Theorem (19.3)

An exact differential equation $P(x, y) + Q(x, y)y' = 0$ has a solution of the form $F(x, y) = C$, where C is a constant and F is any function of x and y such that $F_x = P$ and $F_y = Q$.

Proof By Theorem (19.2) there exists at least one function F such that $F_x = P$ and $F_y = Q$. We shall complete the proof by showing that $y = f(x)$ is a solution of the exact differential equation if and only if $F(x, f(x)) = C$ for some constant C .

By the Chain Rule,

$$D_x F(x, f(x)) = F_x(x, f(x)) + F_y(x, f(x))f'(x)$$

or

$$D_x F(x, f(x)) = P(x, f(x)) + Q(x, f(x))f'(x).$$

If $y = f(x)$ is a solution of the differential equation, then $D_x F(x, f(x)) = 0$ and, therefore, $F(x, f(x)) = C$ for some C . Conversely, if $F(x, f(x)) = C$, then $D_x F(x, f(x)) = 0$, that is,

$$0 = P(x, f(x)) + Q(x, f(x))f'(x)$$

which means that $f(x)$ is a solution of the given differential equation. $\square$

The following examples illustrate a technique for finding the solution $F(x, y) = C$ in Theorem (19.3).

Example 2 Solve the differential equation

$$(3x^2y - 2y^3 + 3) dx + (x^3 - 6xy^2 + 2y) dy = 0.$$

Solution It was shown in Example 1 that the equation is exact. Consequently, by Theorem (19.3), there is a function F such that

$$F_x(x, y) = 3x^2y - 2y^3 + 3$$

$$F_y(x, y) = x^3 - 6xy^2 + 2y.$$

Integrating $F_x(x, y)$ with respect to x gives us

$$F(x, y) = x^3y - 2xy^3 + 3x + g(y)$$

where g is a function of y . It follows that

$$F_y(x, y) = x^3 - 6xy^2 + g'(y).$$

Comparing this equation with the previous formula for $F_y(x, y)$ we see that $g'(y) = 2y$, and hence $g(y) = y^2 + C_1$. Substituting for $g(y)$ in the formula for $F(x, y)$ gives us

$$F(x, y) = x^3y - 2xy^3 + 3x + y^2 + C_1.$$

Hence, by Theorem (19.3), a solution of the differential equation is

$$x^3y - 2xy^3 + 3x + y^2 = C$$

where the constant C_1 has been absorbed by the constant C . This solution may be checked by implicit differentiation. ■

Example 3 Solve the differential equation

$$2x \sin y - y \sin x + (x^2 \cos y + \cos x)y' = 0$$

subject to the boundary conditions $y = \pi/6$ if $x = \pi/2$.

Solution If $P(x, y) = 2x \sin y - y \sin x$ and $Q(x, y) = x^2 \cos y + \cos x$ we see that

$$P_y(x, y) = 2x \cos y - \sin x = Q_x(x, y)$$

and hence the differential equation is exact. By Theorem (19.2) there is a function F such that

$$F_x(x, y) = 2x \sin y - y \sin x$$

$$F_y(x, y) = x^2 \cos y + \cos x.$$

Integrating $F_x(x, y)$ with respect to x gives us

$$F(x, y) = x^2 \sin y + y \cos x + g(y)$$

where g is a function of y . Consequently,

$$F_y(x, y) = x^2 \cos y + \cos x + g'(y).$$

Comparing with the other form of $F_y(x, y)$ we see that $g'(y) = 0$ and therefore $g(y) = C_1$. Applying Theorem (19.3), a solution of the given differential equation is

$$x^2 \sin y + y \cos x = C$$

where again C_1 has been absorbed by C . Finally, the boundary conditions imply that

$$\left(\frac{\pi}{2}\right)^2 \left(\frac{1}{2}\right) + \left(\frac{\pi}{6}\right) \cdot 0 = C$$

or $C = \pi^2/8$. Thus the required particular solution is

$$x^2 \sin y + y \cos x = \frac{\pi^2}{8}. \quad \blacksquare$$

19.2 Exercises

Solve the differential equations in Exercises 1–14.

- 1 $(2x + y) dx + (2y + x) dy = 0$
- 2 $(2xy + y^3) dx + (x^2 + 3y^2x) dy = 0$
- 3 $(1 - 2xy + 3x^2y^2) dx - (x^2 + 3y^2 - 2x^3y) dy = 0$
- 4 $(2x - 3x^2y) dx + (1 + 2y - x^3) dy = 0$
- 5 $(x^2 \cos y + 4y)y' + 2x \sin y = -5$
- 6 $y^2e^x + 4y + (2ye^x + 4x)y' = 0$
- 7 $e^{xy}(xy + 1) dx + (x^2e^{xy} + 2y) dy = 0$
- 8 $(v \ln v + 2u \ln u) du + (u \ln v + u) dv = 0$
- 9 $(\sin^2 \theta + 2r \cos \theta) dr + r(\sin 2\theta - r \sin \theta) d\theta = 0$
- 10 $(x^2 + 2xy - y^2)y' = -y^2 - 2xy + x^2$
- 11 $(e^{-x} \cos y - 2x) dx + e^{-x} \sin y dy = 0$
- 12 $y \cos x dx + (\sin x - \sin y) dy = 0$
- 13 $(x - yx^{-1} + \ln y) + (xy^{-1} - \ln x)y' = 0$

$$14 \left(\frac{x}{1+y^2} + y^3 - 1 \right) y' + \left(\frac{1}{1+x^2} + \tan^{-1} y \right) = 0$$

In Exercises 15–18 find a solution of the differential equation that satisfies the given boundary conditions.

- 15 $(2xy^3 + 8x) dx + (3x^2y^2 + 5) dy = 0$;
 $y = -1$ when $x = 2$
- 16 $(x^2e^y + 3e^x)y' + (2xe^y + 3ye^x) = 0$; $y = 0$ when $x = 1$
- 17 $\left(ye^{2x} + \frac{y}{1+4y^2} \right) dy + (y^2e^{2x} - 1) dx = 0$; $x = 0, y = 1/2$
- 18 $(x \cos x + \sin x + y) dx + (\sin y + x) dy = 0$;
 $x = \pi/6, y = \pi/3$
- 19 Complete the proof of Theorem (19.2).
- 20 Prove that the separable differential equation
 $M(x) + N(y)y' = 0$ is a special case of
$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0.$$

19.3 Homogeneous Differential Equations

Definition (19.4)

A function f of two variables is said to be **homogeneous of degree n** if

$$f(tx, ty) = t^n f(x, y)$$

for every $t > 0$ such that (tx, ty) is in the domain of f .

Example 1 Show that the following define homogeneous functions, and find their degrees.

$$(a) f(x, y) = 2x^4 - x^2y^2 + 5xy^3 \quad (b) g(x, y) = \frac{1}{x^2 + y^2} e^{x/y}$$

Solution

(a) Since

$$\begin{aligned} f(tx, ty) &= 2(tx)^4 - (tx)^2(ty)^2 + 5(tx)(ty)^3 \\ &= t^4(2x^4 - x^2y^2 + 5xy^3) = t^4 f(x, y), \end{aligned}$$

it follows from Definition (19.4) that f is homogeneous of degree 4.

(b) Since

$$g(tx, ty) = \frac{1}{t^2x^2 + t^2y^2} e^{tx/ty} = t^{-2}g(x, y),$$

we see from (19.4) that g is homogeneous of degree -2 . ■

Definition (19.5)

A **homogeneous differential equation** is an equation the form

$$P(x, y)dx + Q(x, y)dy = 0$$

where P and Q are homogeneous functions of the same degree.

The next result is the principal theorem for homogeneous differential equations.

Theorem (19.6)

Let $P(x, y)dx + Q(x, y)dy = 0$ be a homogeneous differential equation. If g is a differentiable function, then either of the following substitutions leads to a separable differential equation:

- (i) $y = xv$, where $v = g(x)$.
- (ii) $x = yu$, where $u = g(y)$.

Proof We shall only prove (i), since the proof of (ii) is similar. If $y = xv$, where $v = g(x)$, then

$$dy = v dx + x dv.$$

Thus, substitution of xv for y in $P(x, y)dx + Q(x, y)dy = 0$ yields

$$P(x, xv)dx + Q(x, xv)(v dx + x dv) = 0.$$

If P and Q are homogeneous of degree n , then

$$P(x, xv) = x^n P(1, v) \quad \text{and} \quad Q(x, xv) = x^n Q(1, v).$$

Substituting in the preceding differential equation and dividing both sides by x^n , we obtain

$$P(1, v)dx + Q(1, v)(v dx + x dv) = 0.$$

This equation may be written in the separable form

$$\frac{1}{x} dx + \frac{Q(1, v)}{P(1, v) + vQ(1, v)} dv = 0$$

provided no zero denominators occur. We have proved that if $y = xv$ is a solution of the homogeneous differential equation, then v is a solution of the preceding separable differential equation. Conversely, if v is a solution of the separable differential equation, then we may reverse the argument to show that $y = xv$ is a solution of $P(x, y) dx + Q(x, y) dy = 0$. It is not advisable to memorize the final form of the separable differential equation. Instead, remember the substitution $y = xv$. $\square$

Example 2 Solve the differential equation

$$(y^2 - xy) dx + x^2 dy = 0.$$

Solution If $P(x, y) = y^2 - xy$ and $Q(x, y) = x^2$, then the functions P and Q are both homogeneous of degree 2. (Why?) Hence the differential equation is homogeneous and as in (i) of Theorem (19.6) we substitute

$$y = xv, \quad dy = v dx + x dv.$$

This leads to the following chain of equations.

$$\begin{aligned} (x^2v^2 - x^2v) dx + x^2(v dx + x dv) &= 0 \\ x^2(v^2 - v) dx + x^2(v dx + x dv) &= 0 \\ (v^2 - v) dx + v dx + x dv &= 0 \\ v^2 dx + x dv &= 0 \\ \frac{1}{x} dx + \frac{1}{v^2} dv &= 0 \end{aligned}$$

Integrating each term we obtain

$$\ln |x| - \frac{1}{v} = C_1.$$

Since $v = y/x$, this gives us

$$\ln |x| - \frac{x}{y} = C_1.$$

If we let $C_1 = \ln |C|$, then

$$\ln |x| - \ln |C| = \frac{x}{y}, \quad \text{or} \quad \ln \left| \frac{x}{C} \right| = \frac{x}{y}.$$

This may be written

$$\frac{x}{C} = e^{x/y}, \quad \text{or} \quad x = Ce^{x/y}.$$

An explicit form for the solution may be obtained by solving for y . $\blacksquare$

Example 3 Solve the differential equation

$$\left(y - x \cot \frac{x}{y} \right) dy + y \cot \frac{x}{y} dx = 0.$$

Solution The equation is homogeneous of degree 1. (Why?) It is convenient to let $x = vy$, since in this case $\cot(x/y)$ reduces to $\cot v$. Thus we substitute

$$x = vy, \quad dx = v dy + y dv$$

obtaining

$$(y - vy \cot v) dy + y \cot v(v dy + y dv) = 0.$$

Dividing both sides by y and simplifying leads to

$$(1 - v \cot v) dy + \cot v(v dy + y dv) = 0$$

$$dy + y \cot v dv = 0.$$

Separating the variables and integrating, we obtain

$$\frac{1}{y} dy + \cot v dv = 0$$

$$\ln |y| + \ln |\sin v| = C_1 = \ln |C|$$

$$\ln |y \sin v| = \ln |C|$$

$$y \sin v = C.$$

Since $v = x/y$, a solution to the given equation is

$$y \sin \frac{x}{y} = C. \quad \blacksquare$$

19.3 Exercises

Solve the differential equations in Exercises 1–10.

- 1 $(x + 3y) dx + x dy = 0$
- 2 $(2x - y) dx + (x + 2y) dy = 0$
- 3 $\left(y \sin \frac{y}{x} + x \cos \frac{y}{x}\right) dx - x \sin \frac{y}{x} dy = 0$
- 4 $x^2 dy + (y^2 - xy) dx = 0$
- 5 $(x^2 + y^2) dx - x^2 dy = 0$
- 6 $(y^2 - xy + x^2) dx + xy dy = 0$
- 7 $y' = x^3/(4x^3 - 3x^2y)$
- 8 $y' = x^{-1}y - \cos x^{-1}y$
- 9 $(y + \sqrt{x^2 - y^2}) dx = x dy$
- 10 $(x + y) dx = (x - y) dy$

Solve the differential equations in Exercises 11–14 by using the substitution $x = vy$.

- 11 $2xy dx + (y^2 - x^2) dy = 0$
- 12 $y^2 dx + x(x - y) dy = 0$
- 13 $3y dx + (x + 2y) dy = 0$
- 14 $dx - (y^{-1}x + \tan y^{-1}x) dy = 0$

In Exercises 15 and 16 show that the differential equation is both homogeneous and exact and solve it by the corresponding method.

- 15 $(x + 3y) dx + (3x - 2y) dy = 0$
- 16 $(x^2 + y^2) dx + 2xy dy = 0$

19.4 First-Order Linear Differential Equations

Definition (19.7)

A **first-order linear differential equation** is an equation of the form

$$y' + P(x)y = Q(x)$$

where P and Q are continuous functions.

If, in Definition (19.7), $Q(x) = 0$ for all x , we obtain $y' + P(x)y = 0$, which is separable. Specifically, we may write

$$\frac{1}{y} y' = -P(x)$$

provided $y \neq 0$. Integration gives us

$$\ln |y| = - \int P(x) dx + \ln |C|.$$

We have expressed the constant of integration as $\ln |C|$ in order to change the form of the last equation as follows:

$$\ln |y| - \ln |C| = - \int P(x) dx$$

$$\ln \left| \frac{y}{C} \right| = - \int P(x) dx$$

$$\frac{y}{C} = e^{-\int P(x) dx}$$

$$ye^{\int P(x) dx} = C.$$

We next observe that

$$\begin{aligned} D_x[ye^{\int P(x) dx}] &= y'e^{\int P(x) dx} + P(x)ye^{\int P(x) dx} \\ &= e^{\int P(x) dx}[y' + P(x)y]. \end{aligned}$$

Consequently, if we multiply both sides of $y' + P(x)y = Q(x)$ by $e^{\int P(x) dx}$, then the resulting equation may be written

$$D_x[ye^{\int P(x) dx}] = Q(x)e^{\int P(x) dx}.$$

This gives us the following (implicit) solution of the first-order differential equation:

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + D.$$

Solving this equation for y leads to an explicit solution. The expression $e^{\int P(x) dx}$ is called an **integrating factor** of the given equation. We have proved the following result.

Theorem (19.8)

The first-order linear differential equation $y' + P(x)y = Q(x)$ may be transformed into a separable differential equation by multiplying both sides by the integrating factor $e^{\int P(x) dx}$.

Example 1 Solve the differential equation $x^2y' + 5xy + 3x^5 = 0$ where $x \neq 0$.

Solution To find an integrating factor we begin by expressing the given differential equation in the “standardized” form (19.7), where the coefficient of y' is 1. Thus, dividing both sides by x^2 , we obtain

$$y' + \frac{5}{x}y = -3x^3$$

which has the form given in (19.7) with $P(x) = 5/x$ and $Q(x) = -3x^3$. By Theorem (19.8) an integrating factor is

$$e^{\int P(x) dx} = e^{5 \ln |x|} = e^{\ln |x|^5} = |x|^5.$$

If $x > 0$ then $|x|^5 = x^5$, whereas if $x < 0$ then $|x|^5 = -x^5$. In either case, multiplying both sides of the standardized form by $|x|^5$ gives us

$$x^5y' + 5x^4y = -3x^8, \quad \text{or} \quad D_x(x^5y) = -3x^8.$$

Thus a solution is

$$x^5y = -\frac{x^9}{3} + C$$

or
$$y = -\frac{x^4}{3} + \frac{C}{x^5}.$$
 ■

A generalization of a first-order linear differential equation is the following **Bernoulli equation**

$$(19.9) \quad y' + P(x)y = Q(x)y^n$$

where $n \neq 0$. Evidently $y = 0$ is a solution. If $y \neq 0$ we may divide both sides by y^n , obtaining

$$(*) \quad y^{-n}y' + P(x)y^{1-n} = Q(x).$$

If we let $w = y^{1-n}$, then

$$w' = D_x w = (1 - n)y^{-n}y'$$

or
$$y^{-n}y' = \frac{1}{1-n}w'.$$

Replacing $y^{-n}y'$ in (*) by the last expression gives us

$$\frac{1}{1-n}w' + P(x)w = Q(x).$$

This first-order linear differential equation may be solved for w using the integrating factor technique. After w has been found, the solution of (19.9) is given by $y^{1-n} = w$ (and $y = 0$).

Example 2 Solve the differential equation $y' + 2x^{-1}y = x^6y^3$.

Solution The equation has the Bernoulli form (19.9) with $n = 3$. If, as in the preceding discussion, we multiply both sides by y^{-3} and substitute $w = y^{1-n} = y^{-2}$ we obtain

$$\begin{aligned} y^{-3}y' + 2x^{-1}y^{-2} &= x^6 \\ -\frac{1}{2}w' + 2x^{-1}w &= x^6 \\ w' - 4x^{-1}w &= -2x^6. \end{aligned}$$

By Theorem (19.8), an integrating factor for the last equation is

$$e^{\int (-4/x) dx} = e^{-4 \ln |x|} = e^{\ln |x|^{-4}} = |x|^{-4} = x^{-4}$$

and we proceed as follows:

$$\begin{aligned} x^{-4}w' - 4x^{-5}w &= -2x^2, \\ x^{-4}w &= -\frac{2}{3}x^3 + C \\ w &= -\frac{2}{3}x^7 + Cx^4. \end{aligned}$$

Finally, since $w = y^{-2}$, the solution of the given equation is

$$y^{-2} = -\frac{2}{3}x^7 + Cx^4$$

or
$$\left(-\frac{2}{3}x^7 + Cx^4\right)y^2 = 1. \quad \blacksquare$$

19.4 Exercises

Solve the differential equations in Exercises 1–26.

1 $y' + 2y = e^{2x}$

2 $y' - 3y = 2$

3 $xy' - 3y = x^5$

4 $y' + y \cot x = \csc x$

5 $xy' + y + x = e^x$

- 6 $xy' + (1 + x)y = 5$
- 7 $x^2 dy + (2xy - e^x) dx = 0$
- 8 $x^2 dy + (x - 3xy + 1) dx = 0$
- 9 $y' + y \cot x = 4x^2 \csc x$
- 10 $y' + y \tan x = \sin x$
- 11 $(y \sin x - 2) dx + \cos x dy = 0$
- 12 $(x^2 y - 1) dx + x^3 dy = 0$
- 13 $(x^2 \cos x + y) dx - x dy = 0$
- 14 $y' + y = \sin x$
- 15 $xy' + (2 + 3x)y = xe^{-3x}$
- 16 $(x + 4)y' + 5y = x^2 + 8x + 16$
- 17 $xy' - y = x^3 y^4$
- 18 $xy' + 2y = 4x^4 y^4$
- 19 $(2xy - x^2 y^2) dx + (1 + x^2) dy = 0$

- 20 $y' + y = y^2 e^x$
- 21 $x^{-1} y' + 2y = 3$
- 22 $y' - 5y = e^{5x}$
- 23 $\tan x dy + (y - \sin x) dx = 0$
- 24 $\cos x dy - (y \sin x + e^{-x}) dx = 0$
- 25 $y' + 3x^2 y = x^2 + e^{-x^3}$
- 26 $y' + y \tan x = \cos^3 x$

In Exercises 27–30 find the particular solution of the differential equation that satisfies the given boundary conditions.

- 27 $xy' - y = x^2 + x; x = 1, y = 2$
- 28 $y' + 2y = e^{-3x}; x = 0, y = 2$
- 29 $xy' + y + xy = e^{-x}; x = 1, y = 0$
- 30 $y' + 2xy - e^{-x^2} = x; x = 0, y = 1$

19.5 Applications

Differential equations are indispensable in the physical sciences. Earlier in the text we considered elementary applications involving laws of exponential growth, motion, geometry, economics, and several other concepts. As problems become more difficult the corresponding differential equations become more complicated, and computer approximations are often employed.

In this section we shall consider several other applications of differential equations. Additional examples may be found in texts on differential equations or in advanced books on engineering or physics.

We have previously used antidifferentiation to derive laws of motion for falling bodies, assuming that air resistance could be neglected (see Example 6 in Section 4.9). This is a valid assumption for small, slowly moving objects; however, in many cases air resistance must be taken into account. Indeed, this frictional force often increases as the speed of the object increases. In the following example we shall derive the law of motion for a falling body under the assumption that the resistance due to the air is directly proportional to the speed of the body.

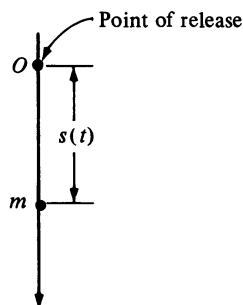


FIGURE 19.3

Example 1 An object of mass m is released from a balloon. Find the distance it falls in t seconds, if the force of resistance due to the air is directly proportional to the speed of the object.

Solution Let us introduce a vertical axis with positive direction downward and origin at the point of release as illustrated in Figure 19.3. We wish to find the distance $s(t)$ from the origin to the object at time t .

From our earlier work, the speed of the object is $v = s'(t)$ and the magnitude of the acceleration is $a = dv/dt = s''(t)$. If g is the gravitational constant, then the object is attracted toward the earth with a force of magnitude mg . By hypothesis, the force of resistance due to the air is kv for some constant k , and this force is directed opposite to the motion. It follows that the downward force F on the object is $mg - kv$. Since Newton's Second Law of Motion states that $F = ma = m(dv/dt)$, we arrive at the following differential equation:

$$m \frac{dv}{dt} = mg - kv$$

or
$$dv = \left(g - \frac{k}{m} v \right) dt.$$

If we denote the constant k/m by c , then the last equation may be written in the separable form

$$\frac{dv}{g - cv} = dt.$$

Integrating both sides we obtain

$$-\frac{1}{c} \ln(g - cv) = t + D$$

for some constant D . This may also be written as

$$\ln(g - cv) = -c(t + D)$$

or
$$g - cv = e^{-c(t+D)} = e^{-cD} e^{-ct}.$$

If we let $t = 0$, then $v = 0$ and hence $g = e^{-cD}$. Consequently,

$$g - cv = g e^{-ct}.$$

Solving for v gives us

$$v = \frac{g - g e^{-ct}}{c}$$

or, since $v = s'(t)$,

$$s'(t) = \frac{g}{c} - \frac{g}{c} e^{-ct}.$$

Integrating both sides of this equation we see that

$$s(t) = \frac{g}{c} t + \frac{g}{c^2} e^{-ct} + E.$$

The constant E may be found by letting $t = 0$. Since $s(0) = 0$, we obtain

$$0 = 0 + \frac{g}{c^2} + E, \quad \text{or} \quad E = -\frac{g}{c^2}.$$

Consequently, the distance the object falls in t seconds is

$$s(t) = \frac{g}{c} t + \frac{g}{c^2} e^{-ct} - \frac{g}{c^2}.$$

It is interesting to compare this formula for $s(t)$ with that obtained when the air resistance is neglected. In the latter case the differential equation $m(dv/dt) = mg - kv$ reduces to $dv/dt = g$ and it follows that $s'(t) = v = gt$. Integrating both sides leads to the much simpler formula $s(t) = \frac{1}{2}gt^2$. ■

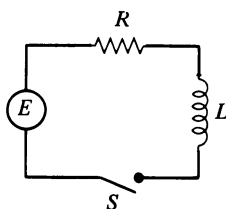


FIGURE 19.4

Example 2 A simple electrical circuit consists of a resistance R and an inductance L connected in series, as illustrated schematically in Figure 19.4, where E is a constant electromotive force. If the switch S is closed at $t = 0$, then it follows from one of Kirchhoff's rules for electrical circuits that if $t > 0$, the current i satisfies the differential equation

$$L \frac{di}{dt} + Ri = E.$$

Express i as a function of t .

Solution The given equation may be written

$$\frac{di}{dt} + \frac{R}{L} i = \frac{E}{L}$$

which is a first-order linear differential equation (see Definition (19.7)). Applying Theorem (19.8) we multiply both sides by the integrating factor

$$e^{\int (R/L) dt} = e^{(R/L)t}$$

obtaining
$$e^{(R/L)t} \frac{di}{dt} + \frac{R}{L} e^{(R/L)t} i = \frac{E}{L} e^{(R/L)t}.$$

Consequently, as in the previous section,

$$ie^{(R/L)t} = \int \frac{E}{L} e^{(R/L)t} dt = \frac{E}{R} e^{(R/L)t} + C.$$

Since $i = 0$ when $t = 0$, it follows that $C = -E/R$. Substituting for C leads to

$$ie^{(R/L)t} = \frac{E}{R} e^{(R/L)t} - \frac{E}{R}.$$

Finally, multiplying both sides by $e^{-(R/L)t}$ gives us

$$i = \frac{E}{R} [1 - e^{-(R/L)t}].$$

Observe that as t increases without bound i approaches E/R , which is the current when no inductance is present. In Exercise 11 you are asked to find a formula for i if the electromotive force is given by $E \sin \omega t$ where ω is a constant. ■

An **orthogonal trajectory** of a family of curves is a curve that intersects each curve of the family orthogonally. We shall restrict our discussion to curves in a coordinate plane. To illustrate, given the family $y = 2x + b$ of all lines of slope 2, every line $y = (-1/2)x + c$ of slope $-1/2$ is an orthogonal trajectory. We sometimes call two such families of curves **mutually orthogonal**. As another example, the family of all lines through the origin and the family of all concentric circles with centers at the origin are mutually orthogonal.

Pairs of mutually orthogonal families of curves occur frequently in physical applications of mathematics. In the theory of electricity and magnetism, the lines of force associated with a given field are orthogonal trajectories of the corresponding equipotential curves. Similarly, the stream lines studied in aerodynamics and hydrodynamics are orthogonal trajectories of the so-called *velocity-equipotential* curves. As a final illustration, in the study of thermodynamics the flow of heat across a plane surface is shown to be orthogonal to the isothermal curves.

The next example illustrates a technique for finding the orthogonal trajectories of a family of curves.

Example 3 Find the orthogonal trajectories of the family of ellipses $x^2 + 3y^2 = c$ and sketch several members of each family.

Solution Differentiating the given equation implicitly we obtain

$$2x + 6yy' = 0, \quad \text{or} \quad y' = -\frac{x}{3y}.$$

Hence the slope of the tangent line at any point (x, y) on one of the ellipses is $y' = -x/3y$. If dy/dx is the slope of the tangent line on a corresponding orthogonal trajectory, then it must equal the negative reciprocal of y' . This gives us the following differential equation for the family of orthogonal trajectories:

$$\frac{dy}{dx} = \frac{3y}{x}.$$

Separating the variables,

$$\frac{dy}{y} = 3 \frac{dx}{x}.$$

Integrating, and writing the constant of integration as $\ln |k|$ gives us

$$\ln |y| = 3 \ln |x| + \ln |k| = \ln |kx^3|.$$

It follows that $y = kx^3$ is an equation for the family of orthogonal trajectories. Several members of the given family and corresponding orthogonal trajectories are sketched (with dashes) in Figure 19.5. ■

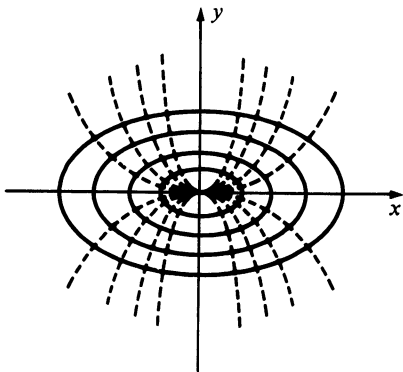


FIGURE 19.5 Orthogonal trajectories

19.5 Exercises

Find the orthogonal trajectories of the families of curves in Exercises 1–8 and sketch several members of each family.

- 1 $x^2 - y^2 = c$
- 2 $xy = c$
- 3 $y^2 = cx$
- 4 $y = cx^2$
- 5 $y^2 = cx^3$
- 6 $y = ce^{-x}$
- 7 $x^2 + y^2 = 2cx$
- 8 $x^2 + y^2 - 2cy = 0$

9 The differential equation $p \, dv + cv \, dp = 0$ describes the adiabatic change of state of air, where p and v are the pressure and volume, respectively, and c is a constant. Solve for p as a function of v .

10 The equation

$$R \frac{dQ}{dt} + \frac{Q}{C} = E$$

describes the charge Q on a condenser of capacity C during a charging process involving a resistance R and electromotive force E . If the charge is 0 when $t = 0$, express Q as a function of t .

11 Solve Example 2 for the case where the electromotive force is $E \sin \omega t$, where E and ω are constants.

12 The equation

$$R \frac{di}{dt} + \frac{i}{C} = \frac{dE}{dt}$$

describes an electrical circuit consisting of a resistance R and capacity C in series. Express i as a function of t if $i = i_0$ when $t = 0$, and (a) E is a constant;

(b) $E = E_0 \sin \omega t$ where E_0 and ω are constants.

13 Solve Example 1 if the force of resistance due to the air is directly proportional to the square of the speed of the object.

14 An object of mass m is moving rectilinearly subject to a force given by $F(t) = e^{-t}$, where t is time. The motion is resisted by a frictional force that is numerically equal to twice the speed of the object. If $v = 0$ at $t = 0$, find a formula for v at any time $t > 0$.

15 At time $t = 0$ a tank contains K lb of salt dissolved in 80 gal of water. Suppose that water containing $\frac{1}{3}$ lb of salt per gallon is being added to the tank at a rate of 6 gal/min, and that the well-stirred solution is being drained from the tank at the same rate. Find a formula for the amount $f(t)$ of salt in the tank at time t .

6 A room having dimensions 10 ft $\times$ 15 ft $\times$ 8 ft originally contains 0.001 percent carbon monoxide. Suppose that at

time $t = 0$, fumes containing 5 percent carbon monoxide begin entering the room at a rate of 0.12 ft³/min, and that the well-circulated mixture is eliminated from the room at the same rate.

(a) Find a formula for the volume $f(t)$ of carbon monoxide in the room at time t .

(b) After approximately how many minutes will the room contain the “dangerous-to-your-health” level of 0.015 percent carbon monoxide?

17 **Learning curves** are used to describe how certain skills are acquired. To illustrate, suppose a manufacturer estimates that a new employee will produce A items the first day on the job, and that as the proficiency of the employee increases, items will be produced more rapidly, until a maximum of M items per day is reached. Let $f(t)$ denote the number produced on day t , where $t \geq 1$. Suppose that the rate of production $f'(t)$ is proportional to $M - f(t)$.

(a) Find a formula for $f(t)$.

(b) If $M = 30$, $f(1) = 5$, and $f(2) = 8$, estimate the number of items produced on day 20.

18 In the study of lung physiology the following differential equation is used to describe the transport of a substance across a capillary wall:

$$\frac{dh}{dt} = -\frac{V}{Q} \left(\frac{h}{k+h} \right)$$

where h is the hormone concentration in the bloodstream, t is time, V is the maximum transport rate, Q is the volume of the capillary, and k is a constant that measures the affinity between the hormones and enzymes that assist with the transport process. Find the general solution of the differential equation.

19 In chemistry, the notation $A + B \rightarrow Y$ is used to denote the production of a substance Y due to the interaction of two substances A and B . Let a and b be the initial amounts of A and B , respectively. If, at time t , the concentration of Y is $y = f(t)$, then the concentrations of A and B are $a - f(t)$ and $b - f(t)$, respectively. If the rate at which the production of Y takes place is given by $dy/dt = k(a - y)(b - y)$, where k is a positive constant, and if $f(0) = 0$, find $f(t)$.

20 Let $y = f(t)$ be the population, at time t , of a collection of objects such as human beings, insects, or bacteria. If the rate of growth dy/dt is proportional to y , that is, if $dy/dt = cy$ for some positive constant c , then $f(t) = f(0)e^{ct}$ (see Theorem (7.25)). In most cases the rate of growth is dependent on available resources such as food

supplies, and as t becomes large, $f'(t)$ begins to decrease. An equation that is often used to describe this type of variation in population is the **Law of Logistic Growth** $dy/dt = y(c - by)$ where c and b are positive constants (cf. Exercise 15 of Section 7.6).

(a) If $f(0) = k$, find $f(t)$.

(b) Find $\lim_{t \rightarrow \infty} f(t)$.

(c) Show that $f'(t)$ is increasing if $f(t) < c/2b$ and is decreasing if $f(t) > c/2b$.

(d) Sketch a typical graph of f . (A graph of this type is called a **logistic curve**.)

19.6 Second-Order Linear Differential Equations

Definition (19.10)

Let $f_1, f_2, \dots, f_n$ and k be functions of one variable that have the same domain. A **linear differential equation of order n** is an equation of the form

$$y^{(n)} + f_1(x)y^{(n-1)} + \cdots + f_{n-1}(x)y' + f_n(x)y = k(x).$$

If $k(x) = 0$ for all x , the equation is said to be **homogeneous**.

Notice that the meaning of the word *homogeneous* is different from that in Section 19.3. If $k(x) \neq 0$ for some x , the equation is said to be **nonhomogeneous**. A thorough analysis of (19.10) may be found in textbooks on differential equations. We shall restrict our work to second-order equations in which f_1 and f_2 are constant functions. In this section we shall consider the homogeneous case. Nonhomogeneous equations will be discussed in the next section.

The general second-order homogeneous linear differential equation with constant coefficients has the form

$$y'' + by' + cy = 0$$

where b and c are constants. Before attempting to find particular solutions let us establish the following result.

Theorem (19.11)

If $y = f(x)$ and $y = g(x)$ are solutions of $y'' + by' + cy = 0$, then

$$y = C_1 f(x) + C_2 g(x)$$

is a solution for all real numbers C_1 and C_2 .

Proof By hypothesis,

$$f''(x) + bf'(x) + cf(x) = 0$$

$$g''(x) + bg'(x) + cg(x) = 0.$$

If we multiply the first of these equations by C_1 , the second by C_2 , and add, the result is

$$[C_1 f''(x) + C_2 g''(x)] + b[C_1 f'(x) + C_2 g'(x)] + c[C_1 f(x) + C_2 g(x)] = 0.$$

Thus $C_1 f(x) + C_2 g(x)$ is a solution. $\square$

It can be shown that if the solutions f and g in Theorem (19.11) have the property that $f(x) \neq Cg(x)$ for every real number C , and if $g(x)$ is not identically 0, then $y = C_1 f(x) + C_2 g(x)$ is the general solution of the differential equation $y'' + by' + cy = 0$. Thus, to determine the general solution it is sufficient to find two such functions f and g and employ (19.11).

In our search for a solution of $y'' + by' + cy = 0$ we shall use $y = e^{mx}$ as a trial solution. Since $y' = me^{mx}$ and $y'' = m^2 e^{mx}$, it follows that $y = e^{mx}$ is a solution if and only if

$$m^2 e^{mx} + bme^{mx} + ce^{mx} = 0$$

or, since $e^{mx} \neq 0$, if and only if

$$(19.12) \quad m^2 + bm + c = 0.$$

Equation (19.12) is called the **auxiliary equation** of $y'' + by' + cy = 0$. It can be obtained from this differential equation by replacing y'' by m^2 , y' by m , and y by 1. In simple cases the roots of the auxiliary equation can be found by factoring. If a factorization is not evident, then applying the quadratic formula, we see that the roots are

$$m = \frac{-b \pm \sqrt{b^2 - 4c}}{2}.$$

Thus the auxiliary equation has unequal real roots m_1 and m_2 , a double real root m , or two complex conjugate roots according as $b^2 - 4c$ is positive, zero, or negative, respectively. The next theorem is a consequence of the remark following the proof of Theorem (19.11).

Theorem (19.13)

If the roots m_1, m_2 of the auxiliary equation are real and unequal, then the general solution of $y'' + by' + cy = 0$ is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}.$$

Example 1 Solve the differential equation $y'' - 3y' - 10y = 0$.

Solution The auxiliary equation is $m^2 - 3m - 10 = 0$, or equivalently, $(m - 5)(m + 2) = 0$. Since the roots $m_1 = 5$ and $m_2 = -2$ are real and unequal, it follows from Theorem (19.13) that the general solution is

$$y = C_1 e^{5x} + C_2 e^{-2x}. \quad \blacksquare$$

Theorem (19.14)

If the auxiliary equation has a double root m , then the general solution of $y'' + by' + cy = 0$ is

$$y = C_1 e^{mx} + C_2 x e^{mx}.$$

Proof The roots of $m^2 + bm + c = 0$ are $m = (-b \pm \sqrt{b^2 - 4c})/2$. If $b^2 - 4c = 0$, we obtain $m = -b/2$ or $2m + b = 0$. Since m satisfies the auxiliary equation, $y = e^{mx}$ is a solution of the differential equation. According to the remark following the proof of Theorem (19.11), it is sufficient to show that $y = x e^{mx}$ is also a solution. Substitution of $x e^{mx}$ for y in the equation $y'' + by' + cy = 0$ gives us

$$\begin{aligned} (2me^{mx} + m^2 x e^{mx}) + b(mx e^{mx} + e^{mx}) + c x e^{mx} \\ = (m^2 + bm + c)x e^{mx} + (2m + b)e^{mx} \\ = 0x e^{mx} + 0e^{mx} = 0 \end{aligned}$$

which is what we wished to show. □

Example 2 Solve the differential equation $y'' - 6y' + 9y = 0$.

Solution The auxiliary equation $m^2 - 6m + 9 = 0$, or equivalently $(m - 3)^2 = 0$, has a double root 3. Hence, by Theorem (19.14), the general solution is

$$y = C_1 e^{3x} + C_2 x e^{3x} = e^{3x}(C_1 + C_2 x). \quad \blacksquare$$

We may also consider second-order differential equations of the form

$$ay'' + by' + cy = 0$$

where $a \neq 1$. It is possible to obtain the form stated in Theorems (19.13) and (19.14) by dividing both sides by a ; however, it is usually simpler to employ the auxiliary equation

$$am^2 + bm + c = 0$$

as illustrated in the next example.

Example 3 Solve the differential equation $6y'' - 7y' + 2y = 0$.

Solution The auxiliary equation $6m^2 - 7m + 2 = 0$ can be factored as follows:

$$(2m - 1)(3m - 2) = 0.$$

Hence the roots are $m_1 = \frac{1}{2}$ and $m_2 = \frac{2}{3}$. Applying Theorem (19.13), the general solution of the given equation is

$$y = C_1 e^{x/2} + C_2 e^{2x/3}. \quad \blacksquare$$

The final case to consider is that in which the roots of the auxiliary equation $m^2 + bm + c = 0$ of $y'' + by' + cy = 0$ are complex numbers. Recall that complex numbers may be represented by expressions of the form $a + bi$, where a and b are real numbers, and i is a symbol that may be manipulated in the same manner as a real number, but has the additional property that $i^2 = -1$. Two complex numbers $a + bi$ and $c + di$ are said to be **equal**, and we write $a + bi = c + di$, if and only if $a = c$ and $b = d$. Operations of addition, subtraction, multiplication, and division are defined *just as though* all letters denote real numbers, with the additional stipulation that whenever i^2 occurs, it may be replaced by -1 . For example, the formulas for addition and multiplication of two complex numbers $a + bi$ and $c + di$ are

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

We may regard the real numbers as a subset of the complex numbers by identifying the real number a with the complex number $a + 0i$. A complex number of the form $0 + bi$ is abbreviated bi .

Complex numbers are often required for solving equations of the form $f(x) = 0$, where $f(x)$ is a polynomial. For example, if only real numbers are allowed, then the equation $x^2 = -4$ has no solutions. However, if complex numbers are available, then the equation has a solution $2i$, since

$$(2i)^2 = 2^2 i^2 = 4(-1) = -4.$$

Similarly, $-2i$ is a solution of $x^2 = -4$.

Since $i^2 = -1$, we sometimes use the symbol $\sqrt{-1}$ in place of i and write

$$\sqrt{-13} = \sqrt{13}i, \quad 2 + \sqrt{-25} = 2 + \sqrt{25}i = 2 + 5i$$

etc. A *quadratic equation* $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$, has roots given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

If $b^2 - 4ac < 0$, then the roots are complex numbers. To illustrate, if we apply the quadratic formula to the equation $x^2 - 4x + 13 = 0$, we obtain

$$x = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i.$$

Thus the equation has the two complex roots $2 + 3i$ and $2 - 3i$.

The complex number $a - bi$ is called the **conjugate** of the complex number $a + bi$. We see from the quadratic formula that if a quadratic equation with real coefficients has complex roots, then they are necessarily conjugates of one another.

It follows from the preceding discussion that if the auxiliary equation $m^2 + bm + c = 0$ of $y'' + by' + cy = 0$ has complex roots, then they are of the form

$$z_1 = s + ti \quad \text{and} \quad z_2 = s - ti$$

where s and t are real numbers. We may anticipate, from Theorem (19.13), that the general solution of the differential equation is $y = C_1 e^{s_1 x} + C_2 e^{s_2 x}$, or

$$(19.15) \quad y = C_1 e^{(s+ti)x} + C_2 e^{(s-ti)x}.$$

To handle such complex exponents it is necessary to extend some of the concepts of calculus to include functions whose domains include complex numbers. Since a complete development is beyond the scope of our work, we shall merely outline the main ideas.

In Section 11.8 we discussed how certain functions can be represented by power series. It is not difficult to extend the definitions and theorems of Chapter 11 to infinite series that involve complex numbers. Since this is true, we shall use the power series representations (11.42) to *define* e^z , $\sin z$, and $\cos z$ for every complex number z as follows:

$$(19.16) \quad \begin{aligned} e^z &= 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!} + \cdots \\ \sin z &= z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots + (-1)^n \frac{z^{2n+1}}{(2n+1)!} + \cdots \\ \cos z &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots + (-1)^n \frac{z^{2n}}{(2n)!} + \cdots \end{aligned}$$

Using the first formula in (19.16) gives us

$$\begin{aligned} e^{iz} &= 1 + (iz) + \frac{(iz)^2}{2!} + \frac{(iz)^3}{3!} + \frac{(iz)^4}{4!} + \frac{(iz)^5}{5!} + \cdots \\ &= 1 + iz + i^2 \frac{z^2}{2!} + i^3 \frac{z^3}{3!} + i^4 \frac{z^4}{4!} + i^5 \frac{z^5}{5!} + \cdots \end{aligned}$$

Since $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, etc., we see that

$$e^{iz} = 1 + iz - \frac{z^2}{2!} - i \frac{z^3}{3!} + \frac{z^4}{4!} + i \frac{z^5}{5!} - \cdots$$

which may also be written in the form

$$e^{iz} = \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots\right) + i \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right).$$

If we now use the formulas for $\cos z$ and $\sin z$ in (19.16), we obtain the following important result.

Euler's Formula (19.17)

For every complex number z ,

$$e^{iz} = \cos z + i \sin z.$$

It can be shown that the Laws of Exponents are true for complex numbers. In addition, formulas for derivatives developed earlier in this text may be

extended to functions of a *complex* variable z . For example, $D_z e^{kz} = k e^{kz}$, where k is a complex number. It can then be proved that the general solution of $y'' + by' + c = 0$, where the roots of the auxiliary equation are the complex numbers $s \pm ti$, is given by (19.15). The form of this solution may be changed as follows:

$$\begin{aligned} y &= C_1 e^{(s+ti)x} + C_2 e^{(s-ti)x} \\ &= C_1 e^{sx+txi} + C_2 e^{sx-txi} \\ &= C_1 e^{sx} e^{txi} + C_2 e^{sx} e^{-txi} \end{aligned}$$

or equivalently,

$$(19.18) \quad y = e^{sx}(C_1 e^{itx} + C_2 e^{-itx}).$$

This can be further simplified by using Euler's Formula. Specifically, we see from (19.17) that

$$\begin{aligned} e^{itx} &= \cos tx + i \sin tx, \\ e^{-itx} &= \cos tx - i \sin tx \end{aligned}$$

from which it follows that

$$\cos tx = \frac{e^{itx} + e^{-itx}}{2}, \quad \sin tx = \frac{e^{itx} - e^{-itx}}{2i}.$$

If we let $C_1 = C_2 = \frac{1}{2}$ in (19.18) we obtain

$$y = \frac{1}{2} e^{sx}(e^{itx} + e^{-itx}).$$

Applying the preceding formula for $\cos tx$ gives us

$$y = \frac{1}{2} e^{sx}(2 \cos tx) = e^{sx} \cos tx.$$

Thus, $y = e^{sx} \cos tx$ is a particular solution of $y'' + by' + cy = 0$. Letting $C_1 = -C_2 = i/2$ leads to the particular solution $y = e^{sx} \sin tx$. This is a partial proof of the next theorem.

Theorem (19.19)

If the auxiliary equation $m^2 + bm + c = 0$ has distinct complex roots $s \pm ti$, then the general solution of $y'' + by' + cy = 0$ is

$$y = e^{sx}(C_1 \cos tx + C_2 \sin tx).$$

Example 4 Solve the differential equation $y'' - 10y' + 41y = 0$.

Solution The roots of the auxiliary equation $m^2 - 10m + 41 = 0$ are

$$m = \frac{10 \pm \sqrt{100 - 164}}{2} = \frac{10 \pm \sqrt{-64}}{2} = \frac{10 \pm 8i}{2} = 5 \pm 4i.$$

Hence by Theorem (19.19), the general solution of the differential equation is

$$y = e^{5x}(C_1 \cos 4x + C_2 \sin 4x). \quad \blacksquare$$

19.6 Exercises

Solve the differential equations in Exercises 1–22.

- | | |
|--|--|
| 1 $y'' - 5y' + 6y = 0$ | 2 $y'' - y' - 2y = 0$ |
| 3 $y'' - 3y' = 0$ | 4 $y'' + 6y' + 8y = 0$ |
| 5 $y'' + 4y' + 4y = 0$ | 6 $y'' - 4y' + 4y = 0$ |
| 7 $y'' - 4y' + y = 0$ | 8 $6y'' - 7y' - 3y = 0$ |
| 9 $y'' + 2\sqrt{2}y' + 2y = 0$ | 10 $4y'' + 20y' + 25y = 0$ |
| 11 $8y'' + 2y' - 15y = 0$ | 12 $y'' + 4y' + y = 0$ |
| 13 $9y'' - 24y' + 16y = 0$ | 14 $4y'' - 8y' + 7y = 0$ |
| 15 $2y'' - 4y' + y = 0$ | 16 $2y'' + 7y' = 0$ |
| 17 $y'' - 2y' + 2y = 0$ | 18 $y'' - 2y' + 5y = 0$ |
| 19 $y'' - 4y' + 13y = 0$ | 20 $y'' + 4 = 0$ |
| 21 $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 2y = 0$ | 22 $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 6y = 0$ |

In Exercises 23–30 find the particular solution of the differential equation that satisfies the stated boundary conditions.

- 23 $y'' - 3y' + 2y = 0$; $y = 0$ and $y' = 2$ when $x = 0$
- 24 $y'' - 2y' + y = 0$; $y = 1$ and $y' = 2$ when $x = 0$
- 25 $y'' + y = 0$; $y = 1$ and $y' = 2$ when $x = 0$
- 26 $y'' - y' - 6y = 0$; $y = 0$ and $y' = 1$ when $x = 0$
- 27 $y'' + 8y' + 16y = 0$; $y = 2$ and $y' = 1$ when $x = 0$
- 28 $y'' + 5y = 0$; $y = 4$ and $y' = 2$ when $x = 0$
- 29 $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 0$; $y = 0$ and $\frac{dy}{dx} = 1$ when $x = 0$
- 30 $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$; $y = 2$ and $\frac{dy}{dx} = 3$ when $x = 0$

19.7 Nonhomogeneous Linear Differential Equations

In this section we shall consider second-order nonhomogeneous linear differential equations with constant coefficients, that is, equations of the form

$$y'' + by' + cy = k(x)$$

where b and c are constants and the function k is continuous.

It is convenient to use the differential operator symbols D and D^2 where if $y = f(x)$, then

$$Dy = y' = f'(x) \quad \text{and} \quad D^2y = y'' = f''(x).$$

Definition (19.20)

If $y = f(x)$ and f'' exists, then the **linear differential operator** $L = D^2 + bD + c$ is defined by

$$L(y) = (D^2 + bD + c)y = D^2y + bDy + cy = y'' + by' + cy.$$

Using L , the differential equation $y'' + by' + cy = k(x)$ may be written in the compact form $L(y) = k(x)$. In Exercises 19 and 20 you are asked to verify that for every real number C ,

$$L(Cy) = CL(y)$$

and that if $y_1 = f_1(x)$ and $y_2 = f_2(x)$, then

$$L(y_1 \pm y_2) = L(y_1) \pm L(y_2).$$

Given the differential equation $y'' + by' + cy = k(x)$, that is, $L(y) = k(x)$, the corresponding homogeneous equation $L(y) = 0$ is called the **complementary equation**.

Theorem (19.21)

Let $y'' + by' + cy = k(x)$ be a second-order nonhomogeneous linear differential equation. If y_p is a particular solution of $L(y) = k(x)$ and if y_c is the general solution of the complementary equation $L(y) = 0$, then the general solution of $L(y) = k(x)$ is $y = y_p + y_c$.

Proof Since $L(y_p) = k(x)$ and $L(y_c) = 0$,

$$L(y_p + y_c) = L(y_p) + L(y_c) = k(x) + 0 = k(x)$$

which means that $y_p + y_c$ is a solution of $L(y) = k(x)$. Moreover, if $y = f(x)$ is any other solution, then

$$L(y - y_p) = L(y) - L(y_p) = k(x) - k(x) = 0.$$

Consequently, $y - y_p$ is a solution of the complementary equation. Thus $y - y_p = y_c$ for some y_c , which is what we wished to prove. $\square$

If we use the results of the previous section to find the general solution y_c of $L(y) = 0$, then according to Theorem (19.21) all that is needed to determine the general solution of $L(y) = k(x)$ is *one* particular solution y_p .

Example 1 Solve the differential equation $y'' - 4y = 6x - 4x^3$.

Solution We see by inspection that $y_p = x^3$ is a particular solution of the given equation. The complementary equation is $y'' - 4y = 0$, which by Theorem (19.13) has the general solution

$$y_c = C_1 e^{2x} + C_2 e^{-2x}.$$

Applying Theorem (19.21), the general solution of the given nonhomogeneous equation is

$$y = C_1 e^{2x} + C_2 e^{-2x} + x^3. \quad \blacksquare$$

If a particular solution of $y'' + by' + cy = k(x)$ cannot be found by inspection, then the following technique, called *variation of parameters*, may be employed. Given the differential equation $L(y) = k(x)$, let y_1 and y_2 be the expressions that appear in the general solution $y = C_1 y_1 + C_2 y_2$ of

the complementary equation $L(y) = 0$. For example we might have, as in (19.13), $y_1 = e^{m_1 x}$ and $y_2 = e^{m_2 x}$. Let us now attempt to find a particular solution of $L(y) = k(x)$ which has the form

$$y_p = uy_1 + vy_2$$

where $u = g(x)$ and $v = h(x)$ for some functions g and h . The first and second derivatives of y_p are

$$y'_p = (uy'_1 + vy'_2) + (u'y_1 + v'y_2)$$

$$y''_p = (uy''_1 + vy''_2) + (u'y'_1 + v'y'_2) + (u'y_1 + v'y_2)'.$$

Substituting these into $L(y_p) = y''_p + by'_p + cy_p$ and rearranging terms, we obtain

$$\begin{aligned} L(y_p) &= u(y''_1 + by'_1 + cy_1) + v(y''_2 + by'_2 + cy_2) \\ &\quad + b(u'y_1 + v'y_2) + (u'y_1 + v'y_2)' + (u'y_1 + v'y_2)'. \end{aligned}$$

Since y_1 and y_2 are solutions of $y'' + by' + cy = 0$, the first two terms on the right are 0. Hence, to obtain $L(y_p) = k(x)$, it is sufficient to choose u and v such that

$$\begin{aligned} u'y_1 + v'y_2 &= 0 \\ u'y_1 + v'y_2 &= k(x). \end{aligned}$$

It can be shown that this system of equations always has a unique solution u' and v' . We may then determine u and v by integration and use the fact that $y_p = uy_1 + vy_2$ to find a particular solution of the differential equation. Our discussion may be summarized as follows.

Variation of Parameters (19.22)

If $y = C_1y_1 + C_2y_2$ is the general solution of the complementary equation $L(y) = 0$ of $y'' + by' + cy = k(x)$, then a particular solution of $L(y) = k(x)$ is $y_p = uy_1 + vy_2$, where $u = g(x)$ and $v = h(x)$ satisfy the following system of equations:

$$\begin{aligned} u'y_1 + v'y_2 &= 0 \\ u'y_1 + v'y_2 &= k(x). \end{aligned}$$

Example 2 Solve the differential equation $y'' + y = \cot x$.

Solution The complementary equation is $y'' + y = 0$. Since the auxiliary equation $m^2 + 1 = 0$ has roots $\pm i$, we see from Theorem (19.19) that the general solution of the homogeneous differential equation $y'' + y = 0$ is $y = C_1 \cos x + C_2 \sin x$. As in the preceding discussion, we let $y_1 = \cos x$ and $y_2 = \sin x$. The system of equations in Theorem (19.22) is, therefore,

$$\begin{aligned} u' \cos x + v' \sin x &= 0 \\ -u' \sin x + v' \cos x &= \cot x. \end{aligned}$$

Solving for u' and v' gives us

$$u' = -\cos x, \quad v' = \csc x - \sin x.$$

If we integrate each of these expressions (and drop the constants of integration) we obtain

$$u = -\sin x, \quad v = \ln |\csc x - \cot x| + \cos x.$$

Applying Theorem (19.22), a particular solution of the given equation is

$$y_p = -\sin x \cos x + \sin x \ln |\csc x - \cot x| + \sin x \cos x$$

or
$$y_p = \sin x \ln |\csc x - \cot x|.$$

Finally, by Theorem (19.21), the general solution of $y'' + y = \cot x$ is

$$y = C_1 \cos x + C_2 \sin x + \sin x \ln |\csc x - \cot x|. \quad \blacksquare$$

Given the differential equation

$$L(y) = y'' + by' + cy = e^{nx}$$

where e^{nx} is not a solution of $L(y) = 0$, it is reasonable to expect that there exists a particular solution of the form $y_p = Ae^{nx}$, since e^{nx} is the result of finding $L(Ae^{nx})$. This suggests that we use Ae^{nx} as a trial solution in the given equation and attempt to find the value of the coefficient A . This technique is called the **method of undetermined coefficients**, and is illustrated in the next example.

Example 3 Solve the differential equation $y'' + 2y' - 8y = e^{3x}$.

Solution Since the auxiliary equation $m^2 + 2m - 8 = 0$ of the differential equation $y'' + 2y' - 8y = 0$ has roots 2 and -4 , it follows from the previous section that the general solution of the complementary equation is

$$y_c = C_1 e^{2x} + C_2 e^{-4x}.$$

From the preceding remarks we seek a particular solution of the form $y_p = Ae^{3x}$. Since $y_p' = 3Ae^{3x}$ and $y_p'' = 9Ae^{3x}$, substitution in the given equation leads to

$$9Ae^{3x} + 6Ae^{3x} - 8Ae^{3x} = e^{3x}.$$

Dividing both sides by e^{3x} , we obtain

$$9A + 6A - 8A = 1, \quad \text{or} \quad A = \frac{1}{7}.$$

Thus $y_p = \frac{1}{7}e^{3x}$ and by Theorem (19.21), the general solution is

$$y = C_1 e^{2x} + C_2 e^{-4x} + \frac{1}{7}e^{3x}. \quad \blacksquare$$

Three rules for arriving at trial solutions of second-order nonhomogeneous differential equations with constant coefficients are stated without proof in the next theorem. The reader is referred to texts on differential equations for a more extensive treatment of this topic.

Theorem (19.23)

- (i) If $y'' + by' + cy = e^{nx}$, and n is not a root of the auxiliary equation $m^2 + bm + c = 0$, then there is a particular solution of the form $y_p = Ae^{nx}$.
- (ii) If $y'' + by' + cy = xe^{nx}$, and n is not a solution of the auxiliary equation $m^2 + bm + c = 0$, then there is a particular solution of the form $y_p = (A + Bx)e^{nx}$.
- (iii) If either

$$y'' + by' + cy = e^{sx} \sin tx$$

or $y'' + by' + cy = e^{sx} \cos tx$

and the complex number $s + ti$ is not a solution of the auxiliary equation $m^2 + bm + c = 0$, then there is a particular solution of the form

$$y_p = Ae^{sx} \cos tx + Be^{sx} \sin tx.$$

Rule (i) of Theorem (19.23) was used in the solution of Example 3. Illustrations of rules (ii) and (iii) are given in the following examples.

Example 4 Solve $y'' - 3y' - 18y = xe^{4x}$.

Solution Since the auxiliary equation $m^2 - 3m - 18 = 0$ has roots 6 and -3 , it follows from the preceding section that the general solution of $y'' - 3y' - 18y = 0$ is

$$y = C_1 e^{6x} + C_2 e^{-3x}.$$

Since 4 is not a root of the auxiliary equation, we see from (ii) of Theorem (19.23) that there is a particular solution of the form

$$y_p = (A + Bx)e^{4x}.$$

Differentiating, we obtain

$$y'_p = (4A + 4Bx + B)e^{4x}$$

$$y''_p = (16A + 16Bx + 8B)e^{4x}.$$

Substitution in the given differential equation produces

$$(16A + 16Bx + 8B)e^{4x} - 3(4A + 4Bx + B)e^{4x} - 18(A + Bx)e^{4x} = xe^{4x}$$

which reduces to

$$-14A + 5B - 14Bx = x.$$

Thus y_p is a solution provided

$$-14A + 5B = 0 \quad \text{and} \quad -14B = 1.$$

This gives us $B = -\frac{1}{14}$ and $A = -\frac{5}{196}$. Consequently,

$$y_p = \left(-\frac{5}{196} - \frac{1}{14}x \right) e^{4x} = -\frac{1}{196} (5 + 14x) e^{4x}.$$

Applying Theorem (19.21), the general solution is

$$y = C_1 e^{6x} + C_2 e^{-3x} - \frac{1}{196} (5 + 14x) e^{4x}. \quad \blacksquare$$

Example 5 Solve $y'' - 10y' + 41y = \sin x$.

Solution The general solution $y_c = e^{5x}(C_1 \cos 4x + C_2 \sin 4x)$ of the complementary equation was found in Example 4 of the preceding section. Referring to (iii) of Theorem (19.23) with $s = 0$ and $t = 1$, we seek a particular solution of the form

$$y_p = A \cos x + B \sin x.$$

Since

$$y'_p = -A \sin x + B \cos x \quad \text{and} \quad y''_p = -A \cos x - B \sin x,$$

substitution in the given equation produces

$$\begin{aligned} -A \cos x - B \sin x + 10A \sin x - 10B \cos x \\ + 41A \cos x + 41B \sin x = \sin x \end{aligned}$$

which can be written

$$(40A - 10B) \cos x + (10A + 40B) \sin x = \sin x.$$

Consequently, y_p is a solution provided

$$40A - 10B = 0 \quad \text{and} \quad 10A + 40B = 1.$$

The solution of this system of equations is $A = \frac{1}{170}$ and $B = \frac{4}{170}$. Hence

$$y_p = \frac{1}{170} \cos x + \frac{4}{170} \sin x = \frac{1}{170} (\cos x + 4 \sin x)$$

and the general solution is

$$y = e^{5x}(C_1 \cos 4x + C_2 \sin 4x) + \frac{1}{170} (\cos x + 4 \sin x). \quad \blacksquare$$

19.7 Exercises

Solve the differential equations in Exercises 1–10 by using the method of variation of parameters.

- 1 $y'' + y = \tan x$
- 2 $y'' + y = \sec x$
- 3 $y'' - 6y' + 9y = x^2 e^{3x}$
- 4 $y'' + 3y' = e^{-3x}$
- 5 $y'' - y = e^x \cos x$
- 6 $y'' - 4y' + 4y = x^{-2} e^{2x}$
- 7 $y'' - 9y = e^{3x}$
- 8 $y'' + y = \sin x$
- 9 $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} - 4y = 2$
- 10 $\frac{d^2 y}{dx^2} - \frac{dy}{dx} = x + 1$

Solve the differential equations in Exercises 11–18 by using undetermined coefficients.

- 11 $y'' - 3y' + 2y = 4e^{-x}$
- 12 $y'' + 6y' + 9y = e^{2x}$
- 13 $y'' + 2y' = \cos 2x$
- 14 $y'' + y = 5 \sin x$
- 15 $y'' - y = x e^{2x}$
- 16 $y'' + 3y' - 4y = x e^{-x}$
- 17 $\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 13y = e^x \cos x$
- 18 $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 2y = e^{-x} \sin 2x$

Prove the identities in Exercises 19 and 20, where L is the linear differential operator (19.20) and y, y_1, y_2 represent functions of x .

- 19 $L(Cy) = CL(y)$
- 20 $L(y_1 \pm y_2) = L(y_1) \pm L(y_2)$

19.8 Vibrations

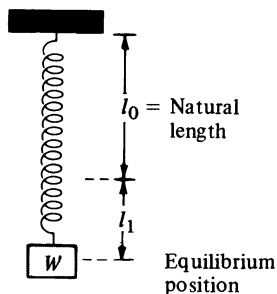


FIGURE 19.6

Vibrations in mechanical systems are caused by external forces. Thus, a violin string vibrates if bowed, a steel beam vibrates if struck by a hammer, and a bridge vibrates if a marching band crosses it in cadence. In this section we shall use differential equations to analyze vibrations that may occur in a spring.

According to Hooke's Law, the force required to stretch a spring y units beyond its natural length is ky , here k is a positive real number called the **spring constant** (see (6.11)). The **restoring force** of the spring is $-ky$. Suppose that a weight W is attached to the spring and, at the equilibrium position, the spring is stretched a distance l_1 beyond its natural length l_0 , as illustrated in Figure 19.6. If g is the gravitational constant and m is the mass of the weight, then $W = mg$, and at the equilibrium position,

$$mg = kl_1, \quad \text{or} \quad mg - kl_1 = 0$$

where we have assumed that the mass of the spring is negligible compared to m .

Suppose the weight is pulled down and released. Consider the coordinate line shown in Figure 19.7, where y denotes the directed distance from the equilibrium point to the center of mass of the weight after t seconds. By

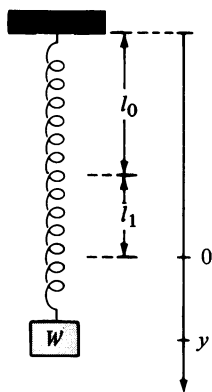


FIGURE 19.7

Newton's Second Law of Motion, the force acting on the weight is $F = ma$, where a is its acceleration. Assuming there is no external retarding force and the weight moves in a frictionless medium, we see that

$$F = mg - k(l_1 + y) = mg - kl_1 - ky = -ky.$$

Since $a = d^2y/dt^2$, this implies that

$$m \frac{d^2y}{dt^2} = -ky, \quad \text{or} \quad \frac{d^2y}{dt^2} + \frac{k}{m}y = 0.$$

If, as is customary, we denote k/m by ω^2 , the preceding equation may be written

$$(19.24) \quad \frac{d^2y}{dt^2} + \omega^2y = 0.$$

Since the solutions of the auxiliary equation are $\pm \omega i$ (Why?), it follows from Theorem (19.19) that the general solution of the differential equation is

$$y = C_1 \cos \omega t + C_2 \sin \omega t.$$

Thus the weight is in simple harmonic motion (see Exercise 38 of Section 4.7).

Suppose, for example, that the weight is pulled down a distance l_2 and is released with zero velocity. In this case, at $t = 0$,

$$l_2 = C_1(1) + C_2(0) \quad \text{or} \quad C_1 = l_2.$$

Since

$$\frac{dy}{dt} = -\omega C_1 \sin \omega t + \omega C_2 \cos \omega t$$

we also have (at $t = 0$),

$$0 = -\omega C_1(0) + \omega C_2(1) \quad \text{or} \quad C_2 = 0.$$

Hence the displacement y of the weight at time t is given by

$$y = l_2 \cos \omega t.$$

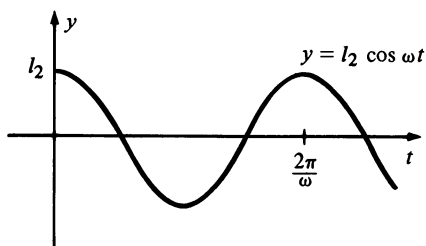


FIGURE 19.8 Simple harmonic motion

This type of motion was discussed earlier in the text (see Definition (4.29)). The **amplitude** (the maximum displacement of W) is l_2 , and the **period** (the time for one complete vibration) is $2\pi/\omega = 2\pi\sqrt{m/k}$. A typical graph indicating this type of motion is illustrated in Figure 19.8.

Example 1 An 8-pound weight stretches a spring 2 ft beyond its natural length. The weight is then pulled down another $\frac{1}{2}$ foot and released with an initial upward velocity of 6 ft/sec. Find a formula for the displacement of the weight at any time t .

Solution From Hooke's Law, $8 = k(2)$, or $k = 4$. If y is the displacement of the weight from the equilibrium position at time t , then

$$m \frac{d^2 y}{dt^2} = -4y.$$

Since $W = mg$ it follows that $m = W/g = \frac{8}{32} = \frac{1}{4}$. Hence

$$\frac{1}{4} \frac{d^2 y}{dt^2} = -4y, \quad \text{or} \quad \frac{d^2 y}{dt^2} + 16y = 0$$

which has the form of (19.24). As in our discussion, this implies that

$$y = C_1 \cos 4t + C_2 \sin 4t.$$

At $t = 0$ we have $y = \frac{1}{2}$ and therefore

$$\frac{1}{2} = C_1(1) + C_2(0), \quad \text{or} \quad C_1 = \frac{1}{2}.$$

Since

$$\frac{dy}{dt} = -4C_1 \sin 4t + 4C_2 \cos 4t$$

and $dy/dt = -6$ at $t = 0$, we obtain

$$-6 = -4C_1(0) + 4C_2(1), \quad \text{or} \quad C_2 = -\frac{3}{2}.$$

Hence the displacement at time t is given by

$$y = \frac{1}{2} \cos 4t - \frac{3}{2} \sin 4t. \quad \blacksquare$$

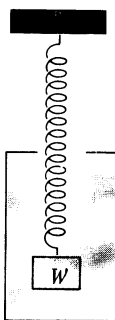


FIGURE 19.9 Damping force

Let us next consider the motion of the spring if a damping (or frictional) force is present, as is the case when the weight moves through a fluid (see Figure 19.9). A shock absorber in an automobile is a good illustration of this situation. We shall assume that the direction of the damping force is opposite to that of the motion and that it is directly proportional to the velocity of the weight. Thus, the damping force is given by $-c(dy/dt)$, where c is a positive constant. According to Newton's Second Law, the differential equation that describes the motion is

$$m \frac{d^2 y}{dt^2} = -ky - c \frac{dy}{dt}, \quad \text{or} \quad \frac{d^2 y}{dt^2} + \frac{c}{m} \frac{dy}{dt} + \frac{k}{m} y = 0.$$

This equation, called the differential equation of **free, damped vibration**, is often written in the form

$$(19.25) \quad \frac{d^2 y}{dt^2} + 2p \frac{dy}{dt} + \omega^2 y = 0$$

where $2p = c/m$ and $\omega^2 = k/m$. The roots of the auxiliary equation are

$$\frac{-2p \pm \sqrt{4p^2 - 4\omega^2}}{2} = -p \pm \sqrt{p^2 - \omega^2}.$$

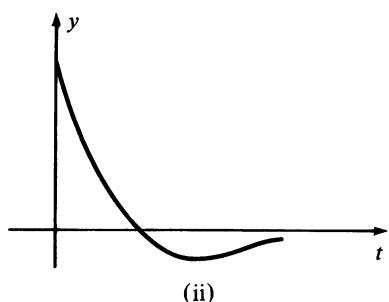
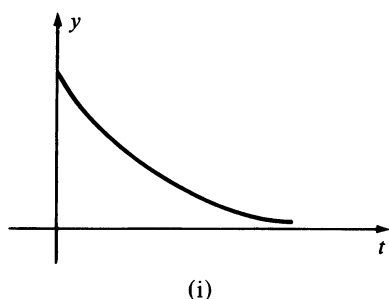


FIGURE 19.10 Overdamped

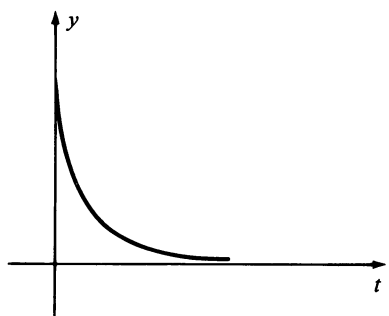


FIGURE 19.11 Critically damped

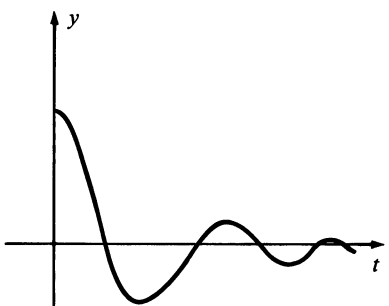


FIGURE 19.12 Underdamped

The following three possibilities may arise:

$$p^2 - \omega^2 > 0, \quad p^2 - \omega^2 = 0, \quad \text{or} \quad p^2 - \omega^2 < 0.$$

Let us classify the corresponding types of motion as follows:

Case (i) $p^2 - \omega^2 > 0$: Overdamped.

By Theorem (19.13), the general solution of the differential equation may be written

$$y = e^{-pt}(C_1 e^{\sqrt{p^2 - \omega^2}t} + C_2 e^{-\sqrt{p^2 - \omega^2}t}).$$

In the case under consideration,

$$p^2 - \omega^2 = \frac{c^2}{4m^2} - \frac{k}{m} = \frac{c^2 - 4mk}{4m^2} > 0$$

so that $c^2 > 4mk$. This shows that c is usually large compared to k , that is, the damping force dominates the restoring force of the spring, and the weight returns to the equilibrium position relatively quickly. In particular, this happens if the fluid has a high viscosity, as is true for heavy oil or grease.

The manner in which y approaches 0 depends on the constants C_1 and C_2 in the general solution. If the constants are both positive, the graph has the general shape shown in (i) of Figure 19.10, whereas if they have opposite signs, the graph resembles that shown in (ii) of the figure.

Case (ii) $p^2 - \omega^2 = 0$: Critically damped.

In this case the auxiliary equation has a double root $-p$, and by Theorem (19.14), the general solution of the differential equation is

$$y = e^{-pt}(C_1 + C_2 t).$$

The typical graph sketched in Figure 19.11 is similar to that in (i) of Figure 19.9; however, any decrease in the damping force leads to the oscillatory motion discussed in the following case.

Case (iii) $p^2 - \omega^2 < 0$: Underdamped.

The roots of the auxiliary equation are complex conjugates $a \pm bi$, and by Theorem (19.19) the general solution of the differential equation is

$$y = e^{-at}(C_1 \sin bt + C_2 \cos bt).$$

In this case c is usually smaller than k , and the spring oscillates while returning to the equilibrium position, as illustrated in Figure 19.12. This type of motion could occur in a worn shock absorber in an automobile.

Example 2 A 24-pound weight stretches a spring 1 ft beyond its natural length. If the weight is pushed downward from its equilibrium position with an initial velocity of 2 ft/sec, and if the damping force is $-9(dy/dt)$, find a formula for the displacement y of the weight at any time t .

Solution By Hooke's Law, $24 = k(1)$, or $k = 24$. The mass of the weight is $m = W/g = \frac{24}{32} = \frac{3}{4}$. Using the notation introduced in (19.25),

$$2p = \frac{c}{m} = \frac{9}{\frac{3}{4}} = 12 \quad \text{and} \quad \omega^2 = \frac{k}{m} = \frac{24}{\frac{3}{4}} = 32.$$

Substitution in (19.25) gives us

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 32y = 0.$$

It may be verified that the roots of the auxiliary equation are -8 and -4 . Thus the motion is overdamped and the general solution of the differential equation is

$$y = C_1e^{-4t} + C_2e^{-8t}.$$

Letting $t = 0$,

$$0 = C_1 + C_2 \quad \text{or} \quad C_2 = -C_1.$$

Since

$$\frac{dy}{dt} = -4C_1e^{-4t} - 8C_2e^{-8t}$$

we have, at $t = 0$,

$$2 = -4C_1 - 8C_2 = -4C_1 - 8(-C_1) = 4C_1.$$

Consequently, $C_1 = \frac{1}{2}$ and $C_2 = -C_1 = -\frac{1}{2}$. Thus the displacement y of the weight at time t is

$$y = \frac{1}{2}e^{-4t} - \frac{1}{2}e^{-8t} = \frac{1}{2}e^{-4t}(1 - e^{-4t}).$$

The graph is similar in appearance to that in (i) of Figure 19.10. In Exercise 9 you are asked to find the damping force that will produce critical damping. ■

19.8 Exercises

- 1 A 5-pound weight stretches a spring 6 in. beyond its natural length. If the weight is raised 4 in. and then released with zero velocity, find a formula for the displacement of the weight at any time t .
- 2 A 10-pound weight stretches a spring 8 in. beyond its natural length. If the weight is pulled down another 3 in. and then released with a downward velocity of 6 in./sec, find a formula for the displacement at any time t .
- 3 A 10-pound weight stretches a spring 1 ft beyond its natural length. If the weight is released from its equilibrium position with a downward velocity of 2 ft/sec, and if the frictional force is $-5(dy/dt)$, find a formula for the displacement of the weight.
- 4 A 6-pound weight is suspended from a spring whose spring constant is 48 lb/ft. Initially the weight has a downward velocity of 4 ft/sec from a position 5 in. below the equilibrium point. Find a formula for the displacement.
- 5 A 4-pound weight stretches a spring 3 in. beyond its natural length. If the weight is pulled down another 4 in. and released with zero velocity in a medium where the damping force is $-2(dy/dt)$, find the displacement y at any time t .

- 6 A spring stretches $\frac{1}{2}$ ft when an 8-pound weight is attached. If the weight is released from the equilibrium position with an upward velocity of 1 ft/sec, and if the damping force is $-4(dy/dt)$, find a formula for the displacement y at any time t .
- 7 Describe a spring/weight system that leads to the following differential equation and initial values:
- $$\frac{1}{4} \frac{d^2 y}{dt^2} + \frac{dy}{dt} + 6y = 0;$$
- $$y = -2 \quad \text{and} \quad \frac{dy}{dt} = -1 \quad \text{at } t = 0.$$
- 8 A 4-pound weight stretches a spring 1 ft. If the weight moves in a medium whose damping force is $-c(dy/dt)$ where $c > 0$, for what values of c is the motion (a) overdamped? (b) critically damped? (c) underdamped?
- 9 In Example 2 of this section, find the damping force that will produce critical damping.
- 10 Suppose an undamped weight of mass m is attached to a spring having spring constant k and that an external periodic force given by $F \sin \alpha t$ is applied to the weight.
- (a) Show that the differential equation that describes the motion of the weight is
- $$\frac{d^2 y}{dt^2} + \omega^2 y = \frac{F}{m} \sin \alpha t$$
- where $\omega^2 = k/m$.
- (b) Prove that the displacement y of the weight is given by
- $$y = C_1 \cos \omega t + C_2 \sin \omega t + C \sin \alpha t,$$
- where $C = F/m(\omega^2 - \alpha^2)$. This type of motion is called **forced vibration**.

19.9 Series Solutions of Differential Equations

As shown in Chapter 11, a power series $\sum a_n x^n$ determines a function f , such that

$$y = f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$$

for every x in the interval of convergence. Moreover, series representations for the derivatives of f may be obtained by differentiating each term. Thus

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots = \sum_{n=1}^{\infty} n a_n x^{n-1},$$

$$y'' = 2a_2 + 3 \cdot 2a_3 x + 4 \cdot 3a_4 x^2 + \cdots = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2},$$

etc. Power series may be used to solve certain differential equations. In this case, the solution is often expressed as an infinite series, and is called a **series solution** of the differential equation.

Example 1 Find a series solution of the differential equation $y' = 2xy$.

Solution If the solution is given by $y = \sum a_n x^n$, then $y' = \sum n a_n x^{n-1}$, and substitution in the differential equation gives us

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = 2x \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2a_n x^{n+1}.$$

It is convenient to change the summation on the left so that the same power of x appears as in the summation on the right. This may be accomplished by replacing n by $n + 2$ and beginning the summation at $n = -1$. Thus

$$\sum_{n=-1}^{\infty} (n+2)a_{n+2}x^{n+1} = \sum_{n=0}^{\infty} 2a_nx^{n+1}, \quad \text{or}$$

$$a_1 + 2a_2x + \cdots + (n+2)a_{n+2}x^{n+1} + \cdots = 2a_0x + \cdots + 2a_nx^{n+1} + \cdots$$

Comparing coefficients we see that $a_1 = 0$ and $(n+2)a_{n+2} = 2a_n$ if $n \geq 0$. Consequently, the coefficients are given by

$$a_1 = 0 \quad \text{and} \quad a_{n+2} = \frac{2}{n+2}a_n \quad \text{if} \quad n \geq 0.$$

In particular,

$$a_1 = 0, \quad a_2 = a_0, \quad a_3 = \frac{2}{3}a_1 = 0, \quad a_4 = \frac{1}{2}a_2 = \frac{1}{2}a_0, \quad a_5 = \frac{2}{5}a_3 = 0,$$

$$a_6 = \frac{1}{3}a_4 = \frac{1}{2 \cdot 3}a_0, \quad a_7 = \frac{2}{7}a_5 = 0, \quad a_8 = \frac{1}{4}a_6 = \frac{1}{2 \cdot 3 \cdot 4}a_0,$$

etc. It can be shown that if n is odd, then $a_n = 0$, whereas $a_{2n} = (1/n!)a_0$ for every positive integer n . The series solution is, therefore,

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 \left(1 + x^2 + \frac{1}{2!}x^4 + \cdots + \frac{1}{n!}x^{2n} + \cdots \right). \quad \blacksquare$$

It follows from (19.16) that the series solution in Example 1 can be written as $y = a_0 e^{(x^2)}$. Indeed, this form may be found directly from $y' = 2xy$ by using the separation of variables technique discussed in Section 19.1. The objective in Example 1, however, was to illustrate series solutions and not to find the solution in the simplest manner. In many instances it is impossible to find the sum of $\sum a_n x^n$ and the solution must be left in series form.

Example 2 Solve the differential equation $y'' - xy' - 2y = 0$.

Solution Substituting

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad \text{and} \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

we obtain

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - x \sum_{n=1}^{\infty} n a_n x^{n-1} - 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\text{or} \quad \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} 2 a_n x^n.$$

We next adjust the summation on the left so that the power x^n appears instead of x^{n-2} . This can be accomplished by replacing n by $n + 2$ and starting the summation at $n = 0$. This gives us

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n = \sum_{n=0}^{\infty} (n+2)a_nx^n.$$

Comparing coefficients we see that $(n+2)(n+1)a_{n+2} = (n+2)a_n$, that is,

$$a_{n+2} = \frac{1}{n+1} a_n.$$

Letting $n = 0, 1, 2, \dots, 7$ leads to the following form for the coefficients a_k :

$$\begin{aligned} a_2 &= a_0 & a_3 &= \frac{1}{2} a_1 \\ a_4 &= \frac{1}{3} a_2 = \frac{1}{3} a_0 & a_5 &= \frac{1}{4} a_3 = \frac{1}{2 \cdot 4} a_1 \\ a_6 &= \frac{1}{5} a_4 = \frac{1}{3 \cdot 5} a_0 & a_7 &= \frac{1}{6} a_5 = \frac{1}{2 \cdot 4 \cdot 6} a_1 \\ a_8 &= \frac{1}{7} a_6 = \frac{1}{3 \cdot 5 \cdot 7} a_0 & a_9 &= \frac{1}{8} a_7 = \frac{1}{2 \cdot 4 \cdot 6 \cdot 8} a_1 \end{aligned}$$

In general,

$$a_{2n} = \frac{1}{1 \cdot 3 \cdots (2n-1)} a_0, \quad a_{2n+1} = \frac{1}{2 \cdot 4 \cdots (2n)} a_1 = \frac{1}{2^n n!} a_1.$$

The solution $y = \sum a_n x^n$ may, therefore, be expressed as a sum of two infinite series:

$$y = a_0 \left[1 + \sum_{n=1}^{\infty} \frac{1}{1 \cdot 3 \cdots (2n-1)} x^{2n} \right] + a_1 \sum_{n=0}^{\infty} \frac{1}{2^n n!} x^{2n+1}. \quad \blacksquare$$

19.9 Exercises

Find series solutions for the differential equations in Exercises 1–6.

1 $y'' + y = 0$

2 $y'' - 4y = 0$

3 $y'' - 2xy = 0$

4 $y'' + 2xy' + y = 0$

5 $\frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 2y = 0$

6 $\frac{d^2 y}{dx^2} + x^2 y = 0$

7 $(x+1)y' = 3y$

8 $y' = 4x^3 y$

9 $y'' - y = 5x$

10 $y'' - xy = x^4$

11 $(x^2 - 1)y'' + 6xy' + 4y = -4$

12 $y'' + y = e^x$

19.10 Review: Concepts

Discuss methods for solving the following types of differential equations.

- | | |
|-------------|-----------------------|
| 1 Separable | 3 Homogeneous |
| 2 Exact | 4 First-order linear |
| | 5 Bernoulli |
| | 6 Second-order linear |

Review: Exercises

Solve the differential equations in Exercises 1–40.

- | | |
|---|---|
| 1 $xe^y dx - \csc x dy = 0$ | 20 $y'' - y' - 6y = e^{2x}$ |
| 2 $(2xy - 1) dx + (x^2 + 2y) dy = 0$ | 21 $\sec^2 y dx = \sqrt{1 - x^2} dy - x \sec^2 y dx$ |
| 3 $(3x - y) dx + (x + y) dy = 0$ | 22 $(2x - yx^{-1} + \ln y) dx + (xy^{-1} - \ln x + 1) dy = 0$ |
| 4 $y' + 4y = e^{-x}$ | 23 $y' + y = e^{4x}$ |
| 5 $y^2 - ye^{-x} + (e^{-x} + 2xy + 3)y' = 0$ | 24 $y'' + 2y' = 0$ |
| 6 $(x^2y + x^2) dy + y dx = 0$ | 25 $y'' - 3y' + 2y = e^{5x}$ |
| 7 $y \tan x + y' = 2 \sec x$ | 26 $xe^y dx - (x + 1)y dy = 0$ |
| 8 $(x^2 + y^2) - xyy' = 0$ | 27 $xy' + y = (x - 2)^2$ |
| 9 $y\sqrt{1 - x^2} y' = \sqrt{1 - y^2}$ | 28 $(3x^2 - 2xy^2 + 1) dx + (y^2 - 2x^2y) dy = 0$ |
| 10 $(2y + x^3) dx - x dy = 0$ | 29 $y'' - y' - 20y = xe^{-x}$ |
| 11 $\left(2x \sin \frac{y}{x} - y \cos \frac{y}{x}\right) dx + x \cos \frac{y}{x} dy = 0$ | 30 $(x^2 - y^2)y' + 3xy = 0$ |
| 12 $(y \cos x - 2x) + (\sin x + 2y)y' = 0$ | 31 $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 7y = 0$ |
| 13 $xy' - 2y = x^3y^3$ | 32 $\frac{d^2y}{dx^2} + y = \csc x$ |
| 14 $y'' + y' - 6y = 0$ | 33 $e^{x+y} dx - \csc x dy = 0$ |
| 15 $y'' - 8y' + 16y = 0$ | 34 $y'' + 10y' + 25y = 0$ |
| 16 $y'' - 6y' + 25y = 0$ | 35 $\cot x dy = (y - \cos x) dx$ |
| 17 $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$ | 36 $y'' + y' + y = e^x \cos x$ |
| 18 $\frac{d^2y}{dx^2} = y + \sin x$ | 37 $(y - 2e^{-2x} \sin y) dx + (e^{-2x} \cos y + x) dy = 0$ |
| 19 $y'' - y = e^x \sin x$ | 38 $y''' = 0$ |
| | 39 $y' + y \csc x = \tan x$ |
| | 40 $xy^2y' = x^3 + y^3$ |

Appendices

I Mathematical Induction

The method of proof known as **mathematical induction** may be used to show that certain statements or formulas are true for all positive integers. For example, if n is a positive integer, let P_n denote the statement

$$(xy)^n = x^n y^n$$

where x and y are real numbers. Thus, P_1 represents the statement $(xy)^1 = x^1 y^1$, P_2 denotes $(xy)^2 = x^2 y^2$, P_3 is $(xy)^3 = x^3 y^3$, etc. It is easy to show that P_1 , P_2 , and P_3 are *true* statements. However, since the set of positive integers is infinite, it is impossible to check the validity of P_n for every positive integer n . To give a proof, the method provided by (I.2) is required. This method is based on the following fundamental axiom.

Axiom of Mathematical Induction (I.1)

Suppose a set S of positive integers has the following two properties:

- (i) S contains the integer 1.
- (ii) Whenever S contains a positive integer k , S also contains $k + 1$.

Then S contains every positive integer.

The reader should have little reluctance about accepting (I.1). If S is a set of positive integers satisfying property (ii), then whenever S contains an

arbitrary positive integer k , it must also contain the next positive integer, $k + 1$. If S also satisfies property (i), then S contains 1 and hence by (ii), S contains $1 + 1$, or 2. Applying (ii) again, we see that S contains $2 + 1$, or 3. Once again, S must contain $3 + 1$, or 4. If we continue in this manner, it can be argued that if n is any *specific* positive integer, then n is in S , since we can proceed a step at a time as above, eventually reaching n . Although this argument does not *prove* (I.1), it certainly makes it plausible.

We shall use (I.1) to establish the following fundamental principle.

Principle of Mathematical Induction
(I.2)

If with each positive integer n there is associated a statement P_n , then all the statements P_n are true provided the following two conditions hold:

- (i) P_1 is true.
- (ii) Whenever k is a positive integer such that P_k is true, then P_{k+1} is also true.

Proof Assume that (i) and (ii) of (I.2) hold and let S denote the set of all positive integers n such that P_n is true. By assumption, P_1 is true and consequently 1 is in S . Thus S satisfies property (i) of (I.1). Whenever S contains a positive integer k , then by the definition of S , P_k is true and hence from assumption (ii) of (I.2), P_{k+1} is also true. This means that S contains $k + 1$. We have shown that whenever S contains a positive integer k , then S also contains $k + 1$. Consequently, property (ii) of (I.1) is true. Hence by (I.1), S contains every positive integer; that is, P_n is true for every positive integer n . $\square$

There are other variations of the principle of mathematical induction. One variation appears in (I.5). In most of our work the statement P_n will usually be given in the form of an equation involving the arbitrary positive integer n , as in our illustration $(xy)^n = x^n y^n$.

When applying (I.2), the following two steps should always be used.

(I.3)

Step (i) Prove that P_1 is true.

Step (ii) Assume that P_k is true and prove that P_{k+1} is true.

Step (ii) is usually the most confusing for the beginning student. We do not *prove* that P_k is true (except for $k = 1$). Instead, we show that *if* P_k is true, then the statement P_{k+1} is true. That is all that is necessary according to (I.2). The assumption that P_k is true is referred to as the **induction hypothesis**.

Example 1 Prove that for every positive integer n , the sum of the first n positive integers is $n(n + 1)/2$.

Solution For any positive integer n , let P_n denote the statement

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

where by convention, when $n \leq 4$, the left side is adjusted so that there are precisely n terms in the sum. The following are some special cases of P_n :

If $n = 2$, then P_2 is

$$1 + 2 = \frac{2(2+1)}{2}, \quad \text{or} \quad 3 = 3.$$

If $n = 3$, then P_3 is

$$1 + 2 + 3 = \frac{3(3+1)}{2}, \quad \text{or} \quad 6 = 6.$$

If $n = 5$, then P_5 is

$$1 + 2 + 3 + 4 + 5 = \frac{5(5+1)}{2}, \quad \text{or} \quad 15 = 15.$$

We wish to show that P_n is true for every n . Although it is instructive to check P_n for several values of n as we have done, it is unnecessary to do so. We need only follow steps (i) and (ii) of (I.3).

(i) If we substitute $n = 1$ in P_n then by convention the left side collapses to 1 and the right side is $\frac{1(1+1)}{2}$, which also equals 1. This proves that P_1 is true.

(ii) Assume that P_k is true. Thus the induction hypothesis is

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}.$$

Our goal is to prove that P_{k+1} is true, that is,

$$1 + 2 + 3 + \cdots + (k+1) = \frac{(k+1)[(k+1)+1]}{2}.$$

By the induction hypothesis we already have a formula for the sum of the first k positive integers. Hence a formula for the sum of the first $k+1$ positive integers may be found simply by adding $(k+1)$ to both sides of the induction hypothesis. Doing so and simplifying, we obtain

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{k^2 + 3k + 2}{2} \\ &= \frac{(k+1)(k+2)}{2} \\ &= \frac{(k+1)[(k+1)+1]}{2}. \end{aligned}$$

We have shown that P_{k+1} is true, and therefore the proof by mathematical induction is complete. ■

The Laws of Exponents can be proved by mathematical induction. To apply (I.3), we shall use the following definition of exponents.

Definition (I.4)

If x is any real number, then

- (i) $x^1 = x$.
- (ii) whenever k is a positive integer for which x^k is defined, let $x^{k+1} = x^k \cdot x$.

A definition such as (I.4) is called a **recursive definition**. In general, if a concept is defined for every positive integer n in such a way that the case corresponding to $n = 1$ is given, and if it is also stated how any case after the first is obtained from the preceding one, then the definition is a recursive definition. For example, by (i) of (I.4) we have $x^1 = x$. Next, applying (ii) of (I.4) we obtain

$$x^2 = x^{1+1} = x^1 \cdot x = x \cdot x.$$

Since x^2 is now defined, we may employ (ii) again (with $k = 2$), obtaining

$$x^3 = x^{2+1} = x^2 \cdot x = (x \cdot x) \cdot x.$$

This defines x^3 , and hence (ii) of (I.4) may be used again to obtain x^4 . Thus,

$$x^4 = x^{3+1} = x^3 \cdot x = [(x \cdot x) \cdot x] \cdot x.$$

Observe that this agrees with the formulation of x^n as a product of x by itself n times. It can be shown (by mathematical induction) that (I.4) defines x^n for every positive integer n .

Example 2 If x is a real number, prove that $x^m \cdot x^n = x^{m+n}$ for all positive integers m and n .

Solution Let m be an arbitrary positive integer. For each positive integer n , let P_n denote the statement

$$x^m \cdot x^n = x^{m+n}.$$

We shall use (I.3) to prove that P_n is true for every positive integer n .

(i) To show that P_1 is true we may use (i) and (ii) of (I.4) as follows:

$$x^m \cdot x^1 = x^m \cdot x = x^{m+1}.$$

Hence P_1 is true.

(ii) Assume that P_k is true. Thus the induction hypothesis is

$$x^m \cdot x^k = x^{m+k}.$$

We wish to prove that P_{k+1} is true, that is,

$$x^m \cdot x^{k+1} = x^{m+(k+1)}.$$

The proof may be arranged as follows, where reasons are stated to the right of each step.

$$\begin{aligned} x^m \cdot x^{k+1} &= x^m \cdot (x^k \cdot x) && \text{(ii) of (I.4)} \\ &= (x^m \cdot x^k) \cdot x && \text{(associative law in } \mathbb{R}) \\ &= x^{m+k} \cdot x && \text{(induction hypothesis)} \\ &= x^{(m+k)+1} && \text{(ii) of (I.4)} \\ &= x^{m+(k+1)} && \text{(associative law for integers).} \end{aligned}$$

By (I.3) the proof by induction is complete. ■

Consider a positive integer j and suppose that with each integer $n \geq j$ there is associated a statement P_n . For example, if $j = 6$, then the statements are numbered P_6, P_7, P_8 , and so on. The principle of mathematical induction may be extended to cover this situation. Just as before, two steps are used. Specifically, to prove that the statements S_n are true for $n \geq j$, we use the following two steps.

(I.5)

- (i') Prove that S_j is true,
(ii') Assume that S_k is true for $k \geq j$ and prove that S_{k+1} is true.

Example 3 Let a be a nonzero real number such that $a > -1$. Prove that $(1 + a)^n > 1 + na$ for every integer $n \geq 2$.

Solution For each positive integer n , let P_n denote $(1 + a)^n > 1 + na$. Note that P_1 is false since $(1 + a)^1 = 1 + (1)(a)$. However, we can show that P_n is true for $n \geq 2$ by using (I.5) with $j = 2$.

(i') We first note that $(1 + a)^2 = 1 + 2a + a^2$. Since $a \neq 0$, we have $a^2 > 0$ and therefore $1 + 2a + a^2 > 1 + 2a$. Thus $(1 + a)^2 > 1 + 2a$, and hence P_2 is true.

(ii') Assume that P_k is true for $k \geq 2$. Thus the induction hypothesis is

$$(1 + a)^k > 1 + ka.$$

We wish to show that P_{k+1} is true, that is,

$$(1 + a)^{k+1} > 1 + (k + 1)a.$$